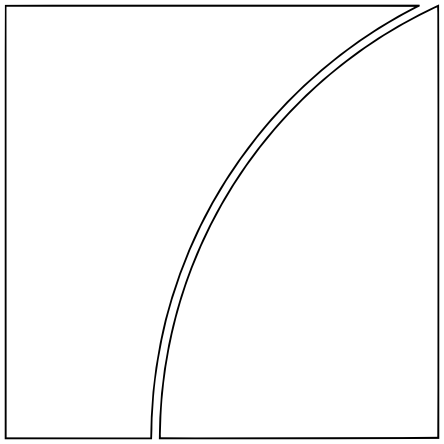
Basel Committee on Banking Supervision

MAR

Calculation of RWA for market risk

This standard describes how to calculate capital requirements for market risk and credit valuation adjustment risk.



BANK FOR INTERNATIONAL SETTLEMENTS

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MAR10

Market risk terminology

Reflects revisions in the standardised and internal models approach for market risk, including the shift to an expected shortfall measure. Also reflects the revised implementation date announced on 27 March 2020.

Version effective as of 01 Jan 2023

Reflects revisions in the standardised and internal models approach for market risk, including the shift to an expected shortfall measure. Also reflects the revised implementation date announced on 27 March 2020.

General terminology

10.1 Market risk: the risk of losses in on- and off-balance sheet risk positions arising from movements in market prices.

10.2 Notional value: the notional value of a derivative instrument is equal to the number of units underlying the instrument multiplied by the current market value of each unit of the underlying.

10.3 Trading desk: a group of traders or trading accounts in a business line within a bank that follows defined trading strategies with the goal of generating revenues or maintaining market presence from assuming and managing risk.

10.4 Pricing model: a model that is used to determine the value of an instrument (mark-to-market or mark-to-model) as a function of pricing parameters or to determine the change in the value of an instrument as a function of risk factors. A pricing model may be the combination of several calculations; eg a first valuation technique to compute a price, followed by valuation adjustments for risks that are not incorporated in the first step.

Terminology for financial instruments

10.5 Financial instrument: any contract that gives rise to both a financial asset of one entity and a financial liability or equity instrument of another entity. Financial instruments include both primary financial instruments (or cash instruments) and derivative financial instruments.

10.6 Instrument: the term used to describe financial instruments, instruments on foreign exchange (FX) and commodities.

10.7 Embedded derivative: a component of a financial instrument that includes a non-derivative host contract. For example, the conversion option in a convertible bond is an embedded derivative.

10.8 Look-through approach: an approach in which a bank determines the relevant capital requirements for a position that has underlyings (such as an index instrument, multi-underlying option, or an equity investment in a fund) as if the underlying positions were held directly by the bank.

Terminology for market risk capital requirement calculations

10.9 Risk factor: a principal determinant of the change in value of an instrument (eg an exchange rate or interest rate).

10.10 Risk position: the portion of the current value of an instrument that may be subject to losses due to movements in a risk factor. For example, a bond denominated in a currency different to a bank's reporting currency has risk positions in general interest rate risk, credit spread risk (non-securitisation) and FX risk, where the risk positions are the potential losses to the current value of the instrument that could occur due to a change in the relevant underlying risk factors (interest rates, credit spreads, or exchange rates).

10.11 Risk bucket: a defined group of risk factors with similar characteristics.

10.12 Risk class: a defined list of risks that are used as the basis for calculating market risk capital requirements: general interest rate risk, credit spread risk (non-securitisation), credit spread risk (securitisation: non-correlation trading portfolio), credit spread risk (securitisation: correlation trading portfolio), FX risk, equity risk and commodity risk.

Terminology for risk metrics

10.13 Sensitivity: a bank's estimate of the change in value of an instrument due to a small change in one of its underlying risk factors. Delta and vega risks are sensitivities.

10.14 Delta risk: the linear estimate of the change in value of a financial instrument due to a movement in the value of a risk factor. The risk factor could be the price of an equity or commodity, or a change in an interest rate, credit spread or FX rate.

10.15 Vega risk: the potential loss resulting from the change in value of a derivative due to a change in the implied volatility of its underlying.

10.16 Curvature risk: the additional potential loss beyond delta risk due to a change in a risk factor for financial instruments with optionality. In the standardised approach in the market risk framework, it is based on two stress scenarios involving an upward shock and a downward shock to each regulatory risk factor.

10.17 Value at risk (VaR): a measure of the worst expected loss on a portfolio of instruments resulting from market movements over a given time horizon and a pre-defined confidence level.

10.18 Expected shortfall (ES): a measure of the average of all potential losses exceeding the VaR at a given confidence level.

10.19 Jump-to-default (JTD): the risk of a sudden default. JTD exposure refers to the loss that could be incurred from a JTD event.

10.20 Liquidity horizon: the time assumed to be required to exit or hedge a risk position without materially affecting market prices in stressed market conditions.

Terminology for hedging and diversification

10.21 Basis risk: the risk that prices of financial instruments in a hedging strategy are imperfectly correlated, reducing the effectiveness of the hedging strategy.

10.22 Diversification: the reduction in risk at a portfolio level due to holding risk positions in different instruments that are not perfectly correlated with one another.

10.23 Hedge: the process of counterbalancing risks from exposures to long and short risk positions in correlated instruments.

10.24 Offset: the process of netting exposures to long and short risk positions in the same risk factor.

10.25 Standalone: being capitalised on a stand-alone basis means that risk positions are booked in a discrete, non-diversifiable trading book portfolio so that the risk associated with those risk positions cannot diversify, hedge or offset risk arising from other risk positions, nor be diversified, hedged or offset by them.

Terminology for risk factor eligibility and modellability

10.26 Real prices: a term used for assessing whether risk factors pass the risk factor eligibility test. A price will be considered real if it is (i) a price from an actual transaction conducted by the bank, (ii) a price from an actual transaction between other arm's length parties (eg at an exchange), or (iii) a price taken from a firm quote (ie a price at which the bank could transact with an arm's length party).

10.27 Modellable risk factor: risk factors that are deemed modellable, based on the number of representative real price observations and additional qualitative principles related to the data used for the calibration of the ES model. Risk factors that do not meet the requirements for the risk factor eligibility test are deemed as non-modellable risk factors (NMRF).

Terminology for internal model validation

10.28 Backtesting: the process of comparing daily actual and hypothetical profits and losses with model-generated VaR measures to assess the conservatism of risk measurement systems.

10.29 Profit and loss (P&L) attribution (PLA): a method for assessing the robustness of banks' risk management models by comparing the risk-theoretical P&L predicted by trading desk risk management models with the hypothetical P&L.

10.30 Trading desk risk management model: the trading desk risk management model (pertaining to in-scope desks) includes all risk factors that are included in the bank's ES model with supervisory parameters and any risk factors deemed not modellable, which are therefore not included in the ES model for calculating the respective regulatory capital requirement, but are included in NMRFs.

10.31 Actual P&L (APL): the actual P&L derived from the daily P&L process. It includes intraday trading as well as time effects and new and modified deals, but excludes fees and commissions as well as valuation adjustments for which separate regulatory capital approaches have been otherwise specified as part of the rules or which are deducted from Common Equity Tier 1. Any other valuation adjustments that are market risk-related must be included in the APL. As is the case for the hypothetical P&L, the APL should include FX and commodity risks from positions held in the banking book.

10.32 Hypothetical P&L (HPL): the daily P&L produced by revaluing the positions held at the end of the previous day using the market data at the end of the current day. Commissions, fees, intraday trading and new/modified deals, valuation adjustments for which separate regulatory capital approaches have been otherwise specified as part of the rules and valuation adjustments which are deducted from CET1 are excluded from the HPL. Valuation adjustments updated daily should usually be included in the HPL. Time effects should be treated in a consistent manner in the HPL and risk-theoretical P&L.

10.33 Risk-theoretical P&L (RTPL): the daily desk-level P&L that is predicted by the valuation engines in the trading desk risk management model using all risk factors used in the trading desk risk management model (ie including the NMRFs).

Terminology for credit valuation adjustment risk

10.34 Credit valuation adjustment (CVA): an adjustment to the valuation of a derivative transaction to account for the credit risk of contracting parties.

10.35 CVA risk: the risk of changes to CVA arising from changes in credit spreads of the contracting parties, compounded by changes to the value or variability in the value of the underlying of the derivative transaction.

MAR11

Definitions and application of market risk

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

Version effective as of 01 Jan 2023

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

Definition and scope of application

11.1 Market risk is defined as the risk of losses arising from movements in market prices. The risks subject to market risk capital requirements include but are not limited to:

(1) default risk, interest rate risk, credit spread risk, equity risk, foreign exchange (FX) risk and commodities risk for trading book instruments; and

(2) FX risk and commodities risk for banking book instruments.

11.2 All transactions, including forward sales and purchases, shall be included in the calculation of capital requirements as of the date on which they were entered into. Although regular reporting will in principle take place only at intervals (quarterly in most countries), banks are expected to manage their market risk in such a way that the capital requirements are being met on a continuous basis, including at the close of each business day. Supervisory authorities have at their disposal a number of effective measures to ensure that banks do not window-dress by showing significantly lower market risk positions on reporting dates. Banks will also be expected to maintain strict risk management systems to ensure that intraday exposures are not excessive. If a bank fails to meet the capital requirements at any time, the national authority shall ensure that the bank takes immediate measures to rectify the situation.

11.3 A matched currency risk position will protect a bank against loss from movements in exchange rates, but will not necessarily protect its capital adequacy ratio. If a bank has its capital denominated in its domestic currency and has a portfolio of foreign currency assets and liabilities that is completely matched, its capital/asset ratio will fall if the domestic currency depreciates. By running a short risk position in the domestic currency, the bank can protect its capital adequacy ratio, although the risk position would lead to a loss if the domestic currency were to appreciate. Supervisory authorities are free to allow banks to protect their capital adequacy ratio in this way and exclude certain currency risk positions from the calculation of net open currency risk positions, subject to meeting each of the following conditions:

(1) The risk position is taken or maintained for the purpose of hedging partially or totally against the potential that changes in exchange rates could have an adverse effect on its capital ratio.

(2) The risk position is of a structural (ie non-dealing) nature such as positions stemming from:

(a) investments in affiliated but not consolidated entities denominated in foreign currencies; or

(b) investments in consolidated subsidiaries or branches denominated in foreign currencies.

(3) The exclusion is limited to the amount of the risk position that neutralises the sensitivity of the capital ratio to movements in exchange rates.

(4) The exclusion from the calculation is made for at least six months.

(5) The establishment of a structural FX position and any changes in its position must follow the bank's risk management policy for structural FX positions. This policy must be pre-approved by the national supervisor.

(6) Any exclusion of the risk position needs to be applied consistently, with the exclusionary treatment of the hedge remaining in place for the life of the assets or other items.

(7) The bank is subject to a requirement by the national supervisor to document and have available for supervisory review the positions and amounts to be excluded from market

risk capital requirements.

11.4 No FX risk capital requirement need apply to positions related to items that are deducted from a bank's capital when calculating its capital base.

11.5 Holdings of capital instruments that are deducted from a bank's capital or risk weighted at 1250% are not allowed to be included in the market risk framework. This includes:

(1) holdings of the bank's own eligible regulatory capital instruments; and

(2) holdings of other banks', securities firms' and other financial entities' eligible regulatory capital instruments, as well as intangible assets, where the national supervisor requires that such assets are deducted from capital.

(3) Where a bank demonstrates that it is an active market-maker, then a national supervisor may establish a dealer exception for holdings of other banks', securities firms', and other financial entities' capital instruments in the trading book. In order to qualify for the dealer exception, the bank must have adequate systems and controls surrounding the trading of financial institutions' eligible regulatory capital instruments.

11.6 In the same way as for credit risk and operational risk, the capital requirements for market risk apply on a worldwide consolidated basis.

(1) Supervisory authorities may permit banking and financial entities in a group which is running a global consolidated trading book and whose capital is being assessed on a global basis to include just the net short and net long risk positions no matter where they are booked.[1]

(2) Supervisory authorities may grant this treatment only when the standardised approach in MAR20 to MAR23 permits a full offset of the risk position (ie risk positions of the opposite sign do not attract a capital requirement).

(3) Nonetheless, there will be circumstances in which supervisory authorities demand that the individual risk positions be taken into the measurement system without any offsetting or netting against risk positions in the remainder of the group. This may be needed, for example, where there are obstacles to the quick repatriation of profits from a foreign subsidiary or where there are legal and procedural difficulties in carrying out the timely management of risks on a consolidated basis.

(4) Moreover, all supervisory authorities will retain the right to continue to monitor the market risks of individual entities on a non-consolidated basis to ensure that significant imbalances within a group do not escape supervision. Supervisory authorities will be especially vigilant in ensuring that banks do not conceal risk positions on reporting dates in such a way as to escape measurement.

Footnotes

[1] *The positions of less than wholly owned subsidiaries would be subject to the generally accepted accounting principles in the country where the parent company is supervised.*

Methods of measuring market risk

11.7 In determining its market risk for regulatory capital requirements, a bank may choose between two broad methodologies: the standardised approach and internal models approach (IMA) for market risk, described in MAR20 to MAR23 and MAR30 to MAR33, respectively, subject to the approval of the national authorities. Supervisors may allow banks that maintain smaller or simpler trading books to use the simplified alternative to the

standardised approach as set out in MAR40.

(1) To determine the appropriateness of the simplified alternative for use by a bank for the purpose of its market risk capital requirements, supervisors may wish to consider the following indicative criteria:

(a) The bank should not be a global systemically important bank (G-SIB).

(b) The bank should not use the IMA for any of its trading desks.

(c) The bank should not hold any correlation trading positions.

(2) The use of the simplified alternative is subject to supervisory approval and oversight. Supervisors can mandate that banks with relatively complex or sizeable risks in particular risk classes apply the full standardised approach instead of the simplified alternative, even if those banks meet the indicative eligibility criteria referred to above.

11.8 All banks, except for those that are allowed to use the simplified alternative as set out in MAR11.7, must calculate the capital requirements using the standardised approach. Banks that are approved by the supervisor to use the IMA for market risk capital requirements must also calculate and report the capital requirement values calculated as set out below.

(1) A bank that uses the IMA for any of its trading desks must also calculate the capital requirement under the standardised approach for all instruments across all trading desks, regardless of whether those trading desks are eligible for the IMA.

(2) In addition, a bank that uses the IMA for any of its trading desks must calculate the standardised approach capital requirement for each trading desk that is eligible for the IMA as if that trading desk were a standalone regulatory portfolio (ie with no offsetting across trading desks). This will:

(a) serve as an indication of the fallback capital requirement for those desks that fail the eligibility criteria for inclusion in the bank's internal model as outlined in MAR30, MAR32 and MAR33;

(b) generate information on the capital outcomes of the internal models relative to a consistent benchmark and facilitate comparison in implementation between banks and/or across jurisdictions;

(c) monitor over time the relative calibration of standardised and modelled approaches, facilitating adjustments as needed; and

(d) provide macroprudential insight in an ex ante consistent format.

11.9 All banks must calculate the market risk capital requirement using the standardised approach for the following:

(1) securitisation exposures; and

(2) equity investments in funds that cannot be looked through but are assigned to the trading book in accordance to the conditions set out in RBC25.8(5)(b).

MAR12

Definition of trading desk

First version in the format of the consolidated framework updated to take account of the revised implementation date announced on 27 March 2020.

Version effective as of 01 Jan 2023

First version in the format of the consolidated framework updated to take account of the revised implementation date announced on 27 March 2020.

12.1 For the purposes of market risk capital calculations, a trading desk is a group of traders or trading accounts that implements a well defined business strategy operating within a clear risk management structure.

12.2 Trading desks are defined by the bank but subject to the regulatory approval of the supervisor for capital purposes.

(1) A bank should be allowed to propose the trading desk structure per their organisational structure, consistent with the requirements set out in MAR12.4.

(2) A bank must prepare a policy document for each trading desk it defines, documenting how the bank satisfies the key elements in MAR12.4.

(3) Supervisors will treat the definition of the trading desk as part of the initial model approval for the trading desk, as well as ongoing approval:

(a) Supervisors may determine, based on the size of the bank's overall trading operations, whether the proposed trading desk definitions are sufficiently granular.

(b) Supervisors should check that the bank's proposed definition of trading desk meets the criteria listed in key elements set out in MAR12.4.

12.3 Within this supervisory approved trading desk structure, banks may further define operational subdesks without the need for supervisory approval. These subdesks would be for internal operational purposes only and would not be used in the market risk capital framework.

12.4 The key attributes of a trading desk are as follows:

(1) A trading desk for the purposes of the regulatory capital charge is an unambiguously defined group of traders or trading accounts.

(a) A trading account is an indisputable and unambiguous unit of observation in accounting for trading activity.

(b) The trading desk must have one head trader and can have up to two head traders provided their roles, responsibilities and authorities are either clearly separated or one has ultimate oversight over the other.

(i) The head trader must have direct oversight of the group of traders or trading accounts.

(ii) Each trader or each trading account in the trading desk must have a clearly defined specialty (or specialities).

(c) Each trading account must only be assigned to a single trading desk. The desk must have a clearly defined risk scope consistent with its pre-established objectives. The scope should include specification of the desk's overall risk class and permitted risk factors.

(d) There is a presumption that traders (as well as head traders) are allocated to one trading desk. A bank can deviate from this presumption and may assign an individual trader to work across several trading desks provided it can be justified to the supervisor on the basis of sound management, business and/or resource allocation reasons. Such assignments must not be made for the only purpose of avoiding other trading desk requirements (eg to optimise the likelihood of success in the backtesting and profit and loss attribution tests).

(e) The trading desk must have a clear reporting line to bank senior management, and should have a clear and formal compensation policy clearly linked to the pre-established objectives of the trading desk.

(2) A trading desk must have a well defined and documented business strategy, including an annual budget and regular management information reports (including revenue, costs and risk-weighted assets).

(a) There must be a clear description of the economics of the business strategy for the trading desk, its primary activities and trading/hedging strategies.

(i) Economics: what is the economics behind the strategy (eg trading on the shape of the yield curve)? How much of the activities are customer driven? Does it entail trade origination and structuring, or execution services, or both?

(ii) Primary activities: what is the list of permissible instruments and, out of this list, which are the instruments most frequently traded?

(iii) Trading/hedging strategies: how would these instruments be hedged, what are the expected slippages and mismatches of hedges, and what is the expected holding period for positions?

(b) The management team at the trading desk (starting from the head trader) must have a clear annual plan for the budgeting and staffing of the trading desk.

(c) A trading desk's documented business strategy must include regular Management Information reports, covering revenue, costs and risk-weighted assets for the trading desk.

(3) A trading desk must have a clear risk management structure.

(a) Risk management responsibilities: the bank must identify key groups and personnel responsible for overseeing the risk-taking activities at the trading desk.

(b) A trading desk must clearly define trading limits based on the business strategy of the trading desk and these limits must be reviewed at least annually by senior management at the bank. In setting limits, the trading desk must have:

(i) well defined trading limits or directional exposures at the trading desk level that are based on the appropriate market risk metric (eg sensitivity of credit spread risk and/or jump-to-default for a credit trading desk), or just overall notional limits; and

(ii) well defined trader mandates.

(c) A trading desk must produce, at least weekly, appropriate risk management reports. This would include, at a minimum: profit and loss reports, which would be periodically reviewed, validated and modified (if necessary) by Product Control; andinternal and regulatory risk measure reports, including trading desk value-at-risk (VaR) / expected shortfall (ES), trading desk VaR/ES sensitivities to risk factors, backtesting and p-value.

12.5 The bank must prepare, evaluate, and have available for supervisors the following for all trading desks:

(1) inventory ageing reports;

(2) daily limit reports including exposures, limit breaches, and follow-up action;

(3) reports on intraday limits and respective utilisation and breaches for banks with active intraday trading; and

(4) reports on the assessment of market liquidity.

12.6 Any foreign exchange or commodity positions held in the banking book must be included in the market risk capital requirement as set out in MAR11.1. For regulatory capital calculation

purposes, these positions will be treated as if they were held on notional trading desks within the trading book.

FAQ

FAQ1 *How should the requirement for a "notional trading desk" be interpreted for banking book FX and commodities positions?*

A "notional trading desk" is a trading desk that need not have traders or trading accounts assigned to it, and need not meet the qualitative trading desk requirements set out in MAR12.

Banks that wish to use the internal models approach (IMA) to measure the FX or commodity risk of such "notional trading desks" must take either or both of the following actions:

- *transfer all or part of banking book FX and commodity risks to another trading desk via intra-trading book internal risk transfers (IRTs) (where trading desk requirements would continue to apply as appropriate for that desk), and/or*
- *apply for IMA approval for the notional trading desk. In this case, the notional desk only needs to meet the quantitative trading desk requirements.*

FAQ2 *Does the standard permit certain traders (ie global treasury desk heads or department heads) to have ownership and responsibilities in both trading book and banking book portfolios?*

Yes.

MAR20

Standardised approach: general provisions and structure

Introduces a revised approach based on expanded use of sensitivities as set out in the January 2019 market risk publication and updated to take account of the revised implementation date announced on 27 March 2020.

Version effective as of 01 Jan 2023

Introduces a revised approach based on expanded use of sensitivities as set out in the January 2019 market risk publication and updated to take account of the revised implementation date announced on 27 March 2020.

General provisions

20.1 The risk-weighted assets for market risk under the standardised approach are determined by multiplying the capital requirements calculated as set out in MAR20 to MAR23 by 12.5.

20.2 The standardised approach must be calculated and reported to the relevant supervisor on a monthly basis. Subject to supervisory approval, the standardised approach for market risks arising from non-banking subsidiaries of a bank may be calculated and reported to the relevant supervisor on a quarterly basis.

20.3 A bank must also determine its regulatory capital requirements for market risk according to the standardised approach for market risk at the demand of its supervisor.

Structure of the standardised approach

20.4 The standardised approach capital requirement is the simple sum of three components: the capital requirement under the sensitivities-based method, the default risk capital (DRC) requirement and the residual risk add-on (RRAO).

(1) The capital requirement under the sensitivities-based method must be calculated by aggregating three risk measures - delta, vega and curvature, as set out in MAR21:

(a) Delta: a risk measure based on sensitivities of an instrument to regulatory delta risk factors.

(b) Vega: a risk measure based on sensitivities to regulatory vega risk factors.

(c) Curvature: a risk measure which captures the incremental risk not captured by the delta risk measure for price changes in an option. Curvature risk is based on two stress scenarios involving an upward shock and a downward shock to each regulatory risk factor.

(d) The above three risk measures specify risk weights to be applied to the regulatory risk factor sensitivities. To calculate the overall capital requirement, the risk-weighted sensitivities are aggregated using specified correlation parameters to recognise diversification benefits between risk factors. In order to address the risk that correlations may increase or decrease in periods of financial stress, a bank must calculate three sensitivities-based method capital requirement values, based on three different scenarios on the specified values for the correlation parameters as set out in MAR21.6 and MAR21.7.

(2) The DRC requirement captures the jump-to-default risk for instruments subject to credit risk as set out in MAR22.2. It is calibrated based on the credit risk treatment in the banking book in order to reduce the potential discrepancy in capital requirements for similar risk exposures across the bank. Some hedging recognition is allowed for similar types of exposures (corporates, sovereigns, and local governments/municipalities).

(3) Additionally, the Committee acknowledges that not all market risks can be captured in the standardised approach, as this might necessitate an unduly complex regime. An RRAO is thus introduced to ensure sufficient coverage of market risks for instruments specified in MAR23.2. The calculation method for the RRAO is set out in MAR23.8.

Definition of correlation trading portfolio

20.5 For the purpose of calculating the credit spread risk capital requirement under the sensitivities based method and the DRC requirement, the correlation trading portfolio is defined as the set of instruments that meet the requirements of (1) or (2) below.

(1) The instrument is a securitisation position that meets the following requirements:

(a) The instrument is not a re-securitisation position, nor a derivative of securitisation exposures that does not provide a pro rata share in the proceeds of a securitisation tranche, where the definition of securitisation positon is identical to that used in the credit risk framework.

(b) All reference entities are single-name products, including single-name credit derivatives, for which a liquid two-way market exists,[1] including traded indices on these reference entities.

(c) The instrument does not reference an underlying that is treated as a retail exposure, a residential mortgage exposure, or a commercial mortgage exposure under the standardised approach to credit risk.

(d) The instrument does not reference a claim on a special purpose entity.

(2) The instrument is a non-securitisation hedge to a position described above.

Footnotes

[1] *A two-way market is deemed to exist where there are independent bona fide offers to buy and sell so that a price reasonably related to the last sales price or current bona fide competitive bid-ask quotes can be determined within one day and the transaction settled at such price within a relatively short time frame in conformity with trade custom.*

MAR21

Standardised approach: sensitivities-based method

Updated to take account of the revised implementation date announced on 27 March 2020, and to incorporate the FAQs published on 5 July 2024 and 23 March 2026.

Version effective as of 01 Jan 2023

Updated to take account of the revised implementation date announced on 27 March 2020, and to incorporate the FAQs published on 5 July 2024 and 23 March 2026.

Main concepts of the sensitivities-based method

21.1 The sensitivities of financial instruments to a prescribed list of risk factors are used to calculate the delta, vega and curvature risk capital requirements. These sensitivities are risk-weighted and then aggregated, first within risk buckets (risk factors with common characteristics) and then across buckets within the same risk class as set out in MAR21.8 to MAR21.14. The following terminology is used in the sensitivities-based method:

(1) Risk class: seven risk classes are defined (in MAR21.39 to MAR21.89).

(a) General interest rate risk (GIRR)

(b) Credit spread risk (CSR): non-securitisations

(c) CSR: securitisations (non-correlation trading portfolio, or non-CTP)

(d) CSR: securitisations (correlation trading portfolio, or CTP)

(e) Equity risk

(f) Commodity risk

(g) Foreign exchange (FX) risk

(2) Risk factor: variables (eg an equity price or a tenor of an interest rate curve) that affect the value of an instrument as defined in MAR21.8 to MAR21.14.

(3) Bucket: a set of risk factors that are grouped together by common characteristics (eg all tenors of interest rate curves for the same currency), as defined in MAR21.39 to MAR21.89.

(4) Risk position: the portion of the risk of an instrument that relates to a risk factor. Methodologies to calculate risk positions for delta, vega and curvature risks are set out in MAR21.3 to MAR21.5 and MAR21.15 to MAR21.26.

(a) For delta and vega risks, the risk position is a sensitivity to a risk factor.

(b) For curvature risk, the risk position is based on losses from two stress scenarios.

(5) Risk capital requirement: the amount of capital that a bank should hold as a consequence of the risks it takes; it is computed as an aggregation of risk positions first at the bucket level, and then across buckets within a risk class defined for the sensitivities-based method as set out in MAR21.3 to MAR21.7.

Instruments subject to each component of the sensitivities-based method

21.2 In applying the sensitivities-based method, all instruments held in trading desks as set out in MAR12 and subject to the sensitivities-based method (ie excluding instruments where the value at any point in time is purely driven by an exotic underlying as set out in MAR23.3), are subject to delta risk capital requirements. Additionally, the instruments specified in (1) to (4) are subject to vega and curvature risk capital requirements:

(1) Any instrument with optionality.[1]

(2) Any instrument with an embedded prepayment option[2] – this is considered an instrument with optionality according to above (1). The embedded option is subject to vega and curvature risk with respect to interest rate risk and CSR (non-securitisation and securitisation) risk classes. When the prepayment option is a behavioural option the instrument may also be subject to the residual risk add-on (RRAO) as per MAR23. The pricing model of the bank must reflect such behavioural patterns where relevant. For securitisation tranches, instruments in the securitised portfolio may have embedded prepayment options as well. In this case the securitisation tranche may be subject to the RRAO.

(3) Instruments whose cash flows cannot be written as a linear function of underlying notional. For example, the cash flows generated by a plain-vanilla option cannot be written as a linear function (as they are the maximum of the spot and the strike). Therefore, all options are subject to vega risk and curvature risk. Instruments whose cash flows can be written as a linear function of underlying notional are instruments without optionality (eg cash flows generated by a coupon bearing bond can be written as a linear function) and are not subject to vega risk nor curvature risk capital requirements.

(4) Curvature risks may be calculated for all instruments subject to delta risk, not limited to those subject to vega risk as specified in (1) to (3) above. For example, where a bank manages the non-linear risk of instruments with optionality and other instruments holistically, the bank may choose to include instruments without optionality in the calculation of curvature risk. This treatment is allowed subject to all of the following restrictions:

(a) Use of this approach shall be applied consistently through time.

(b) Curvature risk must be calculated for all instruments subject to the sensitivities-based method.

Footnotes

[1] *For example, each instrument that is an option or that includes an option (eg an embedded option such as convertibility or rate dependent prepayment and that is subject to the capital requirements for market risk). A non-exhaustive list of example instruments with optionality includes: calls, puts, caps, floors, swaptions, barrier options and exotic options.*

[2] *An instrument with a prepayment option is a debt instrument which grants the debtor the right to repay part of or the entire principal amount before the contractual maturity without having to compensate for any foregone interest. The debtor can exercise this option with a financial gain to obtain funding over the remaining maturity of the instrument at a lower rate in other ways in the market.*

Process to calculate the capital requirement under the sensitivities-based method

21.3 As set out in MAR21.1, the capital requirement under the sensitivities-based method is calculated by aggregating delta, vega and curvature capital requirements. The relevant paragraphs that describe this process are as follows:

(1) The risk factors for delta, vega and curvature risks for each risk class are defined in MAR21.8 to MAR21.14.

(2) The methods to risk weight sensitivities to risk factors and aggregate them to calculate delta and vega risk positions for each risk class are set out in MAR21.4 and MAR21.15 to MAR21.95, which include the definition of delta and vega sensitivities, definition of buckets, risk weights to apply to risk factors, and correlation parameters.

(3) The methods to calculate curvature risk are set out in MAR21.5 and MAR21.96 to MAR21.101, which include the definition of buckets, risk weights and correlation parameters.

(4) The risk class level capital requirement calculated above must be aggregated to obtain the capital requirement at the entire portfolio level as set out in MAR21.6 and MAR21.7.

Calculation of the delta and vega risk capital requirement for each risk class

21.4 For each risk class, a bank must determine its instruments' sensitivity to a set of prescribed

risk factors, risk weight those sensitivities, and aggregate the resulting risk-weighted sensitivities separately for delta and vega risk using the following step-by-step approach:

(1) For each risk factor (as defined in MAR21.8 to MAR21.14), a sensitivity is determined as set out in MAR21.15 to MAR21.38.

(2) Sensitivities to the same risk factor must be netted to give a net sensitivity sk across all instruments in the portfolio to each risk factor k. In calculating the net sensitivity, all sensitivities to the same given risk factor (eg all sensitivities to the one-year tenor point of the three-month Euribor swap curve) from instruments of opposite direction should offset, irrespective of the instrument from which they derive. For instance, if a bank's portfolio is made of two interest rate swaps on three-month Euribor with the same fixed rate and same notional but of opposite direction, the GIRR on that portfolio would be zero.

(3) The weighted sensitivity WSk is the product of the net sensitivity sk and the corresponding risk weight RWk as defined in MAR21.39 to MAR21.95.

$$WS_k = RW_k s_k$$

(4) Within bucket aggregation: the risk position for delta (respectively vega) bucket b, Kb, must be determined by aggregating the weighted sensitivities to risk factors within the same bucket using the prescribed correlation

ρ_{kl}

set out in the following formula, where the quantity within the square root function is floored at zero:

$$K_b = \sqrt{\max(0, \sum_k WS_k^2 + \sum_k \sum_{k \neq l} \rho_{kl} WS_k WS_l)}$$

(5) Across bucket aggregation: The delta (respectively vega) risk capital requirement is calculated by aggregating the risk positions across the delta (respectively vega) buckets within each risk class, using the corresponding prescribed correlations

γ_{bc}

as set out in the following formula, where:

for all risk factors in bucket b, and in bucket c.

$S_b = \sum_k WS_k$

$S_c = \sum_k WS_k$

If these values for and described in above MAR21.4(5)(a) produce a negative number for the overall sum of , the bank is to calculate the delta (respectively vega) risk capital requirement using an alternative specification whereby:

S_b

S_c

$\sum_b K_b^2 + \sum_b \sum_{c \neq b} \gamma_{bc} S_b S_c$

for all risk factors in bucket b; and

$S_b = \max[\min(\sum_k WS_k, K_b), -K_b]$

for all risk factors in bucket c.

$S_c = \max[\min(\sum_k WS_k, K_c), -K_c]$

$$Delta\left(respectively\ vega\right) = \sqrt{\sum_b K_b^2 + \sum_b \sum_{c \neq b} \gamma_{bc} S_b S_c}$$

Calculation of the curvature risk capital requirement for each risk class

21.5 For each risk class, to calculate curvature risk capital requirements a bank must apply an upward shock and a downward shock to each prescribed risk factor and calculate the incremental loss for instruments sensitive to that risk factor above that already captured by the delta risk capital requirement using the following step-by-step approach:

(1) For each instrument sensitive to curvature risk factor k, an upward shock and a downward shock must be applied to k. The size of shock (ie risk weight) is set out in MAR21.98 and MAR21.99.

(a) For example for GIRR, all tenors of all the risk free interest rate curves within a given currency (eg three-month Euribor, six-month Euribor, one year Euribor, etc for the euro) must be shifted upward applying the risk weight as set out in MAR21.99. The resulting potential loss for each instrument, after the deduction of the delta risk positions, is the outcome of the upward scenario. The same approach must be followed on a downward scenario.

(b) If the price of an instrument depends on several risk factors, the curvature risk must be determined separately for each risk factor.

(2) The net curvature risk capital requirement, determined by the values

CVR_k^+

and for a bank's portfolio for risk factor k described in above MAR21.5(1) is calculated by the formula below. It calculates the aggregate incremental loss beyond the delta capital requirement for the prescribed shocks, where

CVR_k^-

i is an instrument subject to curvature risks associated with risk factor k;

is the current level of risk factor k;

x_k

is the price of instrument i at the current level of risk factor k;

$V_i\left(x_k\right)$

and denote the price of instrument i after is shifted (ie "shocked") upward and downward respectively;

$V_i\left(x_k^{\left(RW^{(curvature)}+\right)}\right)$

$V_i\left(x_k^{\left(RW^{(curvature)}-\right)}\right)$

x_k

is the risk weight for curvature risk factor k for instrument i; and

$RW_k^{(curvature)}$

is the delta sensitivity of instrument i with respect to the delta risk factor that corresponds to curvature risk factor k, where:

s_{ik}

for the FX and equity risk classes, is the delta sensitivity of instrument i; and

s_{ik}

for the GIRR, CSR and commodity risk classes, is the sum of delta sensitivities to all tenors of the relevant curve of instrument i with respect to curvature risk factor k.

s_{ik}

$$CVR_k^+ = -\sum_i \left\{ V_i\left(x_k^{RW(Curvature)^+} \right) - V_i\left(x_k \right) - RW_k^{Curvature} \times s_{ik} \right\}$$

$$CVR_k^- = -\sum_i \left\{ V_i\left(x_k^{RW(Curvature)^-} \right) - V_i\left(x_k \right) + RW_k^{Curvature} \times s_{ik} \right\}$$

(3) Within bucket aggregation: the curvature risk exposure must be aggregated within each bucket using the corresponding prescribed correlation

ρ_{kl}

as set out in the following formula, where:

The bucket level capital requirement ( ) is determined as the greater of the capital requirement under the upward scenario ( ) and the capital requirement under the downward scenario ( ). Notably, the selection of upward and downward scenarios is not necessarily the same across the high, medium and low correlations scenarios specified in MAR21.6.

K_b

K_b^+

K_b^-

Where , this shall be termed "selecting the upward scenario".

$K_b = K_b^+$

Where , this shall be termed "selecting the downward scenario".

$K_b = K_b^-$

In the specific case where if , it is deemed that the upward scenario is selected; otherwise the downward scenario is selected.

$K_b^+ = K_b^-,$

$\sum_k CVR_k^+ > \sum_k CVR_k^-$

takes the value 0 if and both have negative signs and the value 1 otherwise.

$\psi\left(CVR_k, CVR_l\right)$

CVR_k

CVR_l

$K_b = \max\left(K_b^+, K_b^-\right),$

$$where \begin{cases} K_b^+ = \sqrt{\max\left(0, \sum_k \max\left(CVR_k^+, 0\right)^2 + \sum_{l \neq k} \sum_k \rho_{kl} CVR_k^+ CVR_l^+ \psi\left(CVR_k^+, CVR_l^+\right)\right)} \\ K_b^- = \sqrt{\max\left(0, \sum_k \max\left(CVR_k^-, 0\right)^2 + \sum_{l \neq k} \sum_k \rho_{kl} CVR_k^- CVR_l^- \psi\left(CVR_k^-, CVR_l^-\right)\right)} \end{cases}$$

(4) Across bucket aggregation: curvature risk positions must then be aggregated across buckets within each risk class, using the corresponding prescribed correlations

γ_{bc}

, where:

for all risk factors in bucket b, when the upward scenario has been selected for bucket b in above (3)(a). otherwise; and

$S_b = \sum_k CVR_k^+$

$S_b = \sum_k CVR_k^-$

takes the value 0 if and both have negative signs and 1 otherwise.

$\psi\left(S_b, S_c\right)$

S_b

S_c

$$Curvature\,risk = \sqrt{\max\left(0, \sum_b K_b^2 + \sum_{c \neq b}\sum_b \gamma_{bc} S_b S_c \psi\left(S_b, S_c\right)\right)}$$

FAQ

FAQ1 *When the delta effect is removed in the calculation of the curvature risk capital requirement, should the delta used in that calculation be the same as the delta used in the delta risk capital requirement? Should the same assumptions that go into the calculation of the delta (ie sticky delta for normal or log-normal volatilities) go into the calculation of the shifted or shocked price of the instrument?*

The delta used for the calculation of the curvature risk capital requirement should be the same as that used for calculating the delta risk capital requirement. The assumptions that are used for the calculation of the delta (ie sticky delta for normal or log-normal volatilities) should also be used for calculating the shifted or shocked price of the instrument.

FAQ2 *Are banks permitted to choose between zero rate and market rate sensitivities for GIRR delta and curvature capital requirements?*

MAR21.17 states that banks must determine each delta sensitivity, vega sensitivity and curvature scenario based on instrument prices or pricing models that an independent risk control unit within a bank uses to report market risks or actual profits and losses to senior management. Banks should use zero rate or market rate sensitivities consistent with the pricing models referenced in that paragraph.

Calculation of aggregate sensitivities-based method capital requirement

21.6 In order to address the risk that correlations increase or decrease in periods of financial stress, the aggregation of bucket level capital requirements and risk class level capital requirements per each risk class for delta, vega, and curvature risks as specified in MAR21.4 to MAR21.5 must be repeated, corresponding to three different scenarios on the specified values for the correlation parameter (correlation between risk factors within a bucket) and (correlation across buckets within a risk class).

ρ_{kl}

γ_{bc}

(1) Under the "medium correlations" scenario, the correlation parameters

ρ_{kl}

and as specified in MAR21.39 to MAR21.101 apply.

γ_{bc}

(2) Under the "high correlations" scenario, the correlation parameters

ρ_{kl}

and that are specified in MAR21.39 to MAR21.101 are uniformly multiplied by 1.25, with and subject to a cap at 100%.

γ_{bc}

ρ_{kl}

γ_{bc}

(3) Under the "low correlations" scenario, the correlation parameters

ρ_{kl} and γ_{bc} that are specified in MAR21.39 to MAR21.101 are replaced by

$$\rho_{kl}^{low} = \max\left(2 \times \rho_{kl} - 100\%; 75\% \times \rho_{kl}\right)$$

$$\gamma_{bc}^{low} = \max\left(2 \times \gamma_{bc} - 100\%; 75\% \times \gamma_{bc}\right)$$

21.7 The total capital requirement under the sensitivities-based method is aggregated as follows:

(1) For each of three correlation scenarios, the bank must simply sum up the separately calculated delta, vega and curvature capital requirements for all risk classes to determine the overall capital requirement for that scenario.

(2) The sensitivities-based method capital requirement is the largest capital requirement from the three scenarios.

(a) For the calculation of capital requirements for all instruments in all trading desks using the standardised approach as set out in MAR11.8(1) and MAR20.2 and MAR33.40, the capital requirement is calculated for all instruments in all trading desks.

(b) For the calculation of capital requirements for each trading desk using the standardised approach as if that desk were a standalone regulatory portfolio as set out in MAR11.8(2), the capital requirements under each correlation scenario are calculated and compared at each trading desk level, and the maximum for each trading desk is taken as the capital requirement.

Sensitivities-based method: risk factor and sensitivity definitions

Risk factor definitions for delta, vega and curvature risks

21.8 GIRR factors

(1) Delta GIRR: the GIRR delta risk factors are defined along two dimensions: (i) a risk-free yield curve for each currency in which interest rate-sensitive instruments are denominated and (ii) the following tenors: 0.25 years, 0.5 years, 1 year, 2 years, 3 years, 5 years, 10 years, 15 years, 20 years and 30 years, to which delta risk factors are assigned.[3]

(a) The risk-free yield curve per currency should be constructed using money market instruments held in the trading book that have the lowest credit risk, such as overnight index swaps (OIS). Alternatively, the risk-free yield curve should be based on one or more market-implied swap curves used by the bank to mark positions to market. For example, interbank offered rate (BOR) swap curves.

(b) When data on market-implied swap curves described in above (1)(a) are insufficient, the risk-free yield curve may be derived from the most appropriate sovereign bond curve for a given currency. In such cases the sensitivities related to sovereign bonds are not exempt from the CSR capital requirement: when a bank cannot perform the decomposition y=r+cs, any sensitivity to y is allocated both to the GIRR and to CSR classes as appropriate with the risk factor and sensitivity definitions in the standardised approach. Applying swap curves to bond-derived sensitivities for GIRR will not change the requirement for basis risk to be captured between bond and credit default swap (CDS) curves in the CSR class.

(c) For the purpose of constructing the risk-free yield curve per currency, an OIS curve (such as Eonia or a new benchmark rate) and a BOR swap curve (such as three-month Euribor or other benchmark rates) must be considered two different curves. Two BOR curves at different maturities (eg three-month Euribor and six-month Euribor) must be considered two different curves. An onshore and an offshore currency curve (eg onshore Indian rupee and offshore Indian rupee) must be considered two different curves.

(2) The GIRR delta risk factors also include a flat curve of market-implied inflation rates for each currency with term structure not recognised as a risk factor.

(a) The sensitivity to the inflation rate from the exposure to implied coupons in an inflation instrument gives rise to a specific capital requirement. All inflation risks for a currency must be aggregated to one number via simple sum.

(b) This risk factor is only relevant for an instrument when a cash flow is functionally dependent on a measure of inflation (eg the notional amount or an interest payment depending on a consumer price index). GIRR risk factors other than for inflation risk will apply to such an instrument notwithstanding.

(c) Inflation rate risk is considered in addition to the sensitivity to interest rates from the same instrument, which must be allocated, according to the GIRR framework, in the term structure of the relevant risk-free yield curve in the same currency.

(3) The GIRR delta risk factors also include one of two possible cross-currency basis risk factors[4] for each currency (ie each GIRR bucket) with the term structure not recognised as a risk factor (ie both cross-currency basis curves are flat).

(a) The two cross-currency basis risk factors are basis of each currency over USD or basis of each currency over EUR. For instance, an AUD-denominated bank trading a JPY/USD cross-currency basis swap would have a sensitivity to the JPY/USD basis but not to the JPY/EUR basis.

(b) Cross-currency bases that do not relate to either basis over USD or basis over EUR must be computed either on "basis over USD" or "basis over EUR" but not both. GIRR risk factors other than for cross-currency basis risk will apply to such an instrument notwithstanding.

(c) Cross-currency basis risk is considered in addition to the sensitivity to interest rates from the same instrument, which must be allocated, according to the GIRR framework, in the term structure of the relevant risk-free yield curve in the same currency.

(4) Vega GIRR: within each currency, the GIRR vega risk factors are the implied volatilities of options that reference GIRR-sensitive underlyings; as defined along two dimensions:[5]

(a) The maturity of the option: the implied volatility of the option as mapped to one or several of the following maturity tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(b) The residual maturity of the underlying of the option at the expiry date of the option: the implied volatility of the option as mapped to two (or one) of the following residual maturity tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(5) Curvature GIRR:

(a) The GIRR curvature risk factors are defined along only one dimension: the constructed risk-free yield curve per currency with no term structure decomposition. For example, the euro, Eonia, three-month Euribor and six-month Euribor curves must be shifted at the same time in order to compute the euro-relevant risk-free

yield curve curvature risk capital requirement. For the calculation of sensitivities, all tenors (as defined for delta GIRR) are to be shifted in parallel.

(b) There is no curvature risk capital requirement for inflation and cross-currency basis risks.

(6) The treatment described in above (1)(b) for delta GIRR also applies to vega GIRR and curvature GIRR risk factors.

Footnotes

[3] *The assignment of risk factors to the specified tenors should be performed by linear interpolation or a method that is most consistent with the pricing functions used by the independent risk control function of a bank to report market risks or P&L to senior management.*

[4] *Cross-currency basis are basis added to a yield curve in order to evaluate a swap for which the two legs are paid in two different currencies. They are in particular used by market participants to price cross-currency interest rate swaps paying a fixed or a floating leg in one currency, receiving a fixed or a floating leg in a second currency, and including an exchange of the notional in the two currencies at the start date and at the end date of the swap.*

[5] *For example, an option with a forward starting cap, lasting 12 months, consists of four consecutive caplets on USD three-month Libor. There are four (independent) options, with option expiry dates in 12, 15, 18 and 21 months. These options are all on underlying USD three-month Libor; the underlying always matures three months after the option expiry date (its residual maturity being three months). Therefore, the implied volatilities for a regular forward starting cap, which would start in one year and last for 12 months should be defined along the following two dimensions: (i) the maturity of the option's individual components (caplets) – 12, 15, 18 and 21 months; and (ii) the residual maturity of the underlying of the option – three months.*

FAQ

FAQ1 *Different results can be produced depending on the bank's curve methodology as diversification will be different for different methodologies. For example, if three-month Euribor is constructed as a "spread to EONIA", this curve will be a spread curve and can be considered a different yield curve for the purpose of computing risk-weighted PV01 and subsequent diversification. In this example, should three-month Euribor and EONIA be considered two distinct yield curves for the purpose of computing the risk capital requirement?*

MAR21.8(1)(c)states that for the purpose of constructing the risk-free yield curve per currency, an overnight index swap curve (such as EONIA) and an interbank offered rate curve (such as three-month Euribor) must be considered two different curves, with distinct risk factors in each tenor bucket, for the purpose of computing the risk capital requirement.

FAQ2 *For GIRR, CSR, equity risk, commodity risk or FX risk, risk factors need to be assigned to prescribed tenors. How should this assignment be performed if the internally used tenors do not match the prescribed ones?*

Banks are not permitted to perform capital computations based on internally used tenors. Risk factors and sensitivities must be assigned to the prescribed tenors. As stated in footnote 3 to MAR21.8 and footnote 8 to MAR21.25, the assignment of risk factors and sensitivities to the specified tenors should be performed by linear interpolation or a method that is most consistent with the pricing functions used by the independent risk control function of the bank to report market risks or profits and

losses to senior management.

FAQ3 *When calculating the cross-currency basis spread (CCBS) capital requirement: since pricing models use a term structure-based CCBS curve, is it acceptable to use sensitivities to individual tenors aggregated by simple sum rather than explicitly modelling the CCBS curve as flat in the pricing model?*

Yes. Banks may use a term structure-based CCBS curve and aggregate sensitivities to individual tenors by simple sum.

FAQ4 *Should inflation and cross-currency bases be included as a risk factor in the vega GIRR capital requirement?*

Yes. Inflation and cross-currency bases are included in the GIRR vega risk capital requirement. As no maturity dimension is specified for the delta capital requirement for inflation or cross-currency bases (ie the possible underlying of the option), the vega risk for inflation and cross-currency bases should be considered only along the single dimension of the maturity of the option.

FAQ5 *Should a bank compute delta, vega and curvature risk for callable bonds, options on sovereign bond futures and bond options?*

For the specified instruments, delta, vega and curvature capital requirements must be computed for both GIRR and CSR.

FAQ6 *The sensitivities-based approach defines the repo risk factor only in the context of equities and not for fixed income funding instruments (to the extent that these instruments fall within the trading book definition as trading-related repo-style transactions). Is it the intention that fixed income funding instruments be excluded from the equity repo treatment? If so, should such funding instruments be subject to the GIRR capital requirement – for example, by considering the repo curve for a given currency as a yield curve subject to interest rate shocks?*

Repo rate risk factors for fixed income funding instruments are subject to the GIRR capital requirement. A relevant repo curve should be considered by currency.

FAQ7 *May risk weights be floored for interest rates when applying the risk weights for GIRR, given that there is a possibility of the interest rates being negative (eg for JPY and EUR curves)?*

No such floor is permitted in the market risk standard for GIRR.

21.9 CSR non-securitisation risk factors

(1) Delta CSR non-securitisation: the CSR non-securitisation delta risk factors are defined along two dimensions:

(a) the relevant issuer credit spread curves (bond and CDS); and

(b) the following tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(2) Vega CSR non-securitisation: the vega risk factors are the implied volatilities of options that reference the relevant credit issuer names as underlyings (bond and CDS); further defined along one dimension - the maturity of the option. This is defined as the implied volatility of the option as mapped to one or several of the following maturity tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(3) Curvature CSR non-securitisation: the CSR non-securitisation curvature risk factors are defined along one dimension: the relevant issuer credit spread curves (bond and CDS). For instance, the bond-inferred spread curve of an issuer and the CDS-inferred spread curve of that same issuer should be considered a single spread curve. For the

calculation of sensitivities, all tenors (as defined for CSR) are to be shifted in parallel.

FAQ

FAQ1 *The second FAQ under MAR21.8 is also relevant to this paragraph.*

FAQ2 *Should a bank compute delta, vega and curvature risk for callable bonds, options on sovereign bond futures and bond options?*

For the specified instruments, delta, vega and curvature capital requirements must be computed for both GIRR and CSR.

FAQ3 *MAR21.9(3) explicitly states that, for CSR curvature, the bond-CDS basis is ignored. Is it correct that, under MAR21.9(1), bond and CDS curves are considered distinct risk factors and the only "basis" taken into account in in MAR21.54 and MAR21.55 is the bond-CDS basis?*

Yes. Bond and CDS credit spreads are considered distinct risk factors under MAR21.9(1), and referenced in MAR21.54 and MAR21.55 is meant to capture only the bond-CDS basis.

FAQ4 *May risk weights be floored for credit spreads when applying the risk weights for the CSR risk classes?*

No such floor is permitted in the market risk standard for the CSR risk classes, except for CSR curvature under the conditions set out in the second FAQ under MAR21.99.

21.10 CSR securitisation: non-CTP risk factors

(1) For securitisation instruments that do not meet the definition of CTP as set out in MAR20.5 (ie, non-CTP), the sensitivities of delta risk factors (ie CS01) must be calculated with respect to the spread of the tranche rather than the spread of the underlying of the instruments.

(2) Delta CSR securitisation (non-CTP): the CSR securitisation delta risk factors are defined along two dimensions:

(a) Tranche credit spread curves; and

(b) The following tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years to which delta risk factors are assigned.

(3) Vega CSR securitisation (non-CTP): Vega risk factors are the implied volatilities of options that reference non-CTP credit spreads as underlyings (bond and CDS); further defined along one dimension - the maturity of the option. This is defined as the implied volatility of the option as mapped to one or several of the following maturity tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(4) Curvature CSR securitisation (non-CTP): the CSR securitisation curvature risk factors are defined along one dimension, the relevant tranche credit spread curves (bond and CDS). For instance, the bond-inferred spread curve of a given Spanish residential mortgage-backed security (RMBS) tranche and the CDS-inferred spread curve of that given Spanish RMBS tranche would be considered a single spread curve. For the calculation of sensitivities, all the tenors are to be shifted in parallel.

FAQ

FAQ1 *The second FAQ under MAR21.8 is also relevant to this paragraph.*

FAQ2 *May risk weights be floored for credit spreads when applying the risk weights for the CSR risk classes?*

No such floor is permitted in the market risk standard for the CSR risk classes,

except for CSR curvature under the conditions set out in the second FAQ under MAR21.99.

21.11 CSR securitisation: CTP risk factors

(1) For securitisation instruments that meet the definition of a CTP as set out in MAR20.5, the sensitivities of delta risk factors (ie CS01) must be computed with respect to the names underlying the securitisation or nth-to-default instrument.

(2) Delta CSR securitisation (CTP): the CSR correlation trading delta risk factors are defined along two dimensions:

(a) the relevant underlying credit spread curves (bond and CDS); and

(b) the following tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years, to which delta risk factors are assigned.

(3) Vega CSR securitisation (CTP): the vega risk factors are the implied volatilities of options that reference CTP credit spreads as underlyings (bond and CDS), as defined along one dimension, the maturity of the option. This is defined as the implied volatility of the option as mapped to one or several of the following maturity tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(4) Curvature CSR securitisation (CTP): the CSR correlation trading curvature risk factors are defined along one dimension, the relevant underlying credit spread curves (bond and CDS). For instance, the bond-inferred spread curve of a given name within an iTraxx series and the CDS-inferred spread curve of that given underlying would be considered a single spread curve. For the calculation of sensitivities, all the tenors are to be shifted in parallel.

FAQ

FAQ1 *The second FAQ under MAR21.8 is also relevant to this paragraph.*

FAQ2 *May risk weights be floored for credit spreads when applying the risk weights for the CSR risk classes?*

No such floor is permitted in the market risk standard for the CSR risk classes, except for CSR curvature under the conditions set out in the second FAQ under MAR21.99.

21.12 Equity risk factors

(1) Delta equity: the equity delta risk factors are:

(a) all the equity spot prices; and

(b) all the equity repurchase agreement rates (equity repo rates).

(2) Vega equity:

(a) The equity vega risk factors are the implied volatilities of options that reference the equity spot prices as underlyings as defined along one dimension, the maturity of the option. This is defined as the implied volatility of the option as mapped to one or several of the following maturity tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(b) There is no vega risk capital requirement for equity repo rates.

(3) Curvature equity:

(a) The equity curvature risk factors are all the equity spot prices.

(b) There is no curvature risk capital requirement for equity repo rates.

FAQ

FAQ1 *The second FAQ under MAR21.8 is also relevant to this paragraph.*

FAQ2 *The sensitivities-based approach defines the repo risk factor only in the context of equities and not for fixed income funding instruments (to the extent that these instruments fall within the trading book definition as trading-related repo-style transactions). Is it the intention that fixed income funding instruments be excluded from the equity repo treatment? If so, should such funding instruments be subject to the GIRR capital requirement – for example, by considering the repo curve for a given currency as a yield curve subject to interest rate shocks?*

Repo rate risk factors for fixed income funding instruments are subject to the GIRR capital requirement. A relevant repo curve should be considered by currency.

21.13 Commodity risk factors

(1) Delta commodity: the commodity delta risk factors are all the commodity spot prices. However for some commodities such as electricity (which is defined to fall within bucket 3 (energy – electricity and carbon trading) in MAR21.82 the relevant risk factor can either be the spot or the forward price, as transactions relating to commodities such as electricity are more frequent on the forward price than transactions on the spot price. Commodity delta risk factors are defined along two dimensions:

(a) legal terms with respect to the delivery location[6] of the commodity; and

(b) time to maturity of the traded instrument at the following tenors: 0 years, 0.25 years, 0.5 years, 1 year, 2 years, 3 years, 5 years, 10 years, 15 years, 20 years and 30 years.

(2) Vega commodity: the commodity vega risk factors are the implied volatilities of options that reference commodity spot prices as underlyings. No differentiation between commodity spot prices by the maturity of the underlying or delivery location is required. The commodity vega risk factors are further defined along one dimension, the maturity of the option. This is defined as the implied volatility of the option as mapped to one or several of the following maturity tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(3) Curvature commodity: the commodity curvature risk factors are defined along only one dimension, the constructed curve (ie no term structure decomposition) per commodity spot prices. For the calculation of sensitivities, all tenors (as defined for delta commodity) are to be shifted in parallel.

Footnotes

[6] *For example, a contract that can be delivered in five ports can be considered having the same delivery location as another contract if and only if it can be delivered in the same five ports. However, it cannot be considered having the same delivery location as another contract that can be delivered in only four (or less) of those five ports.*

FAQ

FAQ1 *The second FAQ under MAR21.8 is also relevant to this paragraph.*

FAQ2 *How are commodity delta risk factors computed for futures and forward contracts?*

The current prices for futures and forward contracts should be used to compute the commodity delta risk factors. Commodity delta should be allocated to the relevant tenor based on the tenor of the futures and forward contract and given that spot commodity price positions should be slotted into the first tenor (0 years).

21.14 FX risk factors

(1) Delta FX: the FX delta risk factors are defined below.

(a) The FX delta risk factors are all the exchange rates between the currency in which an instrument is denominated and the reporting currency. For transactions that reference an exchange rate between a pair of non-reporting currencies, the FX delta risk factors are all the exchange rates between:

(i) the reporting currency; and

(ii) both the currency in which an instrument is denominated and any other currencies referenced by the instrument.[7]

(b) Subject to supervisory approval, FX risk may alternatively be calculated relative to a base currency instead of the reporting currency. In such case the bank must account for not only:

(i) the FX risk against the base currency; but also

(ii) the FX risk between the reporting currency and the base currency (ie translation risk).

(c) The resulting FX risk calculated relative to the base currency as set out in (b) is converted to the capital requirements in the reporting currency using the spot reporting/base exchange rate reflecting the FX risk between the base currency and the reporting currency.

(d) The FX base currency approach may be allowed under the following conditions:

(i) To use this alternative, a bank may only consider a single currency as its base currency; and

(ii) The bank shall demonstrate to the relevant supervisor that calculating FX risk relative to their proposed base currency provides an appropriate risk representation for their portfolio (for example, by demonstrating that it does not inappropriately reduce capital requirements relative to those that would be calculated without the base currency approach) and that the translation risk between the base currency and the reporting currency is taken into account.

(2) Vega FX: the FX vega risk factors are the implied volatilities of options that reference exchange rates between currency pairs; as defined along one dimension, the maturity of the option. This is defined as the implied volatility of the option as mapped to one or several of the following maturity tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(3) Curvature FX: the FX curvature risk factors are defined below.

(a) The FX curvature risk factors are all the exchange rates between the currency in which an instrument is denominated and the reporting currency. For transactions that reference an exchange rate between a pair of non-reporting currencies, the FX risk factors are all the exchange rates between:

(i) the reporting currency; and

(ii) both the currency in which an instrument is denominated and any other currencies referenced by the instrument.

(b) Where supervisory approval for the base currency approach has been granted for delta risks, FX curvature risks shall also be calculated relative to a base currency instead of the reporting currency, and then converted to the capital requirements in the reporting currency using the spot reporting/base exchange rate.

(4) No distinction is required between onshore and offshore variants of a currency for all

FX delta, vega and curvature risk factors.

Footnotes

[7] *For example, for an FX forward referencing USD/JPY, the relevant risk factors for a CAD-reporting bank to consider are the exchange rates USD/CAD and JPY/CAD. If that CAD-reporting bank calculates FX risk relative to a USD base currency, it would consider separate deltas for the exchange rate JPY/USD risk and CAD/USD FX translation risk and then translate the resulting capital requirement to CAD at the USD/CAD spot exchange rate.*

FAQ

FAQ1 *The second FAQ under MAR21.8 is also relevant to this paragraph.*

FAQ2 *MAR21.14(4) states: "No distinction is required between onshore and offshore variants of a currency for all FX delta, vega and curvature risk factors." Does this also apply for deliverable/non-deliverable variants (eg KRO vs KRW, BRO vs BRL, INO vs INR)?*

Yes. No distinction is required between deliverable and non-deliverable variants of a currency.

Sensitivities-based method: definition of sensitivities

21.15 Sensitivities for each risk class must be expressed in the reporting currency of the bank.

21.16 For each risk factor defined in MAR21.8 to MAR21.14, sensitivities are calculated as the change in the market value of the instrument as a result of applying a specified shift to each risk factor, assuming all the other relevant risk factors are held at the current level as defined in MAR21.17 to MAR21.38.

FAQ

FAQ1 *In the context of delta sensitivity calculations, is it acceptable to use alternative formulations of sensitivities calculations that yield results very close to the prescribed formulation of sensitivities calculations?*

Yes, as per MAR21.17, a bank may make use of alternative formulations of sensitivities based on pricing models that the bank's independent risk control unit uses to report market risks or actual profits and losses to senior management. In doing so, the bank is to demonstrate to its supervisor that the alternative formulations of sensitivities yield results very close to the prescribed formulations.

Requirements on instrument price or pricing models for sensitivity calculation

21.17 In calculating the risk capital requirement under the sensitivities-based method in MAR21, the bank must determine each delta and vega sensitivity and curvature scenario based on instrument prices or pricing models that an independent risk control unit within a bank uses to report market risks or actual profits and losses to senior management.

FAQ

FAQ1 *In the context of delta sensitivity calculations, is it acceptable to use alternative formulations of sensitivities calculations that yield results very close to the prescribed formulation of sensitivities calculations?*

Yes, as per MAR21.17, a bank may make use of alternative formulations of sensitivities based on pricing models that the bank's independent risk control unit uses to report market risks or actual profits and losses to senior management. In doing so, the bank is to demonstrate to its supervisor that the alternative

formulations of sensitivities yield results very close to the prescribed formulations.

FAQ2 *Are banks permitted to choose between zero rate and market rate sensitivities for GIRR delta and curvature capital requirements?*

MAR21.17 states that banks must determine each delta sensitivity, vega sensitivity and curvature scenario based on instrument prices or pricing models that an independent risk control unit within a bank uses to report market risks or actual profits and losses to senior management. Banks should use zero rate or market rate sensitivities consistent with the pricing models referenced in that paragraph.

21.18 A key assumption of the standardised approach for market risk is that a bank's pricing models used in actual profit and loss reporting provide an appropriate basis for the determination of regulatory capital requirements for all market risks. To ensure such adequacy, banks must at a minimum establish a framework for prudent valuation practices that include the requirements of CAP50.

Sensitivity definitions for delta risk

21.19 Delta GIRR: the sensitivity is defined as the PV01. PV01 is measured by changing the interest rate r at tenor t (r) of the risk-free yield curve in a given currency by 1 basis point (ie 0.0001 in absolute terms) and dividing the resulting change in the market value of the instrument (V) by 0.0001 (ie 0.01%) as follows, where:

(1) rt is the risk-free yield curve at tenor t;

(2) cst is the credit spread curve at tenor t; and

(3) Vi is the market value of the instrument i as a function of the risk-free interest rate curve and credit spread curve:

$$s_{k,r_t} = \frac{V_i\left(r_t + 0.0001, cs_t\right) - V_i\left(r_t, cs_t\right)}{0.0001}$$

FAQ

FAQ1 *Are banks permitted to choose between zero rate and market rate sensitivities for GIRR delta and curvature capital requirements?*

MAR21.17 states that banks must determine each delta sensitivity, vega sensitivity and curvature scenario based on instrument prices or pricing models that an independent risk control unit within a bank uses to report market risks or actual profits and losses to senior management. Banks should use zero rate or market rate sensitivities consistent with the pricing models referenced in that paragraph.

21.20 Delta CSR non-securitisation, securitisation (non-CTP) and securitisation (CTP): the sensitivity is defined as CS01. The CS01 (sensitivity) of an instrument i is measured by changing a credit spread cs at tenor t (cs) by 1 basis point (ie 0.0001 in absolute terms) and dividing the resulting change in the market value of the instrument (V) by 0.0001 (ie 0.01%) as follows:

$$s_{k,cs_t} = \frac{V_i\left(r_t, cs_t + 0.0001\right) - V_i\left(r_t, cs_t\right)}{0.0001}$$

FAQ

FAQ1 *In cases where the bank does not have counterparty-specific money market curves, can the bank proxy PV01 to CS01?*

Yes. Proxying PV01 to CS01 is permitted for such money market instruments.

21.21 Delta equity spot: the sensitivity is measured by changing the equity spot price by 1 percentage point (ie 0.01 in relative terms) and dividing the resulting change in the market value of the instrument (V) by 0.01 (ie 1%) as follows, where:

(1) k is a given equity;

(2) EQk is the market value of equity k; and

(3) Vi is the market value of instrument i as a function of the price of equity k.

$$s_k = \frac{V_i\left(1.01EQ_k\right) - V_i\left(EQ_k\right)}{0.01}$$

21.22 Delta equity repo rates: the sensitivity is measured by applying a parallel shift to the equity repo rate term structure by 1 basis point (ie 0.0001 in absolute terms) and dividing the resulting change in the market value of the instrument V by 0.0001 (ie 0.01%) as follows, where:

(1) k is a given equity;

(2)

RTS_k

is the repo term structure of equity k; and

(3) Viis the market value of instrument i as a function of the repo term structure of equity k.

$$s_k = \frac{V_i\left(RTS_k + 0.0001\right) - V_i\left(RTS_k\right)}{0.0001}$$

21.23 Delta commodity: the sensitivity is measured by changing the commodity spot price by 1 percentage point (ie 0.01 in relative terms) and dividing the resulting change in the market value of the instrument V by 0.01 (ie 1%) as follows, where:

(1) k is a given commodity;

(2) CTYk is the market value of commodity k; and

(3) Vi is the market value of instrument i as a function of the spot price of commodity k:

$$s_k = \frac{V_i\left(1.01CTY_k\right) - V_i\left(CTY_k\right)}{0.01}$$

FAQ

FAQ1 *In relation to the curvature risk capital requirement for the commodity risk class, MAR21.99 requires that the curvature risk weight is the parallel shift of all the tenors for each curve based on the highest prescribed delta risk weight for each bucket. A parallel shift in MAR21.99 implies that an additive shock (in absolute terms) is applied along the curve. However, MAR21.23 states that the shock applied to delta commodity is a relative shock. How should the shock be applied to commodity curvature?*

The sizes of upward and downward shocks applied to assess the net curvature risk capital requirement for a specific commodityś curvature risk factor should be based on the risk weight connected to the curvature bucket where that commodity is classified, in accordance with MAR21.97 and MAR21.82. The same relative shocks

should be applied to all curvature risk factors classified under the same bucket, defined along the dimension of the constructed curve (ie no term structure decomposition) per each commodity spot price, as described in MAR21.13(3). For example, the constructed curve for gold (with a risk weight of 20%) would be shifted up by multiplying each tenor price by 1.2 and down by multiplying each tenor price by 0.8.

21.24 Delta FX: the sensitivity is measured by changing the exchange rate by 1 percentage point (ie 0.01 in relative terms) and dividing the resulting change in the market value of the instrument V by 0.01 (ie 1%), where:

(1) k is a given currency;

(2) FXk is the exchange rate between a given currency and a bank's reporting currency or base currency, where the FX spot rate is the current market price of one unit of another currency expressed in the units of the bank's reporting currency or base currency; and

(3) Vi is the market value of instrument i as a function of the exchange rate k:

$$s_k = \frac{V_i\left(1.01FX_k\right) - V_i\left(FX_k\right)}{0.01}$$

Sensitivity definitions for vega risk

21.25 The option-level vega risk sensitivity to a given risk factor[8] is measured by multiplying vega by the implied volatility of the option as follows, where:

(1) vega,

$\frac{\partial V_i}{\partial \sigma_i}$

, is defined as the change in the market value of the option as a result of a small amount of change to the implied volatility ; and

V_i

σ_i

(2) the instrument's vega and implied volatility used in the calculation of vega sensitivities must be sourced from pricing models used by the independent risk control unit of the bank.

$s_k = vega \times implied\ volatility$

Footnotes

[8] *As specified in the vega risk factor definitions in MAR21.8 to MAR21.14, the implied volatility of the option must be mapped to one or more maturity tenors.*

21.26 The following sets out how to derive vega risk sensitivities in specific cases:

(1) Options that do not have a maturity, are assigned to the longest prescribed maturity tenor, and these options are also assigned to the RRAO.

(2) Options that do not have a strike or barrier and options that have multiple strikes or barriers, are mapped to strikes and maturity used internally to price the option, and these options are also assigned to the RRAO.

(3) CTP securitisation tranches that do not have an implied volatility, are not subject to vega risk capital requirement. Such instruments may not, however, be exempt from

delta and curvature risk capital requirements.

FAQ

FAQ1 *Under the sensitivities-based method, would a bank need to compute vega risk over the longest maturity for a cancellable swap? Would a bank also be required to compute residual risk for cancellable swaps?*

In the case where options do not have a specified maturity (eg cancellable swaps), the bank must assign those options to the longest prescribed maturity tenor for vega risk sensitivities and also assign such options to the RRAO.

In the case of the bank viewing the optionality of the cancellable swap as a swaption, the bank must assign the swaption to the longest prescribed maturity tenor for vega risk sensitivities (as it does not have a specified maturity) and derive the residual maturity of the underlying of the option accordingly.

Requirements on sensitivity computations

21.27 When computing a first-order sensitivity for instruments subject to optionality, banks should assume that the implied volatility either:

(1) remains constant, consistent with a "sticky strike" approach; or

(2) follows a "sticky delta" approach, such that implied volatility does not vary with respect to a given level of delta.

21.28 For the calculation of vega sensitivities, the distribution assumptions (ie log-normal assumptions or normal assumptions) for pricing models are applied as follows:

(1) For the computation of a vega GIRR or CSR sensitivity, banks may use either the log-normal or normal assumptions.

(2) For the computation of a vega equity, commodity or FX sensitivity, banks must use the log-normal assumption.[9]

Footnotes

[9] *Since vega (of an instrument is multiplied by its implied volatility ( ), the vega risk sensitivity for that instrument will be the same under the log-normal assumption and the normal assumption. As a consequence, banks may use a log-normal or normal assumption for GIRR and CSR (in recognition of the trade-offs between constrained specification and computational burden for a standardised approach). For the other risk classes, banks must only use a log-normal assumption (in recognition that this is aligned with common practices across jurisdictions).*

FAQ

FAQ1 *If banks may use either a log-normal or normal assumption for vega GIRR, does this mean that the same log-normal or normal assumption should be applied to all currencies, or can the application be different for different currencies? For example, is a bank permitted to adopt a normal assumption for EUR and a log-normal assumption for USD?*

To compute vega GIRR, banks may choose a mix of log-normal and normal assumptions for different currencies.

21.29 If, for internal risk management, a bank computes vega sensitivities using different definitions than the definitions set out in this standard, the bank may transform the sensitivities computed for internal risk management purposes to deduce the sensitivities to be used for the calculation of the vega risk measure.

21.30 All vega sensitivities must be computed ignoring the impact of credit valuation adjustments (CVA).

Treatment of index instruments and multi-underlying options

21.31 In the delta and curvature risk context: for index instruments and multi-underlying options, a look-through approach should be used. However, a bank may opt not to apply the look-through approach for instruments referencing any listed and widely recognised and accepted equity or credit index, where:

(1) it is possible to look-through the index (ie the constituents and their respective weightings are known);

(2) the index contains at least 20 constituents;

(3) no single constituent contained within the index represents more than 25% of the total index;

(4) the largest 10% of constituents represents less than 60% of the total index; and

(5) the total market capitalisation of all the constituents of the index is no less than USD 40 billion.

FAQ

FAQ1 *When certain conditions set out in MAR21.31 are satisfied for instruments referencing any listed and widely recognised equity or credit index, a bank may opt not to apply the look-through approach. It is common for funds with diversified constituents to satisfy the conditions set out in MAR21.31. Are positions in funds and instruments that reference them permitted to apply the no look-through approach using index buckets?*

No. Capital requirements for equity investments in funds generally must be calculated in accordance with one of the three ways set out in MAR21.36 – the no look-through approach for equity and credit indices cannot be applied to funds that do not track a listed and widely recognised index even if their holdings meet the criteria set out in MAR21.31 (1) to (5).

Subject to the criteria in MAR21.35, however, equity investment funds that invest purely in either equity or debt instruments to replicate a listed and widely-recognised index may be treated as if they were investments in the those equity or credit indices and apply the no look-through approach available for credit and equity indices on those funds if those investments in funds meet the requirements set out in MAR21.31 to MAR21.34.

21.32 For a given instrument, irrespective of whether a look-through approach is adopted or not, the sensitivity inputs used for the delta and curvature risk calculation must be consistent.

21.33 Where a bank opts not to apply the look-through approach in accordance with MAR21.31, a single sensitivity shall be calculated to each widely recognised and accepted index that an instrument references. The sensitivity to the index should be assigned to the relevant delta risk bucket defined in MAR21.53 and MAR21.72 as follows:

(1) Where more than 75% of constituents in that index (taking into account the weightings of that index) would be mapped to a specific sector bucket (ie bucket 1 to bucket 11 for equity risk, or bucket 1 to bucket 16 for CSR), the sensitivity to the index shall be mapped to that single specific sector bucket and treated like any other single-name sensitivity in that bucket.

(2) In all other cases, the sensitivity may be mapped to an "index" bucket (ie bucket 12 or

bucket 13 for equity risk; or bucket 17 or bucket 18 for CSR). The same principle as above (1) applies when allocating sensitivities to a specific index bucket.

(a) For equity risk, an equity index should be mapped to the large market cap and advanced economy indices bucket (ie bucket 12) if at least 75% of the constituents in that index (taking into account the weightings of that index) are both large cap and advanced economy equities. Otherwise, it should be mapped to the other equity indices bucket (ie bucket 13).

(b) For CSR, a credit index should be mapped to the investment grade indices bucket (ie bucket 17) if at least 75% of the constituents in that index (taking into account the weightings of that index) are investment grade. Otherwise, it should be mapped to the high yield indices bucket (ie bucket 18).

21.34 A look-through approach must always be used for indices that do not meet the criteria set out in MAR21.31(2) to MAR21.31(5), and for any multi-underlying instruments that reference a bespoke set of equities or credit positions.

(1) Where a look-through approach is adopted, for index instruments and multi-underlying options other than the CTP, the sensitivities to constituent risk factors from those instruments or options are allowed to net with sensitivities to single-name instruments without restriction.

(2) Index CTP instruments cannot be broken down into its constituents (ie the index CTP should be considered a risk factor as a whole) and the above-mentioned netting at the issuer level does not apply either.

(3) Where a look-through approach is adopted, it shall be applied consistently through time[10], and shall be used for all identical instruments that reference the same index.

Footnotes

[10] *In other words, a bank can initially not apply a look-through approach, and later decide to apply it. However once applied (for a certain type of instrument referencing a particular index), the bank will require supervisory approval to revert to a "no look-through" approach.*

FAQ

FAQ1 *In accordance with MAR21.58(1), sensitivities to credit spread risk (CSR) arising from the correlation trading portfolio (CTP) should be classified according to the same bucket structure as the one for CSR non-securitisation, as set out in MAR21.51, except for index buckets (bucket 17 and bucket 18). Since an index CTP should be considered a risk factor as a whole and cannot be broken down into its constituents, as stated in MAR21.34(2), how should a bank determine which bucket to assign the delta sensitivity of an index CTP instrument, given the aforementioned bucket structure?*

The delta CSR sensitivity of an index CTP instrument should be assigned to a single specific delta sector bucket consistent with the characteristics of, at least, 75% of the index constituents (taking into account the weightings of that index), in accordance with MAR21.33(1). If this is not possible, then the index should be assigned to bucket 16, "Other sector". The sensitivity to that index CTP instrument should be considered and treated like any other single-name sensitivity assigned to that same sector bucket.

Treatment of equity investments in funds

21.35 For equity investments in funds that can be looked through as set out in RBC25.8(5)(a),

banks must apply a look-through approach and treat the underlying positions of the fund as if the positions were held directly by the bank (taking into account the bank's share of the equity of the fund, and any leverage in the fund structure), except for the funds that meet the following conditions:

(1) For funds that hold an index instrument that meets the criteria set out under MAR21.31, banks must still apply a look-through and treat the underlying positions of the fund as if the positions were held directly by the bank, but the bank may then choose to apply the "no look-through" approach for the index holdings of the fund as set out in MAR21.33.

(2) For funds that track an index benchmark, a bank may opt not to apply the look-through approach and opt to measure the risk assuming the fund is a position in the tracked index only where:

(a) the fund has an absolute value of a tracking difference (ignoring fees and commissions) of less than 1%; and

(b) the tracking difference is checked at least annually and is defined as the annualised return difference between the fund and its tracked benchmark over the last 12 months of available data (or a shorter period in the absence of a full 12 months of data).

21.36 For equity investments in funds that cannot be looked through (ie do not meet the criterion set out in RBC25.8(5)(a)), but that the bank has access to daily price quotes and knowledge of the mandate of the fund (ie meet both the criteria set out in RBC25.8(5)(b)), banks may calculate capital requirements for the fund in one of three ways:

(1) If the fund tracks an index benchmark and meets the requirement set out in MAR21.35(2)(a) and (b), the bank may assume that the fund is a position in the tracked index, and may assign the sensitivity to the fund to relevant sector specific buckets or index buckets as set out in MAR21.33.

(2) Subject to supervisory approval, the bank may consider the fund as a hypothetical portfolio in which the fund invests to the maximum extent allowed under the fund's mandate in those assets attracting the highest capital requirements under the sensitivities-based method, and then progressively in those other assets implying lower capital requirements. If more than one risk weight can be applied to a given exposure under the sensitivities-based method, the maximum risk weight applicable must be used.

(a) This hypothetical portfolio must be subject to market risk capital requirements on a stand-alone basis for all positions in that fund, separate from any other positions subject to market risk capital requirements.

(b) The counterparty credit and CVA risks of the derivatives of this hypothetical portfolio must be calculated using the simplified methodology set out in accordance with CRE60.7(c) of the banking book equity investment in funds treatment.

(3) A bank may treat their equity investment in the fund as an unrated equity exposure to be allocated to the "other sector" bucket (bucket 11). In applying this treatment, banks must also consider whether, given the mandate of the fund, the default risk capital (DRC) requirement risk weight prescribed to the fund is sufficiently prudent (as set out in MAR22.8), and whether the RRAO should apply (as set out in MAR23.6).

21.37 As per the requirement in RBC25.8(5), net long equity investments in a given fund in which the bank cannot look through or does not meet the requirements of RBC25.8(5) for the

fund must be assigned to the banking book. Net short positions in funds, where the bank cannot look through or does not meet the requirements of RBC25.8(5), must be excluded from any trading book capital requirements under the market risk framework, with the net position instead subjected to a 100% capital requirement.

Treatment of vega risk for multi-underlying instruments

21.38 In the vega risk context:

(1) Multi-underlying options (including index options) are usually priced based on the implied volatility of the option, rather than the implied volatility of its underlying constituents and a look-through approach may not need to be applied, regardless of the approach applied to the delta and curvature risk calculation as set out in MAR21.31 through MAR21.34.[11]

(2) For indices, the vega risk with respect to the implied volatility of the multi-underlying options will be calculated using a sector specific bucket or an index bucket defined in MAR21.53 and MAR21.72 as follows:

(a) Where more than 75% of constituents in that index (taking into account the weightings of that index) would be mapped to a single specific sector bucket (ie bucket 1 to bucket 11 for equity risk; or bucket 1 to bucket 16 for CSR), the sensitivity to the index shall be mapped to that single specific sector bucket and treated like any other single-name sensitivity in that bucket.

(b) In all other cases, the sensitivity may be mapped to an "index" bucket (ie bucket 12 or bucket 13 for equity risk or bucket 17 or bucket 18 for CSR).

Footnotes

[11] *As specified in the vega risk factor definitions in MAR21.8 to MAR21.14, the implied volatility of an option must be mapped to one or more maturity tenors.*

Sensitivities-based method: definition of delta risk buckets, risk weights and correlations

21.39 MAR21.41 to MAR21.89 set out buckets, risk weights and correlation parameters for each risk class to calculate delta risk capital requirement as set out in MAR21.4.

21.40 The prescribed risk weights and correlations in MAR21.41 to MAR21.89 have been calibrated to the liquidity adjusted time horizon related to each risk class.

Delta GIRR buckets, risk weights and correlations

21.41 Each currency is a separate delta GIRR bucket, so all risk factors in risk-free yield curves for the same currency in which interest rate-sensitive instruments are denominated are grouped into the same bucket.

21.42 For calculating weighted sensitivities, the risk weights for each tenor in risk-free yield curves are set in Table 1 as follows:

Delta GIRR buckets and risk weights					Table 1
Tenor	0.25 year	0.5 year	1 year	2 year	3 year
Risk weight	1.7%	1.7%	1.6%	1.3%	1.2%
Tenor	5 year	10 year	15 year	20 year	30 year
Risk weight (percentage points)	1.1%	1.1%	1.1%	1.1%	1.1%

21.43 The risk weight for the inflation risk factor and the cross-currency basis risk factors, respectively, is set at 1.6%.

21.44 For specified currencies by the Basel Committee,[12] the above risk weights may, at the discretion of the bank, be divided by the square root of 2.

Footnotes

[12] *Specified currencies by the Basel Committee are: EUR, USD, GBP, AUD, JPY, SEK, CAD as well as the domestic reporting currency of a bank.*

21.45 For aggregating GIRR risk positions within a bucket, the correlation parameter between weighted sensitivities and within the same bucket (ie same currency), same assigned tenor, but different curves is set at 99.90%. In aggregating delta risk positions for cross-currency basis risk for onshore and offshore curves, which must be considered two different curves as set out in MAR21.8, a bank may choose to aggregate all cross-currency basis risk for a currency (ie "Curr/USD" or "Curr/EUR") for both onshore and offshore curves by a simple sum of weighted sensitivities.

ρ_{kl}

WS_k

WS_l

21.46 The delta risk correlation between weighted sensitivities and within the same bucket with different tenor and same curve is set in the following Table 2[13]:

ρ_{kl}

WS_k

WS_l

Delta GIRR correlations (ρ_{kl}) within the same bucket, with different tenor and same curve

Table 2

	0.25 year	0.5 year	1 year	2 year	3 year	5 year	10 year	15 year	20 year	30 year
0.25 year	100.0%	97.0%	91.4%	81.1%	71.9%	56.6%	40.0%	40.0%	40.0%	40.0%
0.5 year	97.0%	100.0%	97.0%	91.4%	86.1%	76.3%	56.6%	41.9%	40.0%	40.0%
1 year	91.4%	97.0%	100.0%	97.0%	94.2%	88.7%	76.3%	65.7%	56.6%	41.9%
2 year	81.1%	91.4%	97.0%	100.0%	98.5%	95.6%	88.7%	82.3%	76.3%	65.7%
3 year	71.9%	86.1%	94.2%	98.5%	100.0%	98.0%	93.2%	88.7%	84.4%	76.3%
5 year	56.6%	76.3%	88.7%	95.6%	98.0%	100.0%	97.0%	94.2%	91.4%	86.1%
10 year	40.0%	56.6%	76.3%	88.7%	93.2%	97.0%	100.0%	98.5%	97.0%	94.2%
15 year	40.0%	41.9%	65.7%	82.3%	88.7%	94.2%	98.5%	100.0%	99.0%	97.0%

20 year	40.0%	40.0%	56.6%	76.3%	84.4%	91.4%	97.0%	99.0%	100.0%	98.5%
30 year	40.0%	40.0%	41.9%	65.7%	76.3%	86.1%	94.2%	97.0%	98.5%	100.0%

ρ_{kl}

Footnotes

13 *The delta GIRR correlation parameters ( ) set out in Table 2 is determined by , where T (respectivel T) is the tenor that relates to (respectively); and is set at 3%. For example, the correlation betwee a sensitivity to the one-year tenor of the Eonia swap curve and the a sensitivity to the five-year tenor of the Eonia swap curve in the same currency is .*

21.47 Between two weighted sensitivities and within the same bucket with different tenor and different curves, the correlation is equal to the correlation parameter specified in MAR21.46 multiplied by 99.90%.[14]

WS_k

WS_l

ρ_{kl}

Footnotes

14 *For example, the correlation between a sensitivity to the one-year tenor of the Eonia swap curve and a sensitivity to the five-year tenor of the three-month Euribor swap curve in the same currency is .*

FAQ

FAQ1 *What should the correlation between two inflation curves in the same currency (eg German vs French, in Euro) be for GIRR?*

Per MAR21.47, a 99.90% correlation should apply to different inflation curves in the same currency.

21.48 The delta risk correlation between a weighted sensitivity to the inflation curve and a weighted sensitivity to a given tenor of the relevant yield curve is 40%.

ρ_{kl}

WS_k

WS_l

21.49 The delta risk correlation between a weighted sensitivity to a cross-currency basis curve and a weighted sensitivity to each of the following curves is 0%:

ρ_{kl}

WS_k

WS_l

(1) a given tenor of the relevant yield curve;

(2) the inflation curve; or

(3) another cross-currency basis curve (if relevant).

21.50 For aggregating GIRR risk positions across different buckets (ie different currencies), the parameter is set at 50%.

γ_{bc}

Delta CSR non-securitisations buckets, risk weights and correlations

21.51 For delta CSR non-securitisations, buckets are set along two dimensions - credit quality and sector - as set out in Table 3. The CSR non-securitisation sensitivities or risk exposures should first be assigned to a bucket defined before calculating weighted sensitivities by applying a risk weight.

Buckets for delta CSR non-securitisations		Table 3
Bucket number	Credit quality	Sector
1	Investment grade (IG)	Sovereigns including central banks, multilateral development banks
2		Local government, government-backed non-financials, education, public administration
3		Financials including government-backed financials
4		Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying
5		Consumer goods and services, transportation and storage, administrative and support service activities
6		Technology, telecommunications
7		Health care, utilities, professional and technical activities
8		Covered bonds[15]
9	High yield (HY) & non-rated (NR)	Sovereigns including central banks, multilateral development banks
10		Local government, government-backed non-financials, education, public administration
11		Financials including government-backed financials
12		Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying
13		Consumer goods and services, transportation and storage, administrative and support service activities
14		Technology, telecommunications
15		Health care, utilities, professional and technical activities
16	Other sector[16]	
17	IG indices	
18	HY indices	

Footnotes

[15] *Covered bonds must meet the definition provided in LEX30.37, LEX30.39 and LEX30.40.*

[16] *Credit quality is not a differentiating consideration for this bucket.*

FAQ

FAQ1 *How are risk weights to be determined when external ratings assigned by credit rating agencies differ and when there are no external ratings available?*

Consistent with the treatment of external ratings under the standardised approach to credit risk (see CRE21.10 and CRE21.11), if there are two ratings which map into

different risk weights, the higher risk weight should be applied. If there are three or more ratings with different risk weights, the ratings corresponding to the two lowest risk weights should be referred to and the higher of those two risk weights will be applied.

Consistent with the treatment where there are no external ratings under the CVA risk chapter (see MAR50.16), where there are no external ratings or where external ratings are not recognised within a jurisdiction, banks may, subject to supervisory approval:

FAQ2 *For the purpose of market risk capital requirements, what are the CSR capital requirements for Fannie Mae and Freddie Mac mortgage-backed security (MBS) bonds? What is the loss-given-default (LGD) for Fannie and Freddie MBS?*

Non-tranched MBS issued by government sponsored-entities (GSEs), such as Fannie and Freddie, are assigned to bucket 2 (local government, government-backed non-financials, education, public administration) for CSR with a risk weight of 1.0%.

In accordance with MAR22.12, the LGD for non-tranched MBS issued by GSEs is 75% (ie the LGD assigned to senior debt instruments) unless the GSE security satisfies the requirements of footnote 15 to MAR21.51 for treatment of the security as a covered bond.

21.52 To assign a risk exposure to a sector, banks must rely on a classification that is commonly used in the market for grouping issuers by industry sector.

(1) The bank must assign each issuer to one and only one of the sector buckets in the table under MAR21.51.

(2) Risk positions from any issuer that a bank cannot assign to a sector in this fashion must be assigned to the other sector (ie bucket 16).

21.53 For calculating weighted sensitivities, the risk weights for buckets 1 to 18 are set out in Table 4. Risk weights are the same for all tenors (ie 0.5 years, 1 year, 3 years, 5 years, 10 years) within each bucket:

Risk weights for buckets for delta CSR non-securitisations	Table 4
Bucket number	Risk weight
1	0.5%
2	1.0%
3	5.0%
4	3.0%
5	3.0%
6	2.0%
7	1.5%
8	2.5%17
9	2.0%
10	4.0%
11	12.0%
12	7.0%
13	8.5%

14	5.5%
15	5.0%
16	12.0%
17	1.5%
18	5.0%

Footnotes

17 *For covered bonds that are rated AA- or higher, the applicable risk weight may at the discretion of the bank be 1.5%.*

21.54 For buckets 1 to 15, for aggregating delta CSR non-securitisations risk positions within a bucket, the correlation parameter between two weighted sensitivities and within the same bucket, is set as follows, where:[18]

ρ_{kl}

WS_k

WS_l

(1)

$\rho_{kl}^{(name)}$

is equal to 1 where the two names of sensitivities k and l are identical, and 35% otherwise;

(2)

$\rho_{kl}^{(tenor)}$

is equal to 1 if the two tenors of the sensitivities k and l are identical, and to 65% otherwise; and

(3)

$\rho_{kl}^{(basis)}$

is equal to 1 if the two sensitivities are related to same curves, and 99.90% otherwise.

$$\rho_{kl} = \rho_{kl}^{(name)} \cdot \rho_{kl}^{(tenor)} \cdot \rho_{kl}^{(basis)}$$

Footnotes

18 *For example, a sensitivity to the five-year Apple bond curve and a sensitivity to the 10-year Google CDS curve would be: .*

FAQ

FAQ1 *MAR21.9(3) explicitly states that, for CSR curvature, the bond-CDS basis is ignored. Is it correct that, under MAR21.9(1), bond and CDS curves are considered distinct risk factors and the only "basis" taken into account in in MAR21.54 and MAR21.55 is the bond-CDS basis?*

Yes. Bond and CDS credit spreads are considered distinct risk factors under MAR21.9(1), and referenced in MAR21.54 and MAR21.55 is meant to capture only the bond-CDS basis.

21.55 For buckets 17 and 18, for aggregating delta CSR non-securitisations risk positions within a bucket, the correlation parameter between two weighted sensitivities and within the same bucket is set as follows, where:

ρ_{kl}

WS_k

WS_l

(1)

$\rho_{kl}^{(name)}$

is equal to 1 where the two names of sensitivities k and l are identical, and 80% otherwise;

(2)

$\rho_{kl}^{(tenor)}$

is equal to 1 if the two tenors of the sensitivities k and l are identical, and to 65% otherwise; and

(3)

$\rho_{kl}^{(basis)}$

is equal to 1 if the two sensitivities are related to same curves, and 99.90% otherwise.

$$\rho_{kl} = \rho_{kl}^{(name)} \cdot \rho_{kl}^{(tenor)} \cdot \rho_{kl}^{(basis)}$$

21.56 The correlations above do not apply to the other sector bucket (ie bucket 16).

(1) The aggregation of delta CSR non-securitisation risk positions within the other sector bucket would be equal to the simple sum of the absolute values of the net weighted sensitivities allocated to this bucket. The same method applies to the aggregation of vega risk positions.

$$K_{b(other\ bucket)} = \sum_k \left| WS_k \right|$$

(2) The aggregation of curvature CSR non-securitisation risk positions within the other sector bucket would be calculated by the formula below.

$$K_{b(other\ bucket)} = \max\left(\sum_k \max\left(CVR_k^+, 0\right), \sum_k \max\left(CVR_k^-, 0\right) \right)$$

21.57 For aggregating delta CSR non-securitisation risk positions across buckets 1 to 18, the correlation parameter is set as follows, where:

γ_{bc}

(1)

$\gamma_{bc}^{(rating)}$

is equal to 50% where the two buckets b and c are both in buckets 1 to 15 and have a different rating category (either IG or HY/NR). is equal to 1 otherwise; and

$\gamma_{bc}^{(rating)}$

(2)

$\gamma_{bc}^{(sector)}$

is equal to 1 if the two buckets belong to the same sector, and to the specified numbers in Table 5 otherwise.

$$\gamma_{bc} = \gamma_{bc}^{(rating)} \cdot \gamma_{bc}^{(sector)}$$

Values of

$\gamma_{bc}^{(sector)}$

where the buckets do not belong to the same sector

Table 5

Bucket	1 / 9	2 / 10	3 / 11	4 / 12	5 / 13	6 / 14	7 / 15	8	16	17	18
1 / 9		75%	10%	20%	25%	20%	15%	10%	0%	45%	45%
2 / 10			5%	15%	20%	15%	10%	10%	0%	45%	45%
3 / 11				5%	15%	20%	5%	20%	0%	45%	45%
4 / 12					20%	25%	5%	5%	0%	45%	45%
5 / 13						25%	5%	15%	0%	45%	45%
6 / 14							5%	20%	0%	45%	45%
7 / 15								5%	0%	45%	45%
8									0%	45%	45%
16										0%	0%
17											75%
18											

$\gamma_{bc}^{(sector)}$

Delta CSR securitisation (CTP) buckets, risk weights and correlations

21.58 Sensitivities to CSR arising from the CTP and its hedges are treated as a separate risk class as set out in MAR21.1. The buckets, risk weights and correlations for the CSR securitisations (CTP) apply as follows:

(1) The same bucket structure and correlation structure apply to the CSR securitisations (CTP) as those for the CSR non-securitisation framework as set out in MAR21.51 to MAR21.57 with an exception of index buckets (ie buckets 17 and 18).

(2) The risk weights and correlation parameters of the delta CSR non-securitisations are modified to reflect longer liquidity horizons and larger basis risk as specified in MAR21.59 to MAR21.61.

FAQ

FAQ1 *In accordance with MAR21.58(1), sensitivities to credit spread risk (CSR) arising from the correlation trading portfolio (CTP) should be classified according to the same bucket structure as the one for CSR non-securitisation, as set out in MAR21.51, except for index buckets (bucket 17 and bucket 18). Since an index CTP should be considered a risk factor as a whole and cannot be broken down into its constituents, as stated in MAR21.34(2), how should a bank determine which bucket to assign the delta sensitivity of an index CTP instrument, given the aforementioned bucket structure?*

The delta CSR sensitivity of an index CTP instrument should be assigned to a single specific delta sector bucket consistent with the characteristics of, at least, 75% of the index constituents (taking into account the weightings of that index), in accordance

with MAR21.33(1). If this is not possible, then the index should be assigned to bucket 16, "Other sector". The sensitivity to that index CTP instrument should be considered and treated like any other single-name sensitivity assigned to that same sector bucket.

21.59 For calculating weighted sensitivities, the risk weights for buckets 1 to 16 are set out in Table 6. Risk weights are the same for all tenors (ie 0.5 years, 1 year, 3 years, 5 years, 10 years) within each bucket:

Risk weights for sensitivities to CSR arising from the CTP	Table 6
Bucket number	Risk weight
1	4.0%
2	4.0%
3	8.0%
4	5.0%
5	4.0%
6	3.0%
7	2.0%
8	6.0%
9	13.0%
10	13.0%
11	16.0%
12	10.0%
13	12.0%
14	12.0%
15	12.0%
16	13.0%

21.60 For aggregating delta CSR securitisations (CTP) risk positions within a bucket, the delta risk correlation is derived the same way as in MAR21.54 and MAR21.55, except that the correlation parameter applying when the sensitivities are not related to same curves, , is modified.

ρ_{kl}

$\rho_{kl}^{(basis)}$

(1)

$\rho_{kl}^{(basis)}$

is now equal to 1 if the two sensitivities are related to same curves, and 99.00% otherwise.

(2) The identical correlation parameters for

$\rho_{kl}^{(name)}$

and to CSR non-securitisation as set out in MAR21.54 and MAR21.55 apply.

$\rho_{kl}^{(tenor)}$

21.61 For aggregating delta CSR securitisations (CTP) risk positions across buckets, the correlation parameters for γ_{bc} are identical to CSR non-securitisation as set out in MAR21.57.

Delta CSR securitisation (non-CTP) buckets, risk weights and correlations

21.62 For delta CSR securitisations not in the CTP, buckets are set along two dimensions – credit quality and sector – as set out in Table 7. The delta CSR securitisation (non-CTP) sensitivities or risk exposures must first be assigned to a bucket before calculating weighted sensitivities by applying a risk weight.

Buckets for delta CSR securitisations (non-CTP)		Table 7
Bucket number	Credit quality	Sector
1	Senior investment grade (IG)	RMBS – Prime
2		RMBS – Mid-prime
3		RMBS – Sub-prime
4		CMBS
5		Asset-backed securities (ABS) – Student loans
6		ABS – Credit cards
7		ABS – Auto
8		Collateralised loan obligation (CLO) non-CTP
9	Non-senior IG	RMBS – Prime
10		RMBS – Mid-prime
11		RMBS – Sub-prime
12		Commercial mortgage-backed securities (CMBS)
13		ABS – Student loans
14		ABS – Credit cards
15		ABS – Auto
16		CLO non-CTP
17	High yield & non-rated	RMBS – Prime
18		RMBS – Mid-prime
19		RMBS – Sub-prime
20		CMBS
21		ABS – Student loans
22		ABS – Credit cards
23		ABS – Auto
24		CLO non-CTP
25	Other sector[19]	

Footnotes

[19] *Credit quality is not a differentiating consideration for this bucket.*

21.63 To assign a risk exposure to a sector, banks must rely on a classification that is commonly used in the market for grouping tranches by type.

(1) The bank must assign each tranche to one of the sector buckets in above Table 7.

(2) Risk positions from any tranche that a bank cannot assign to a sector in this fashion must be assigned to the other sector (ie bucket 25).

21.64 For calculating weighted sensitivities, the risk weights for buckets 1 to 8 (senior IG) are set out in Table 8:

Risk weights for buckets 1 to 8 for delta CSR securitisations (non-CTP)	Table 8
Bucket number	Risk weight (in percentage points)
1	0.9%
2	1.5%
3	2.0%
4	2.0%
5	0.8%
6	1.2%
7	1.2%
8	1.4%

21.65 The risk weights for buckets 9 to 16 (non-senior investment grade) are then equal to the corresponding risk weights for buckets 1 to 8 scaled up by a multiplication by 1.25. For instance, the risk weight for bucket 9 is equal to .

$1.25 \times 0.9\% = 1.125\%$

21.66 The risk weights for buckets 17 to 24 (high yield and non-rated) are then equal to the corresponding risk weights for buckets 1 to 8 scaled up by a multiplication by 1.75. For instance, the risk weight for bucket 17 is equal to .

$1.75 \times 0.9\% = 1.575\%$

21.67 The risk weight for bucket 25 is set at 3.5%.

21.68 For aggregating delta CSR securitisations (non-CTP) risk positions within a bucket, the correlation parameter between two sensitivities and within the same bucket, is set as follows, where:

ρ_{kl}

WS_k

WS_l

(1)

$\rho_{kl}^{(tranche)}$

is equal to 1 where the two names of sensitivities k and l are within the same bucket and related to the same securitisation tranche (more than 80% overlap in notional terms), and 40% otherwise;

(2)

$\rho_{kl}^{(tenor)}$

is equal to 1 if the two tenors of the sensitivities k and l are identical, and to 80% otherwise; and

(3)

$\rho_{kl}^{(basis)}$

is equal to 1 if the two sensitivities are related to same curves, and 99.90% otherwise.

$$\rho_{kl} = \rho_{kl}^{(tranche)} \cdot \rho_{kl}^{(tenor)} \cdot \rho_{kl}^{(basis)}$$

FAQ

FAQ1 *MAR21.68 includes , which equals 1 where the two sensitivities within the same bucket are related to the same securitisation tranche, or 40% otherwise. There is no issuer factor. Does this mean that two sensitivities relating to the same issuer but different tranches require 40% correlation?*

Yes. There is no granularity for issuers in the delta CSR securitisation part as set out in MAR21.10. Where two tranches have exactly the same issuer, same tenor and same basis, but different tranches (ie different credit quality), the correlation must be 40%.

21.69 The correlations above do not apply to the other sector bucket (ie bucket 25).

(1) The aggregation of delta CSR securitisations (non-CTP) risk positions within the other sector bucket would be equal to the simple sum of the absolute values of the net weighted sensitivities allocated to this bucket. The same method applies to the aggregation of vega risk positions.

$$K_{b(other\ bucket)} = \sum_k \left| WS_k \right|$$

(2) The aggregation of curvature CSR risk positions within the other sector bucket would be calculated by the formula below.

$$K_{b(other\ bucket)} = \max\left(\sum_k \max\left(CVR_k^+, 0\right), \sum_k \max\left(CVR_k^-, 0\right) \right)$$

21.70 For aggregating delta CSR securitisations (non-CTP) risk positions across buckets 1 to 24, the correlation parameter is set as 0%.

γ_{bc}

21.71 For aggregating delta CSR securitisations (non-CTP) risk positions between the other sector bucket (ie bucket 25) and buckets 1 to 24, (i) the capital requirements for bucket 25 and (ii) the aggregated capital requirements for buckets 1 to 24 will be simply summed up to the overall risk class level capital requirements. There should be no diversification or hedging effects recognised in aggregating the capital requirements for the other sector bucket (ie bucket 25) with those for buckets 1 to 24.

Equity risk buckets, risk weights and correlations

21.72 For delta equity risk, buckets are set along three dimensions – market capitalisation, economy and sector – as set out in Table 9. The equity risk sensitivities or exposures must first be assigned to a bucket before calculating weighted sensitivities by applying a risk weight.

Buckets for delta sensitivities to equity risk	Table 9

Bucket number	Market cap	Economy	Sector
1	Large	Emerging market economy	Consumer goods and services, transportation and storage, administrative and support service activities, healthcare, utilities
2			Telecommunications, industrials
3			Basic materials, energy, agriculture, manufacturing, mining and quarrying
4			Financials including government-backed financials, real estate activities, technology
5		Advanced economy	Consumer goods and services, transportation and storage, administrative and support service activities, healthcare, utilities
6			Telecommunications, industrials
7			Basic materials, energy, agriculture, manufacturing, mining and quarrying
8			Financials including government-backed financials, real estate activities, technology
9	Small	Emerging market economy	All sectors described under bucket numbers 1, 2, 3 and 4
10		Advanced economy	All sectors described under bucket numbers 5, 6, 7 and 8
11	Other sector[20]		
12	Large market cap, advanced economy equity indices (non-sector specific)		
13	Other equity indices (non-sector specific)		

Footnotes

[20] *Market capitalisation or economy (ie advanced or emerging market) is not a differentiating consideration for this bucket.*

21.73 Market capitalisation (market cap) is defined as the sum of the market capitalisations based on the market value of the total outstanding shares issued by the same listed legal entity or a group of legal entities across all stock markets globally, where the total outstanding shares issued by the group of legal entities refer to cases where the listed entity is a parent company of a group of legal entities. Under no circumstances should the sum of the market capitalisations of multiple related listed entities be used to determine whether a listed entity is "large market cap" or "small market cap".

21.74 Large market cap is defined as a market capitalisation equal to or greater than USD 2 billion and small market cap is defined as a market capitalisation of less than USD 2 billion.

21.75 The advanced economies are Canada, the United States, Mexico, the euro area, the non-euro area western European countries (the United Kingdom, Norway, Sweden, Denmark and Switzerland), Japan, Oceania (Australia and New Zealand), Singapore and Hong Kong SAR.

FAQ

FAQ1 *Are the countries referenced in MAR21.75 to be understood as country of incorporation?*

An equity issuer must be allocated to a particular bucket according to the most

material country or region in which the issuer operates. As stated in MAR21.76: "For multinational multi-sector equity issuers, the allocation to a particular bucket must be done according to the most material region and sector in which the issuer operates.

21.76 To assign a risk exposure to a sector, banks must rely on a classification that is commonly used in the market for grouping issuers by industry sector.

(1) The bank must assign each issuer to one of the sector buckets in the table under MAR21.72 and it must assign all issuers from the same industry to the same sector.

(2) Risk positions from any issuer that a bank cannot assign to a sector in this fashion must be assigned to the other sector (ie bucket 11).

(3) For multinational multi-sector equity issuers, the allocation to a particular bucket must be done according to the most material region and sector in which the issuer operates.

21.77 For calculating weighted sensitivities, the risk weights for the sensitivities to each of equity spot price and equity repo rates for buckets 1 to 13 are set out in Table 10:

Risk weights for buckets 1 to 13 for sensitivities to equity risk		Table 10
Bucket number	Risk weight for equity spot price	Risk weight for equity repo rate
1	55%	0.55%
2	60%	0.60%
3	45%	0.45%
4	55%	0.55%
5	30%	0.30%
6	35%	0.35%
7	40%	0.40%
8	50%	0.50%
9	70%	0.70%
10	50%	0.50%
11	70%	0.70%
12	15%	0.15%
13	25%	0.25%

21.78 For aggregating delta equity risk positions within a bucket, the correlation parameter between two sensitivities and within the same bucket is set at as follows

ρ_{kl}

WS_k

WS_l

(1) The correlation parameter

ρ_{kl}

is set at 99.90%, where:

one is a sensitivity to an equity spot price and the other a sensitivity to an equity repo rates; and

both are related to the same equity issuer name.

(2) The correlation parameter

ρ_{kl}

is set out in (a) to (e) below, where both sensitivities are to equity spot price, and where:
15% between two sensitivities within the same bucket that fall under large market cap, emerging market economy (bucket number 1, 2, 3 or 4).
25% between two sensitivities within the same bucket that fall under large market cap, advanced economy (bucket number 5, 6, 7 or 8).
7.5% between two sensitivities within the same bucket that fall under small market cap, emerging market economy (bucket number 9).
12.5% between two sensitivities within the same bucket that fall under small market cap, advanced economy (bucket number 10).
80% between two sensitivities within the same bucket that fall under either index bucket (bucket number 12 or 13).

(3) The same correlation parameter

ρ_{kl}

as set out in above (2)(a) to (e) apply, where both sensitivities are to equity repo rates.

(4) The correlation parameter

ρ_{kl}

is set as each parameter specified in above (2)(a) to (e) multiplied by 99.90%, where:
One is a sensitivity to an equity spot price and the other a sensitivity to an equity repo rate; and
Each sensitivity is related to a different equity issuer name.

21.79 The correlations set out above do not apply to the other sector bucket (ie bucket 11).

(1) The aggregation of equity risk positions within the other sector bucket capital requirement would be equal to the simple sum of the absolute values of the net weighted sensitivities allocated to this bucket. The same method applies to the aggregation of vega risk positions.

$$K_{b(other\ bucket)} = \sum_k \left| WS_k \right|$$

(2) The aggregation of curvature equity risk positions within the other sector bucket (ie bucket 11) would be calculated by the formula:

$$K_{b(other\ bucket)} = \max\left(\sum_k \max\left(CVR_k^+, 0\right), \sum_k \max\left(CVR_k^-, 0\right) \right)$$

21.80 For aggregating delta equity risk positions across buckets 1 to 13, the correlation parameter is set at:

γ_{bc}

(1) 15% if bucket b and bucket c fall within bucket numbers 1 to 10;

(2) 0% if either of bucket b and bucket c is bucket 11;

(3) 75% if bucket b and bucket c are bucket numbers 12 and 13 (i.e. one is bucket 12, one is bucket 13); and

(4) 45% otherwise.

Commodity risk buckets, risk weights and correlations

21.81 For delta commodity risk, 11 buckets that group commodities by common characteristics are set out in Table 11.

21.82 For calculating weighted sensitivities, the risk weights for each bucket are set out in Table 11:

Delta commodity buckets and risk weights — Table 11

Bucket number	Commodity bucket	Examples of commodities allocated to each commodity bucket (non-exhaustive)	Risk weight
1	Energy - solid combustibles	Coal, charcoal, wood pellets, uranium	30%
2	Energy - liquid combustibles	Light-sweet crude oil; heavy crude oil; West Texas Intermediate (WTI) crude; Brent crude; etc (ie various types of crude oil) Bioethanol; biodiesel; etc (ie various biofuels) Propane; ethane; gasoline; methanol; butane; etc (ie various petrochemicals) Jet fuel; kerosene; gasoil; fuel oil; naphtha; heating oil; diesel etc (ie various refined fuels)	35%
3	Energy - electricity and carbon trading	Spot electricity; day-ahead electricity; peak electricity; off-peak electricity (ie various electricity types) Certified emissions reductions; in-delivery month EU allowance; Regional Greenhouse Gas Initiative CO2 allowance; renewable energy certificates; etc (ie various carbon trading emissions)	60%
4	Freight	Capesize; Panamax; Handysize; Supramax (ie various types of dry-bulk route) Suezmax; Aframax; very large crude carriers (ie various liquid-bulk/gas shipping route)	80%
5	Metals – non-precious	Aluminium; copper; lead; nickel; tin; zinc (ie various base metals) Steel billet; steel wire; steel coil; steel scrap; steel rebar; iron ore; tungsten; vanadium; titanium; tantalum (ie steel raw materials) Cobalt; manganese; molybdenum (ie various minor metals)	40%
6	Gaseous combustibles	Natural gas; liquefied natural gas	45%
7	Precious metals (including gold)	Gold; silver; platinum; palladium	20%
8	Grains and oilseed	Corn; wheat; soybean seed; soybean oil; soybean meal; oats; palm oil; canola; barley; rapeseed seed; rapeseed oil; rapeseed meal; red bean; sorghum; coconut oil; olive oil; peanut oil; sunflower oil; rice	35%
9	Livestock and dairy	Live cattle; feeder cattle; hog; poultry; lamb; fish; shrimp; milk; whey; eggs; butter; cheese	25%
10	Softs and other agriculturals	Cocoa; arabica coffee; robusta coffee; tea; citrus juice; orange juice; potatoes; sugar; cotton; wool; lumber; pulp; rubber	35%
11	Other commodity	Potash; fertilizer; phosphate rocks (ie various industrial materials) Rare earths; terephthalic acid; flat glass	50%

21.83 For the purpose of aggregating commodity risk positions within a bucket using a correlation parameter, the correlation parameter between two sensitivities and within the same bucket, is set as follows, where:[21]

ρ_{kl}

WS_k

WS_l

(1)

$\rho_{kl}^{(cty)}$

is equal to 1 where the two commodities of sensitivities k and l are identical, and to the intra-bucket correlations in Table 12 otherwise, where, any two commodities are considered distinct commodities if in the market two contracts are considered distinct when the only difference between each other is the underlying commodity to be delivered. For example, WTI and Brent in bucket 2 (ie energy - liquid combustibles) would typically be treated as distinct commodities;

(2)

$\rho_{kl}^{(tenor)}$

is equal to 1 if the two tenors of the sensitivities k and l are identical, and to 99.00% otherwise; and

(3)

$\rho_{kl}^{(basis)}$

is equal to 1 if the two sensitivities are identical in the delivery location of a commodity, and 99.90% otherwise.

$\rho_{kl} = \rho_{kl}^{(cty)} \cdot \rho_{kl}^{(tenor)} \cdot \rho_{kl}^{(basis)}$

Values of $\rho_{kl}^{(cty)}$ for intra-bucket correlations	Table 12

Bucket number	Commodity bucket	Correlation ($\rho_{kl}^{(cty)}$)
1	Energy - Solid combustibles	55%
2	Energy - Liquid combustibles	95%
3	Energy - Electricity and carbon trading	40%
4	Freight	80%
5	Metals - non-precious	60%
6	Gaseous combustibles	65%
7	Precious metals (including gold)	55%
8	Grains and oilseed	45%
9	Livestock and dairy	15%
10	Softs and other agriculturals	40%

11	Other commodity	15%

$\rho_{kl}^{(cty)}$

$\rho_{kl}^{(cty)}$

Footnotes

21 *For example, the correlation between the sensitivity to Brent, one-year tenor, for delivery in Le Havre and the sensitivity to WTI, five-year tenor, for delivery in Oklahoma is .*

FAQ

FAQ1 *For instruments with commodity spreads as underlying, are the spreads considered a risk factor, or does the instrument have to be decomposed? For example, if there is a swap on the spread between WTI and Brent, will delta on the spread be reported, or will delta of WTI and delta of Brent be reported individually?*

Instruments with a spread as their underlying are considered sensitive to different risk factors. In the example cited, the swap will be sensitive to both WTI and Brent, each of which require a capital charge at the risk factor level (ie delta of WTI and delta of Brent). The correlation to aggregate capital charges is specified in MAR21.83.

21.84 For determining whether the commodity correlation parameter ( ) as set out in Table 12 in MAR21.83(1)(a) should apply, this paragraph provides non-exhaustive examples of further definitions of distinct commodities as follows:
$(\rho_{kl}^{(cty)})$

(1) For bucket 3 (energy – electricity and carbon trading):

(a) Each time interval (i) at which the electricity can be delivered and (ii) that is specified in a contract that is made on a financial market is considered a distinct electricity commodity (eg peak and off-peak).

(b) Electricity produced in a specific region (eg Electricity NE, Electricity SE or Electricity North) is considered a distinct electricity commodity.

(2) For bucket 4 (freight):

(a) Each combination of freight type and route is considered a distinct commodity.

(b) Each week at which a good has to be delivered is considered a distinct commodity.

FAQ

FAQ1 *For instruments with commodity spreads as underlying, are the spreads considered a risk factor, or does the instrument have to be decomposed? For example, if there is a swap on the spread between WTI and Brent, will delta on the spread be reported, or will delta of WTI and delta of Brent be reported individually?*

Instruments with a spread as their underlying are considered sensitive to different risk factors. In the example cited, the swap will be sensitive to both WTI and Brent, each of which require a capital charge at the risk factor level (ie delta of WTI and delta of Brent). The correlation to aggregate capital charges is specified in MAR21.83.

21.85 For aggregating delta commodity risk positions across buckets, the correlation parameter is set as follows:

(1) 20% if bucket b and bucket c fall within bucket numbers 1 to 10; and

(2) 0% if either bucket b or bucket c is bucket number 11.

Foreign exchange risk buckets, risk weights and correlations

21.86 An FX risk bucket is set for each exchange rate between the currency in which an instrument is denominated and the reporting currency.

21.87 A unique relative risk weight equal to 15% applies to all the FX sensitivities.

21.88 For the specified currency pairs by the Basel Committee,[22] and for currency pairs forming first-order crosses across these specified currency pairs,[23] the above risk weight may at the discretion of the bank be divided by the square root of 2.

Footnotes

[22] *Specified currency pairs by the Basel Committee are: USD/EUR, USD/JPY, USD/GBP, USD/AUD, USD/CAD, USD/CHF, USD/MXN, USD/CNY, USD/NZD, USD/RUB, USD/HKD, USD/SGD, USD/TRY, USD/KRW, USD/SEK, USD/ZAR, USD/INR, USD/NOK, USD/BRL.*

[23] *For example, EUR/AUD is not among the selected currency pairs specified by the Basel Committee, but is a first-order cross of USD/EUR and USD/AUD.*

21.89 For aggregating delta FX risk positions across buckets, the correlation parameter γ_{bc} is uniformly set to 60%.

Sensitivities-based method: definition of vega risk buckets, risk weights and correlations

21.90 MAR21.91 to MAR21.95 set out buckets, risk weights and correlation parameters to calculate vega risk capital requirement as set out in MAR21.4.

21.91 The same bucket definitions for each risk class are used for vega risk as for delta risk.

21.92 For calculating weighted sensitivities for vega risk, the risk of market illiquidity is incorporated into the determination of vega risk, by assigning different liquidity horizons for each risk class as set out in Table 13. The risk weight for each risk class[24] is also set out in Table 13.

Table 13

Regulatory liquidity horizon, $LH_{risk\ class}$, and risk weights per risk class

Risk class	$LH_{risk\ class}$	Risk weights
GIRR	60	100%
CSR non-securitisations	120	100%
CSR securitisations (CTP)	120	100%

CSR securitisations (non-CTP)	120	100%
Equity (large cap and indices)	20	77.78%
Equity (small cap and other sector)	60	100%
Commodity	120	100%
FX	40	100%

$LH_{risk\ class}$

$LH_{risk\ class}$

Footnotes

24 *The risk weight for a given vega risk factor k is determined by ,where is set at 55%; and is specified per risk class in Table 13.*

FAQ

FAQ1 *When applying risk weights for equity vega risk factors, does the 20 days liquidity horizon apply to equities that are both large market cap and indices, or does it apply to equities that are either large market cap or indices? Similarly, does the 60 days liquidity horizon apply to equities that are both small market cap and other sector, or does it apply to equities that are either small market cap or other sector?*

The 20-day liquidity horizon applies to vega risk factors that would be allocated to large market cap buckets (ie buckets 1 to 8) or to index buckets (ie buckets 12 and 13) as set out in MAR21.72. The 60-day liquidity horizon applies to vega risk factors that would be allocated to small market cap buckets (ie buckets 9 and 10) or to the other sector bucket (ie bucket 11) as set out in MAR21.72.

21.93 For aggregating vega GIRR risk positions within a bucket, the correlation parameter is set as follows, where:

ρ_{kl}

(1)

$\rho_{kl}^{(option\ maturity)}$

is equal to , where:

$e^{-\alpha \cdot \frac{|T_k - T_l|}{min\{T_k; T_l\}}}$

is set at 1%;

α

(respectively) is the maturity of the option from which the vega sensitivity ( ) is derived, expressed as a number of years; and

T_k

T_l

VR_k

VR_l

(2)

$\rho_{kl}^{(underlying\ maturity)}$

is equal to , where:

$e^{-\alpha \cdot \frac{|T_k^U - T_l^U|}{min\{T_k^U; T_l^U\}}}$

is set at 1%; and
α

(respectively) is the maturity of the underlying of the option from which the sensitivity ( ) is derived, expressed as a number of years after the maturity of the option.
T_k^U
T_l^U
VR_k
VR_l

$$\rho_{kl} = min\left[\rho_{kl}^{(option\ maturity)} \cdot \rho_{kl}^{(underlying\ maturity)}; 1\right]$$

21.94 For aggregating vega risk positions within a bucket of the other risk classes (ie non-GIRR), the correlation parameter is set as follows, where:
ρ_{kl}

(1)

$\rho_{kl}^{(DELTA)}$
is equal to the correlation that applies between the delta risk factors that correspond to vega risk factors k and l. For instance, if k is the vega risk factor from equity option X and l is the vega risk factor from equity option Y then is the delta correlation applicable between X and Y; and
$\rho_{kl}^{(DELTA)}$

(2)

$\rho_{kl}^{(option\ maturity)}$
is defined as in MAR21.93:

$$\rho_{kl} = min\left[\rho_{kl}^{(DELTA)} \cdot \rho_{kl}^{(option\ maturity)}; 1\right]$$

FAQ

FAQ1 *MAR21.94 defines the vega correlation between risk factors k and l as the product of the option maturity correlation ( ) and the delta correlation ( ) that applies between the delta risk factors that correspond to vega risk factors k and l. Please clarify the meaning of "delta risk factors that correspond to vega risk factors k and l". In particular, besides the option maturity, should banks consider for CSR and commodity risk (i) the correlation across vega risk factors for the dimensions defined for vega for a given risk class only, or (ii) all dimensions of delta risk factors?*

For CSR and commodity risks in MAR21.9 to MAR21.11 and MAR21.13, if the vega risk factors are defined for a smaller number of dimensions than are defined for delta risk factors, only the dimensions that are defined both as a vega risk factor dimension and as a delta risk factor dimension for the relevant risk class need to be considered as a correlation based on delta risk factors ( ) in the calculation of vega risk per MAR21.94. This means that the following dimensions are considered:

21.95 For aggregating vega risk positions across different buckets within a risk class (GIRR and non-GIRR), the same correlation parameters for , as specified for delta correlations for each risk class in MAR21.39 to MAR21.89 are to be used for the aggregation of vega risk (eg = 50% is to be used for the aggregation of vega risk sensitivities across different GIRR buckets).

γ_{bc}

γ_{bc}

Sensitivities-based method: definition of curvature risk buckets, risk weights and correlations

21.96 MAR21.97 to MAR21.101 set out buckets, risk weights and correlation parameters to calculate curvature risk capital requirement as set out in MAR21.5.

21.97 The delta buckets are replicated for the calculation of curvature risk capital requirement, unless specified otherwise in the preceding paragraphs within MAR21.8 to MAR21.89.

21.98 For calculating the net curvature risk capital requirement for risk factor k for FX and equity risk classes, the curvature risk weight, which is the size of a shock to the given risk factor, is a relative shift equal to the respective delta risk weight. For FX curvature, for options that do not reference a bank's reporting currency (or base currency as set out in MAR21.14(b)) as an underlying, net curvature risk charges (and) may be divided by a scalar of 1.5. Alternatively, and subject to supervisory approval, a bank may apply the scalar of 1.5 consistently to all FX instruments provided curvature sensitivities are calculated for all currencies, including sensitivities determined by shocking the reporting currency (or base currency where used) relative to all other currencies.

CVR_k

CVR_k^+

CVR_k^-

21.99 For calculating the net curvature risk capital requirement for curvature risk factor k for GIRR, CSR and commodity risk classes, the curvature risk weight is the parallel shift of all the tenors for each curve based on the highest prescribed delta risk weight for each bucket. For example, in the case of GIRR for a given currency (ie bucket), the risk weight assigned to 0.25-year tenor (ie the most punitive tenor risk weight) is applied to all the tenors simultaneously for each risk-free yield curve (consistent with a "translation", or "parallel shift" risk calculation).

CVR_k

FAQ

FAQ1 *In relation to the curvature risk capital requirement for the commodity risk class, MAR21.99 requires that the curvature risk weight is the parallel shift of all the tenors for each curve based on the highest prescribed delta risk weight for each bucket. A parallel shift in MAR21.99 implies that an additive shock (in absolute terms) is applied along the curve. However, MAR21.23 states that the shock applied to delta commodity is a relative shock. How should the shock be applied to commodity curvature?*

The sizes of upward and downward shocks applied to assess the net curvature risk capital requirement for a specific commodityś curvature risk factor should be based on the risk weight connected to the curvature bucket where that commodity is classified, in accordance with MAR21.97 and MAR21.82. The same relative shocks should be applied to all curvature risk factors classified under the same bucket, defined along the dimension of the constructed curve (ie no term structure decomposition) per each commodity spot price, as described in MAR21.13(3). For example, the constructed curve for gold (with a risk weight of 20%) would be shifted up by multiplying each tenor price by 1.2 and down by multiplying each tenor price by 0.8.

FAQ2 *When calculating curvature capital requirements for the CSR risk classes as described in MAR21.1(1b), (1c) and (1d), are banks allowed to floor the respective CSR curvature risk factors at zero when applying the downward shock to the CSR curvature risk factor?*

When calculating curvature capital requirements for the CSR risk classes in accordance with MAR21.5 that would result in a negative credit spread for , banks may floor the CSR curvature risk factor to zero, after the application of the downward shock (ie). When this approach is applied, banks must cap the risk weight ( ) as the difference between the level of the CSR curvature risk factor and zero in calculating . However, banks must not cap the risk weight when calculating .

21.100 For aggregating curvature risk positions within a bucket, the curvature risk correlations are determined by squaring the corresponding delta correlation parameters . In a case where a curvature risk factor is defined differently than the corresponding delta risk factor for a given risk class (ie for CSR non-securitisations, CSR securitisations (CTP), CSR securitisations (non-CTP) and commodities as defined in MAR21.9 to MAR21.13), banks do not need to consider this delta risk factor dimension. For example, for CSR non-securitisations and CSR securitisations (CTP), consistent with MAR21.9 which defines a bucket along one dimension (ie the relevant credit spread curve), the correlation parameter as defined in MAR21.54 and MAR21.55 is not applicable to the curvature risk capital requirement calculation. Thus, the correlation parameter is determined by whether the two names of weighted sensitivities are the same. In the formula in MAR21.54 and MAR21.55, the correlation parameters and need not apply and only correlation parameter applies between two weighted sensitivities within the same bucket. This correlation parameter should be squared. In applying the high and low correlations scenario set out in MAR21.6, the curvature risk capital requirements are calculated by applying the curvature correlation parameters determined in this paragraph.

ρ_{kl}

ρ_{kl}

ρ_{kl}

$\rho_{kl}^{(basis)}$

$\rho_{kl}^{(tenor)}$

$\rho_{kl}^{(name)}$

ρ_{kl}

21.101 For aggregating curvature risk positions across buckets, the curvature risk correlations are determined by squaring the corresponding delta correlation parameters . For instance, when aggregating and for the GIRR, the correlation should be . In applying the high and low correlations scenario set out in MAR21.6, the curvature risk capital requirements are calculated by applying the curvature correlation parameters , (ie the square of the corresponding delta correlation parameter).

γ_{bc}

γ_{bc}

CVR_{EUR}

CVR_{USD}

$50\%^2 = 25\%$

γ_{bc}

MAR22

Standardised approach: default risk capital requirement

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

Version effective as of 01 Jan 2023

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

Main concepts of default risk capital requirements

22.1 The default risk capital (DRC) requirement is intended to capture jump-to-default (JTD) risk that may not be captured by credit spread shocks under the sensitivities-based method. DRC requirements provide some limited hedging recognition. In this chapter offsetting refers to the netting of exposures to the same obligor (where a short exposure may be subtracted in full from a long exposure) and hedging refers to the application of a partial hedge benefit from the short exposures (where the risk of long and short exposures in distinct obligors do not fully offset due to basis or correlation risks).

Instruments subject to the default risk capital requirement

22.2 The DRC requirement must be calculated for instruments subject to default risk:

(1) Non-securitisation portfolios

(2) Securitisation portfolio (non-correlation trading portfolio, or non-CTP)

(3) Securitisation (correlation trading portfolio, or CTP)

Overview of DRC requirement calculation

22.3 The following step-by-step approach must be followed for each risk class subject to default risk. The specific definition of gross JTD risk, net JTD risk, bucket, risk weight and the method for aggregation of DRC requirement across buckets are separately set out per each risk class in subsections in MAR22.9 to MAR22.26.

(1) The gross JTD risk of each exposure is computed separately.

(2) With respect to the same obligator, the JTD amounts of long and short exposures are offset (where permissible) to produce net long and/or net short exposure amounts per distinct obligor.

(3) Net JTD risk positions are then allocated to buckets.

(4) Within a bucket, a hedge benefit ratio is calculated using net long and short JTD risk positions. This acts as a discount factor that reduces the amount of net short positions to be netted against net long positions within a bucket. A prescribed risk weight is applied to the net positions which are then aggregated.

(5) Bucket level DRC requirements are aggregated as a simple sum across buckets to give the overall DRC requirement.

22.4 No diversification benefit is recognised between the DRC requirements for:

(1) non-securitisations;

(2) securitisations (non-CTP); and

(3) securitisations (CTP).

22.5 For traded non-securitisation credit and equity derivatives, JTD risk positions by individual constituent issuer legal entity should be determined by applying a look-through approach.
FAQ

FAQ1 *What is the JTD equivalent when decomposing multiple underlying positions of a single security or product (eg index options) for purposes of the standardised approach?*

The JTD equivalent is defined as the difference between the value of the security or product assuming that each single name referenced by the security or product, separately from the others, defaults (with zero recovery) and the value of the security

or product assuming that none of the names referenced by the security or product default.

22.6 For the CTP, the capital requirement calculation includes the default risk for non-securitisation hedges. These hedges must be removed from the calculation of default risk non-securitisation.

22.7 Claims on sovereigns, public sector entities and multilateral development banks may, at national discretion, be subject to a zero default risk weight in line with CRE20.7 to CRE20.15 of the credit risk standard. National authorities may apply a non-zero risk weight to securities issued by certain foreign governments, including to securities denominated in a currency other than that of the issuing government.

22.8 For claims on an equity investment in a fund that is subject to the treatment specified in MAR21.36(3) (ie treated as an unrated "other sector" equity), the equity investment in the fund shall be treated as an unrated equity instrument. Where the mandate of that fund allows the fund to invest in primarily high-yield or distressed names, banks shall apply the maximum risk weight per Table 2 in MAR22.24 that is achievable under the fund's mandate (by calculating the effective average risk weight of the fund when assuming that the fund invests first in defaulted instruments to the maximum possible extent allowed under its mandate, and then in CCC-rated names to the maximum possible extent, and then B-rated, and then BB-rated). Neither offsetting nor diversification between these generated exposures and other exposures is allowed.

FAQ

FAQ1 *For equity investments in funds for which sensitivities-based method capital requirements are calculated under MAR21.36(3) (ie the "other sector" equity treatment), may the mandate of the fund be used to determine the jump-to-default (JTD) of the fund for default risk?*

No. In calculating the JTD, the LGD of equity investments in funds for which sensitivities-based method capital requirements are calculated under MAR21.36(3) should be 100%, consistent with the requirement in MAR22.8 to treat the equity investment as a position in an unrated equity instrument.

Default risk capital requirement for non-securitisations

Gross jump-to-default risk positions (gross JTD)

22.9 The gross JTD risk position is computed exposure by exposure. For instance, if a bank has a long position on a bond issued by Apple, and another short position on a bond issued by Apple, it must compute two separate JTD exposures.

22.10 For the purpose of DRC requirements, the determination of the long/short direction of positions must be on the basis of long or short with respect to whether the credit exposure results in a loss or gain in the case of a default.

(1) Specifically, a long exposure is defined as a credit exposure that results in a loss in the case of a default.

(2) For derivative contracts, the long/short direction is also determined by whether the contract will result in a loss in the case of a default (ie long or short position is not determined by whether the option or credit default swap (CDS), is bought or sold). Thus, for the purpose of DRC requirements, a sold put option on a bond is a long credit exposure, since a default results in a loss to the seller of the option.

22.11 The gross JTD is a function of the loss given default (LGD), notional amount (or face value) and the cumulative profit and loss (P&L) already realised on the position, where:

(1) notional is the bond-equivalent notional amount (or face value) of the position; and

(2) P&L is the cumulative mark-to-market loss (or gain) already taken on the exposure. P&L is equal to the market value minus the notional amount, where the market value is the current market value of the position.

$$JTD(long) = \max\left(LGD \times notional + P\&L, 0\right)$$

$$JTD(short) = \min\left(LGD \times notional + P\&L, 0\right)$$

FAQ

FAQ1 *What is the JTD equivalent when decomposing multiple underlying positions of a single security or product (eg index options) for purposes of the standardised approach?*

The JTD equivalent is defined as the difference between the value of the security or product assuming that each single name referenced by the security or product, separately from the others, defaults (with zero recovery) and the value of the security or product assuming that none of the names referenced by the security or product default.

22.12 For calculating the gross JTD, LGD is set as follows:

(1) Equity instruments and non-senior debt instruments are assigned an LGD of 100%.

(2) Senior debt instruments are assigned an LGD of 75%.

(3) Covered bonds, as defined within MAR21.51, are assigned an LGD of 25%.

(4) When the price of the instrument is not linked to the recovery rate of the defaulter (eg a foreign exchange-credit hybrid option where the cash flows are swap of cash flows, long EUR coupons and short USD coupons with a knockout feature that ends cash flows on an event of default of a particular obligor), there should be no multiplication of the notional by the LGD.

FAQ

FAQ1 *For the purpose of market risk capital requirements, what are the credit spread risk capital requirements for Fannie Mae and Freddie Mac mortgage-backed security (MBS) bonds? What is the LGD for Fannie and Freddie MBS?*

Non-tranched MBS issued by government sponsored-entities (GSEs), such as Fannie and Freddie, are assigned to bucket 2 (local government, government-backed non-financials, education, public administration) for credit spread risk with a risk weight of 1.0%.

In accordance with MAR22.12, the LGD for non-tranched MBS issued by GSEs is 75% (ie the LGD assigned to senior debt instruments) unless the GSE security satisfies the requirements of footnote 15 to MAR21.51 for treatment of the security as a covered bond.

22.13 In calculating the JTD as set out in MAR22.11, the notional amount of an instrument that gives rise to a long (short) exposure is recorded as a positive (negative) value, while the P&L loss (gain) is recorded as a negative (positive) value. If the contractual or legal terms of the derivative allow for the unwinding of the instrument with no exposure to default risk, then the JTD is equal to zero.

22.14 The notional amount is used to determine the loss of principal at default, and the mark-to-market loss is used to determine the net loss so as to not double-count the mark-to-market loss already recorded in the market value of the position.

(1) For all instruments, the notional amount is the notional amount of the instrument relative to which the loss of principal is determined. Examples are as follows:

(a) For a bond, the notional amount is the face value.

(b) For credit derivatives, the notional amount of a CDS contract or a put option on a bond is the notional amount of the derivative contract.

(c) In the case of a call option on a bond, the notional amount to be used in the JTD calculation is zero (since, in the event of default, the call option will not be exercised). In this case, a JTD would extinguish the call option's value and this loss would be captured through the mark-to-market P&L term in the JTD calculation.

(2) Table 1 illustrates examples of the notional amounts and market values for a long credit position with a mark-to-market loss to be used in the JTD calculation, where:

(a) the bond-equivalent market value is an intermediate step in determining the P&L for derivative instruments;

(b) the mark-to-market value of CDS or an option takes an absolute value; and

(c) the strike amount of the bond option is expressed in terms of the bond price (not the yield).

Examples of components for a long credit position in the JTD calculation — Table 1

Instrument	Notional	Bond-equivalent market value	P&L
Bond	Face value of bond	Market value of bond	Market value - face value
CDS	Notional of CDS	Notional of CDS + mark-to-market (MtM) value of CDS	- MtM value of CDS
Sold put option on a bond	Notional of option	Strike amount - | MtM value of option |	(Strike - | MtM value of option |) - Notional
Bought call option on a bond	0	MtM value of option	MtM value of option

P&L = bond-equivalent market value - notional. With this representation of the P&L for a sold put option, a lower strike results in a lower JTD loss.

FAQ

FAQ1 *What is the JTD equivalent when decomposing multiple underlying positions of a single security or product (eg index options) for purposes of the standardised approach?*

The JTD equivalent is defined as the difference between the value of the security or product assuming that each single name referenced by the security or product, separately from the others, defaults (with zero recovery) and the value of the security or product assuming that none of the names referenced by the security or product default.

FAQ2 *Are convertible bonds to be treated the same way as vanilla bonds in computing the DRC requirement?*

No. Banks should also consider the P&L of the equity optionality embedded within a convertible bond when computing its DRC requirement. A convertible bond can be decomposed into a vanilla bond and a long equity option. Hence, treating the

convertible bond as a vanilla bond will potentially underestimate the JTD risk of the instrument.

22.15 To account for defaults within the one-year capital horizon, the JTD for all exposures of maturity less than one year and their hedges are scaled by a fraction of a year. No scaling is applied to the JTD for exposures of one year or greater.[1] For example, the JTD for a position with a six month maturity would be weighted by one-half, while the JTD for a position with a one year maturity would have no scaling applied to the JTD.

Footnotes

[1] *Note that this paragraph refers to the scaling of gross JTD (ie not net JTD).*

FAQ

FAQ1 *MAR22.16 states that for the standardised approach DRC requirement, cash equity positions may be attributed a maturity of three months or a maturity of more than one year, at firms' discretion. Such restrictions do not exist in MAR33 for the internal models approach, which allows banks discretion to apply a 60-day liquidity horizon for equity sub-portfolios. Furthermore, MAR22.15 states "… the JTD for all exposures of maturity less than one year and their hedges are scaled by a fraction of a year". Given the above-mentioned paragraphs, for purposes of the standardised approach DRC requirement, is a bank permitted to assign cash equities and equity derivatives such as index futures any maturity between three months and one year on a sub-portfolio basis in order to avoid broken hedges?*

No. Such discretion is not permitted in the standardised approach. As required by MAR22.16, cash equity positions are assigned a maturity of either more than one year or three months. There is no discretion permitted to assign cash equity positions to any maturity between three months and one year. In determining the offsetting criterion, MAR22.17 specifies that the maturity of the derivatives contract be considered, not the maturity of the underlying instrument. MAR22.18 further states that the maturity weighting applied to the JTD for any product with a maturity of less than three months is floored at three months.

*To illustrate how the standardised approach DRC requirement should be calculated with a simple hypothetical portfolio, consider equity index futures with one month to maturity and a negative market value of EUR 10 million (–EUR 10 million, maturity 1M), hedged with the underlying equity positions with a positive market value of EUR 10 million (+EUR 10 million). Both positions in the example should be considered having a three-month maturity. Based on MAR22.15, which requires maturity scaling, defined as a fraction of the year, of positions and their hedge, the JTD for the above trading portfolio would be calculated as follows: 1/4*10 – 1/4*10 = 0.*

22.16 Cash equity positions (ie stocks) are assigned to a maturity of either more than one year or three months, at banks' discretion.

FAQ

FAQ1 *MAR22.16 states that for the standardised approach DRC requirement, cash equity positions may be attributed a maturity of three months or a maturity of more than one year, at firms' discretion. Such restrictions do not exist in MAR33 for the internal models approach, which allows banks discretion to apply a 60-day liquidity horizon for equity sub-portfolios. Furthermore, MAR22.15 states "… the JTD for all exposures of maturity less than one year and their hedges are scaled by a fraction of a year". Given the above-mentioned paragraphs, for purposes of the standardised approach DRC requirement, is a bank permitted to assign cash equities and equity derivatives*

such as index futures any maturity between three months and one year on a sub-portfolio basis in order to avoid broken hedges?

No. Such discretion is not permitted in the standardised approach. As required by MAR22.16, cash equity positions are assigned a maturity of either more than one year or three months. There is no discretion permitted to assign cash equity positions to any maturity between three months and one year. In determining the offsetting criterion, MAR22.17 specifies that the maturity of the derivatives contract be considered, not the maturity of the underlying instrument. MAR22.18 further states that the maturity weighting applied to the JTD for any product with maturity of less than three months is floored at three months.

*To illustrate how the standardised approach DRC requirement should be calculated with a simple hypothetical portfolio, consider equity index futures with one month to maturity and a negative market value of EUR 10 million (–EUR 10 million, maturity 1M), hedged with the underlying equity positions with a positive market value of EUR 10 million (+EUR 10 million). Both positions in the example should be considered having a three-month maturity. Based on MAR22.15, which requires maturity scaling, defined as a fraction of the year, of positions and their hedge, the JTD for the above trading portfolio would be calculated as follows: 1/4*10 – 1/4*10 = 0.*

22.17 For derivative exposures, the maturity of the derivative contract is considered in determining the offsetting criterion, not the maturity of the underlying instrument.

22.18 The maturity weighting applied to the JTD for any sort of product with a maturity of less than three months (such as short term lending) is floored at a weighting factor of one-fourth or, equivalently, three months (that means that the positions having shorter-than-three months remaining maturity would be regarded as having a remaining maturity of three months for the purpose of the DRC requirement).

FAQ

FAQ1 *In the case where a total return swap (TRS) with a maturity of one month is hedged by the underlying equity, would the bank still need to compute a DRC requirement if there were sufficient legal terms on the TRS such that there is no settlement risk at swap maturity as the swap is terminated based on the executed price of the stock/bond hedge and any unwind of the TRS can be delayed (beyond the swap maturity date) in the event of hedge disruption until the stock/bond can be liquidated?*

The net JTD for such a position would be zero. If the contractual/legal terms of the derivative allow for the unwinding of both legs of the position at the time of expiry of the first to mature with no exposure to default risk of the underlying credit beyond that point, then the JTD for the maturity-mismatched position is equal to zero.

Net jump-to-default risk positions (net JTD)

22.19 Exposures to the same obligator may be offset as follows:

(1) The gross JTD risk positions of long and short exposures to the same obligor may be offset where the short exposure has the same or lower seniority relative to the long exposure. For example, a short exposure in an equity may offset a long exposure in a bond, but a short exposure in a bond cannot offset a long exposure in the equity.

(2) For the purposes of determining whether a guaranteed bond is an exposure to the underlying obligor or an exposure to the guarantor, the credit risk mitigation requirements set out in CRE22.71 and CRE22.73 apply.

(3) Exposures of different maturities that meet this offsetting criterion may be offset as

follows.

(a) Exposures with maturities longer than the capital horizon (one year) may be fully offset.

(b) An exposure to an obligor comprising a mix of long and short exposures with a maturity less than the capital horizon (equal to one year) must be weighted by the ratio of the exposure's maturity relative to the capital horizon. For example, with the one-year capital horizon, a three-month short exposure would be weighted so that its benefit against long exposures of longer-than-one-year maturity would be reduced to one quarter of the exposure size.

22.20 In the case of long and short offsetting exposures where both have a maturity under one year, the scaling can be applied to both the long and short exposures.

22.21 Finally, the offsetting may result in net long JTD risk positions and net short JTD risk positions. The net long and net short JTD risk positions are aggregated separately as described below.

Calculation of default risk capital requirement for non-securitisation

22.22 For the default risk of non-securitisations, three buckets are defined as:

(1) corporates;

(2) sovereigns; and

(3) local governments and municipalities.

22.23 In order to recognise hedging relationship between net long and net short positions within a bucket, a hedge benefit ratio is computed as follows.

(1) A simple sum of the net long JTD risk positions (not risk-weighted) must be calculated, where the summation is across the credit quality categories (ie rating bands). The aggregated amount is used in the numerator and denominator of the expression of the hedge benefit ratio (HBR) below.

(2) A simple sum of the net (not risk-weighted) short JTD risk positions must be calculated, where the summation is across the credit quality categories (ie rating bands). The aggregated amount is used in the denominator of the expression of the HBR below.

(3) The HBR is the ratio of net long JTD risk positions to the sum of net long JTD and absolute value of net short JTD risk positions:

$$HBR = \frac{\sum \text{net JTD}_{\text{long}}}{\sum \text{net JTD}_{\text{long}} + \sum \left|\text{net JTD}_{\text{short}}\right|}$$

22.24 For calculating the weighted net JTD, default risk weights are set depending on the credit quality categories (ie rating bands) for all three buckets (ie irrespective of the type of counterparty), as set out in Table 2:

Default risk weights for non-securitisations by credit quality category	Table 2
Credit quality category	Default risk weight
AAA	0.5%
AA	2%
A	3%

BBB	6%
BB	15%
B	30%
CCC	50%
Unrated	15%
Defaulted	100%

FAQ

FAQ1 *How are risk weights to be determined when external ratings assigned by credit rating agencies differ and when there are no external ratings available?*

Consistent with the treatment of external ratings under the standardised approach to credit risk (see CRE21.10 and CRE21.11), if there are two ratings that map into different risk weights, the higher risk weight should be applied. If there are three or more ratings with different risk weights, the ratings corresponding to the two lowest risk weights should be referred to and the higher of those two risk weights will be applied.

Consistent with the treatment where there are no external ratings under the CVA risk chapter (see MAR50.16), where there are no external ratings or where external ratings are not recognised within a jurisdiction, banks may, subject to supervisory approval:

- *for the purpose of assigning delta CSR non-securitisation risk weights, map the internal rating to an external rating, and assign a risk weight corresponding to either "investment grade" or "high-yield" in the MAR21.51;*
- *for the purpose of assigning default risk weights under the DRC requirement, map the internal rating to an external rating, and assign a risk weight corresponding to one of the seven external ratings in the table included in MAR22.24; or*
- *apply the risk weights specified in MAR21.53 and MAR22.24 for unrated/non-rated categories.*

22.25 The capital requirement for each bucket is to be calculated as the combination of the sum of the risk-weighted long net JTD, the HBR, and the sum of the risk-weighted short net JTD, where the summation for each long net JTD and short net JTD is across the credit quality categories (ie rating bands). In the following formula, DRC stands for DRC requirement; and i refers to an instrument belonging to bucket b.

$$DRC_b = \max\left[\left(\sum_{i \in \text{Long}} RW_i \cdot \text{net JTD}_i\right) - HBR \cdot \left(\sum_{i \in \text{Short}} RW_i \cdot |\text{net JTD}_i|\right); 0\right]$$

22.26 No hedging is recognised between different buckets - the total DRC requirement for non-securitisations must be calculated as a simple sum of the bucket level capital requirements.

Default risk capital requirement for securitisations (non-CTP)

Gross jump-to-default risk positions (gross JTD)

22.27 For the computation of gross JTD on securitisations, the same approach must be followed

as for default risk (non-securitisations), except that an LGD ratio is not applied to the exposure. Because the LGD is already included in the default risk weights for securitisations to be applied to the securitisation exposure (see below), to avoid double counting of LGD the JTD for securitisations is simply the market value of the securitisation exposure (ie the JTD for tranche positions is their market value).

22.28 For the purposes of offsetting and hedging recognition for securitisations (non-CTP), positions in underlying names or a non-tranched index position may be decomposed proportionately into the equivalent replicating tranches that span the entire tranche structure. When underlying names are treated in this way, they must be removed from the non-securitisation default risk treatment.

Net jump-to-default risk positions (net JTD)

22.29 For default risk of securitisations (non-CTP), offsetting is limited to a specific securitisation exposure (ie tranches with the same underlying asset pool). This means that:

(1) no offsetting is permitted between securitisation exposures with different underlying securitised portfolio (ie underlying asset pools), even if the attachment and detachment points are the same; and

(2) no offsetting is permitted between securitisation exposures arising from different tranches with the same securitised portfolio.

22.30 Securitisation exposures that are otherwise identical except for maturity may be offset. The same offsetting rules for non-securitisations including scaling down positions of less than one year as set out in MAR22.15 through MAR22.18 apply to JTD risk positions for securitisations (non-CTP). Offsetting within a specific securitisation exposure is allowed as follows.

(1) Securitisation exposures that can be perfectly replicated through decomposition may be offset. Specifically, if a collection of long securitisation exposures can be replicated by a collection of short securitisation exposures, then the securitisation exposures may be offset.

(2) Furthermore, when a long securitisation exposure can be replicated by a collection of short securitisation exposures with different securitised portfolios, then the securitisation exposure with the "mixed" securitisation portfolio may be offset by the combination of replicating securitisation exposures.

(3) After the decomposition, the offsetting rules would apply as in any other case. As in the case of default risk (non-securitisations), long and short securitisation exposures should be determined from the perspective of long or short the underlying credit, eg the bank making losses on a long securitisation exposure in the event of a default in the securitised portfolio.

Calculation of default risk capital requirement for securitisations (non-CTP)

22.31 For default risk of securitisations (non-CTP), the buckets are defined as follows:

(1) Corporates (excluding small and medium enterprises) – this bucket takes into account all regions.

(2) Other buckets – these are defined along two dimensions:

(a) Asset classes: the 11 asset classes are defined as asset-backed commercial paper; auto Loans/Leases; residential mortgage-backed securities (MBS); credit cards; commercial MBS; collateralised loan obligations; collateralised debt obligation (CDO)-squared; small and medium enterprises; student loans, other retail; and

other wholesale.

(b) Regions: the four regions are defined as Asia, Europe, North America and all other.

22.32 To assign a securitisation exposure to a bucket, banks must rely on a classification that is commonly used in the market for grouping securitisation exposures by type and region of underlying.

(1) The bank must assign each securitisation exposure to one and only one of the buckets above and it must assign all securitisations with the same type and region of underlying to the same bucket.

(2) Any securitisation exposure that a bank cannot assign to a type or region of underlying in this fashion must be assigned to the "other bucket".

22.33 The capital requirement for default risk of securitisations (non-CTP) is determined using a similar approach to that for non-securitisations. The DRC requirement within a bucket is calculated as follows:

(1) The hedge benefit discount HBR, as defined in MAR22.23, is applied to net short securitisation exposures in that bucket.

(2) The capital requirement is calculated as in MAR22.25.

22.34 For calculating the weighted net JTD, the risk weights of securitisation exposures are defined by the tranche instead of the credit quality. The risk weight for securitisations (non-CTP) is applied as follows:

(1) The default risk weights for securitisation exposures are based on the corresponding risk weights for banking book instruments as set out in CRE40 to CRE44, with the following modification: the maturity component in the banking book securitisation framework is set to zero (ie a one-year maturity is assumed) to avoid double-counting of risks in the maturity adjustment (of the banking book approach) since migration risk in the trading book will be captured in the credit spread capital requirement.

(2) Following the corresponding treatment in the banking book, the hierarchy of approaches in determining the risk weights should be applied at the underlying pool level.

(3) The capital requirement under the standardised approach for an individual cash securitisation position can be capped at the fair value of the transaction.

22.35 No hedging is recognised between different buckets. Therefore, the total DRC requirement for securitisations (non-CTP) must be calculated as a simple sum of the bucket-level capital requirements.

Default risk capital requirement for securitisations (CTP)

Gross jump-to-default risk positions (gross JTD)

22.36 For the computation of gross JTD on securitisations (CTP), the same approach must be followed as for default risk-securitisations (non-CTP) as described in MAR22.27.

22.37 The gross JTD for non-securitisations (CTP) (ie single-name and index hedges) positions is defined as their market value.

22.38 Nth-to-default products should be treated as tranched products with attachment and detachment points defined below, where "Total names" is the total number of names in the underlying basket or pool:

(1) Attachment point = (N – 1) / Total names

(2) Detachment point = N / Total names

Net jump-to-default risk positions (net JTD)

22.39 Exposures that are otherwise identical except for maturity may be offset. The same concept of long and short positions from a perspective of loss or gain in the event of a default as set out in MAR22.10 and offsetting rules for non-securitisations including scaling down positions of less than one year as set out in MAR22.15 to MAR22.18 apply to JTD risk positions for securitisations (non-CTP).

(1) For index products, for the exact same index family (eg CDX.NA.IG), series (eg series 18) and tranche (eg 0–3%), securitisation exposures should be offset (netted) across maturities (subject to the offsetting allowance as described above).

(2) Long and short exposures that are perfect replications through decomposition may be offset as follows. When the offsetting involves decomposing single name equivalent exposures, decomposition using a valuation model would be allowed in certain cases as follows. Such decomposition is the sensitivity of the security's value to the default of the underlying single name obligor. Decomposition with a valuation model is defined as follows: a single name equivalent constituent of a securitisation (eg tranched position) is the difference between the unconditional value of the securitisation and the conditional value of the securitisation assuming that the single name defaults, with zero recovery, where the value is determined by a valuation model. In such cases, the decomposition into single-name equivalent exposures must account for the effect of marginal defaults of the single names in the securitisation, where in particular the sum of the decomposed single name amounts must be consistent with the undecomposed value of the securitisation. Further, such decomposition is restricted to vanilla securitisations (eg vanilla CDOs, index tranches or bespokes); while the decomposition of exotic securitisations (eg CDO squared) is prohibited.

(3) Moreover, for long and short positions in index tranches, and indices (non-tranched), if the exposures are to the exact same series of the index, then offsetting is allowed by replication and decomposition. For instance, a long securitisation exposure in a 10–15% tranche vs combined short securitisation exposures in 10–12% and 12–15% tranches on the same index/series can be offset against each other. Similarly, long securitisation exposures in the various tranches that, when combined perfectly, replicate a position in the index series (non-tranched) can be offset against a short securitisation exposure in the index series if all the positions are to the exact same index and series (eg CDX.NA.IG series 18). Long and short positions in indices and single-name constituents in the index may also be offset by decomposition. For instance, single-name long securitisation exposures that perfectly replicate an index may be offset against a short securitisation exposure in the index. When a perfect replication is not possible, then offsetting is not allowed except as indicated in the next sentence. Where the long and short securitisation exposures are otherwise equivalent except for a residual component, the net amount must show the residual exposure. For instance, a long securitisation exposure in an index of 125 names, and short securitisation exposures of the appropriate replicating amounts in 124 of the names, would result in a net long securitisation exposure in the missing 125th name of the index.

(4) Different tranches of the same index or series may not be offset (netted), different series of the same index may not be offset, and different index families may not be offset.

Calculation of default risk capital requirement for securitisations (CTP)

22.40 For default risk of securitisations (CTP), each index is defined as a bucket of its own. A non-exhaustive list of indices include: CDX North America IG, iTraxx Europe IG, CDX HY, iTraxx XO, LCDX (loan index), iTraxx LevX (loan index), Asia Corp, Latin America Corp, Other Regions Corp, Major Sovereign (G7 and Western Europe) and Other Sovereign.

22.41 Bespoke securitisation exposures should be allocated to the index bucket of the index they are a bespoke tranche of. For instance, the bespoke tranche 5% - 8% of a given index should be allocated to the bucket of that index.

22.42 The default risk weights for securitisations applied to tranches are based on the corresponding risk weights for the banking book instruments, which is defined in a separate Basel Committee publication - Revisions to the Securitisations framework of 2014, 2016 and 2018, with the following modification: the maturity component in the banking book securitisation framework is set to zero, ie a one-year maturity is assumed to avoid double-counting of risks in the maturity adjustment (of the banking book approach) since migration risk in the trading book will be captured in the credit spread capital requirement.

22.43 For the non-tranched products, the same risk weights for non-securitisations as set out in MAR22.24 apply. For the tranched products, banks must derive the risk weight using the banking book treatment as set out in MAR22.42.

22.44 Within a bucket (ie for each index) at an index level, the capital requirement for default risk of securitisations (CTP) is determined in a similar approach to that for non-securitisations.

(1) The hedge benefit ratio (HBR), as defined in MAR22.23, is modified and applied to net short positions in that bucket as in the formula below, where the subscript ctp for the term HBRctp indicates that the HBR is determined using the combined long and short positions across all indices in the CTP (ie not only the long and short positions of the bucket by itself). The summation of risk-weighted amounts in the formula spans all exposures relating to the index (ie index tranche, bespoke, non-tranche index or single name).

(2) A deviation from the approach for non-securitisations is that no floor at zero applies at the bucket level, and consequently, the DRC requirement at the index level (DRC_b) can be negative.

$$DRC_b = \left(\sum_{i \in Long} RW_i \cdot net\ JTD_i \right) - HBR_{ctp} \cdot \left(\sum_{i \in Short} RW_i \cdot |\ net\ JTD_i\ | \right)$$

22.45 The total DRC requirement for securitisations (CTP) is calculated by aggregating bucket level capital amounts as follows. For instance, if the DRC requirement for the index CDX North America IG is +100 and the DRC requirement for the index Major Sovereign (G7 and Western Europe) is -100, the total DRC requirement for the CTP is $100 - 0.5 \times 100 = 50$.[2]

$$DRC_{CTP} = max[\sum_b \left(max\left[DRC_b, 0 \right] + 0.5 \times min\left[DRC_b, 0 \right] \right), 0]$$

Footnotes

[2] *The procedure for the and terms accounts for the basis risk in cross index hedges,*

as the hedge benefit from cross-index short positions is discounted twice, first by the hedge benefit ratio HBR in , and again by the term 0.5 in the equation.

MAR23

Standardised approach: residual risk add-on

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

Version effective as of 01 Jan 2023

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

Introduction

23.1 The residual risk add-on (RRAO) is to be calculated for all instruments bearing residual risk separately in addition to other components of the capital requirement under the standardised approach.

Instruments subject to the residual risk add-on

23.2 Instruments with an exotic underlying and instruments bearing other residual risks are subject to the RRAO.

23.3 Instruments with an exotic underlying are trading book instruments with an underlying exposure that is not within the scope of delta, vega or curvature risk treatment in any risk class under the sensitivities-based method or default risk capital (DRC) requirements in the standardised approach.[1]

Footnotes

[1] *Examples of exotic underlying exposures include: longevity risk, weather, natural disasters, future realised volatility (as an underlying exposure for a swap).*

FAQ

FAQ1 *Is future realised volatility considered an "exotic underlying" for the purpose of the RRAO?*

Yes, future realised volatility is considered an exotic underlying for the purpose of the RRAO.

23.4 Instruments bearing other residual risks are those that meet criteria (1) and (2) below:

(1) Instruments subject to vega or curvature risk capital requirements in the trading book and with pay-offs that cannot be written or perfectly replicated as a finite linear combination of vanilla options with a single underlying equity price, commodity price, exchange rate, bond price, credit default swap price or interest rate swap; or

(2) Instruments which fall under the definition of the correlation trading portfolio (CTP) in MAR20.5, except for those instruments that are recognised in the market risk framework as eligible hedges of risks within the CTP.

FAQ

FAQ1 *Are bonds with multiple call dates considered instruments bearing other residual risks for the purpose of the RRAO?*

Yes. Bonds with multiple call dates would be considered as instruments bearing other residual risks, as they are path-dependent options.

23.5 A non-exhaustive list of other residual risks types and instruments that may fall within the criteria set out in MAR23.4 include:

(1) Gap risk: risk of a significant change in vega parameters in options due to small movements in the underlying, which results in hedge slippage. Relevant instruments subject to gap risk include all path dependent options, such as barrier options, and Asian options as well as all digital options.

(2) Correlation risk: risk of a change in a correlation parameter necessary for determining the value of an instrument with multiple underlyings. Relevant instruments subject to correlation risk include all basket options, best-of-options, spread options, basis options, Bermudan options and quanto options.

(3) Behavioural risk: risk of a change in exercise/prepayment outcomes such as those that

arise in fixed rate mortgage products where retail clients may make decisions motivated by factors other than pure financial gain (such as demographical features and/or and other social factors). A callable bond may only be seen as possibly having behavioural risk if the right to call lies with a retail client.

23.6 When an instrument is subject to one or more of the following risk types, this by itself will not cause the instrument to be subject to the RRAO:

(1) Risk from a cheapest-to-deliver option;

(2) Smile risk: the risk of a change in an implied volatility parameter necessary for determining the value of an instrument with optionality relative to the implied volatility of other instruments optionality with the same underlying and maturity, but different moneyness;

(3) Correlation risk arising from multi-underlying European or American plain vanilla options, and from any options that can be written as a linear combination of such options. This exemption applies in particular to the relevant index options;

(4) Dividend risk arising from a derivative instrument whose underlying does not consist solely of dividend payments; and

(5) Index instruments and multi-underlying options of which treatment for delta, vega or curvature risk are set out in MAR21.31 and MAR21.32. These are subject to the RRAO if they fall within the definitions set out in this chapter. For funds that are subject to the treatment specified in MAR21.36(3) (ie treated as an unrated "other sector" equity), banks shall assume the fund is exposed to exotic underlying exposures, and to other residual risks, to the maximum possible extent allowed under the fund's mandate.

23.7 In cases where a transaction exactly matches with a third-party transaction (ie a back-to-back transaction), the instruments used in both transactions must be excluded from the RRAO capital requirement. Any instrument that is listed and/or eligible for central clearing must be excluded from the RRAO for other residual risks as defined in MAR23.4. Any instrument that is listed and/or eligible for central clearing with an exotic underlying must be included in the RRAO.

FAQ

FAQ1 *Can hedges (for example, dividend swaps hedging dividend risks) be excluded from the RRAO?*

Hedges may be excluded from the RRAO only if the hedge exactly matches the trade (ie via a back-to-back transaction) as per MAR23.7. For the example cited, dividend swaps should remain within the RRAO.

FAQ2 *Can total return swap (TRS) products be netted with the underlying product(s) that drive the value of the TRS for the purposes of the RRAO?*

As per MAR23.7, a TRS on an underlying product may be excluded from the RRAO capital requirement if there is an equal and opposite exposure in the same TRS. If no exactly matching transaction exists, the entire notional of the TRS would be allocated to the RRAO.

Calculation of the residual risk add-on

23.8 The residual risk add-on must be calculated in addition to any other capital requirements within the standardised approach. The residual risk add-on is to be calculated as follows.

(1) The scope of instruments that are subject to the RRAO must not have an impact in terms of increasing or decreasing the scope of risk factors subject to the delta, vega,

curvature or DRC treatments in the standardised approach.

(2) The RRAO is the simple sum of gross notional amounts of the instruments bearing residual risks, multiplied by a risk weight.

(a) The risk weight for instruments with an exotic underlying specified in MAR23.3 is 1.0%.

(b) The risk weight for instruments bearing other residual risks specified in MAR23.4 is 0.1%.[2]

Footnotes

[2] *Where the bank cannot satisfy the supervisor that the RRAO provides a sufficiently prudent capital charge, the supervisor will address any potentially under-capitalised risks by imposing a conservative additional capital charge under Pillar 2.*

MAR30

Internal models approach: general provisions

More rigorous model approval process that enables supervisors to remove internal modelling permission for individual trading desks. Updated to take account of the revised implementation date announced on 27 March 2020. FAQ on climate related financial risks added on 8 December 2022. Updated to incorporate the FAQ published on 5 July 2024.

Version effective as of 01 Jan 2023

More rigorous model approval process that enables supervisors to remove internal modelling permission for individual trading desks. Updated to take account of the revised implementation date announced on 27 March 2020. FAQ on climate related financial risks added on 8 December 2022. Updated to incorporate the FAQ published on 5 July 2024.

General criteria

30.1 The use of internal models for the purposes of determining market risk capital requirements is conditional upon the explicit approval of the bank's supervisory authority.

30.2 The supervisory authority will only approve a bank's use of internal models to determine market risk capital requirements if, at a minimum:

(1) the supervisory authority is satisfied that the bank's risk management system is conceptually sound and is implemented with integrity;

(2) the bank has, in the supervisory authority's view, a sufficient number of staff skilled in the use of sophisticated models not only in the trading area but also in the risk control, audit and, if necessary, back office areas;

(3) the bank's trading desk risk management model has, in the supervisory authority's judgement, a proven track record of reasonable accuracy in measuring risk;

(4) the bank regularly conducts stress tests along the lines set out in MAR30.19 to MAR30.23; and

(5) the positions included in the bank's internal trading desk risk management models for determining minimum market risk capital requirements are held in trading desks that have been approved for the use of those models and that have passed the required tests described in MAR30.17.

FAQ

FAQ1 *How are the capital requirements for modellable risk factors (internally modelled capital charge, IMCC), stressed expected shortfall (SES) and default risk charge (DRC) calculated for reporting at the end of each quarter? More precisely, how are the results of backtesting at the trading desk level, the profit and loss attribution test (PLAT) at the trading desk level, and the risk factor eligibility test (RFET), as well as a change of a stress period and a reduced set of risk factors, considered in the calculation of the single risk figures used to determine the average capital numbers at the end of the quarter?*

The scope of desks approved and eligible for the internal models approach (IMA) capital requirement calculation at the end of the quarter should be based on the results of backtesting and PLAT at the trading desk level. The PLAT and RFET must be performed quarterly. The Basel Framework does not specify in detail when these tests must be performed during the quarter but requires that "the bank's risk management system is conceptually sound and is implemented with integrity". Therefore, risk measures (Expected Shortfall (ES), SES, DRC) entering the 60-day (or 12-week) averages used for IMA capital computation at the end of the quarter should be calculated based on a stable set of desks. To calculate those risk measures, the results of the backtesting, PLAT and RFET should be used to update the set of eligible trading desks as well as the reduced set and the stressed period at the beginning of the 60-day (or 12-week) calculation period. To ensure representativeness, banks should ensure that the dates on which the backtesting and PLAT conclude and the RFET is performed are sufficiently close to the beginning of the 60-day (or 12-week) calculation period (the end of the previous quarter).

30.3 Supervisory authorities may insist on a period of initial monitoring and live testing of a bank's internal trading desk risk management model before it is used for the purposes of determining the bank's market risk capital requirements.

30.4 The scope of trading portfolios that are eligible to use internal models to determine market

risk capital requirements is determined based on a three-prong approach as follows:

(1) The bank must satisfy its supervisory authority that both the bank's organisational infrastructure (including the definition and structure of trading desks) and its bank-wide internal risk management model meet qualitative evaluation criteria, as set out in MAR30.5 to MAR30.16.

(2) The bank must nominate individual trading desks, as defined in MAR12.1 to MAR12.6, for which the bank seeks model approval in order to use the internal models approach (IMA).

(a) The bank must nominate trading desks that it intends to be in-scope for model approval and trading desks that are out-of-scope for the use of the IMA. The bank must specify in writing the basis for these nominations.

(b) The bank must not nominate trading desks to be out-of-scope for model approval due to capital requirements for a particular trading desk determined using the standardised approach being lower than those determined using the IMA.

(c) The bank must use the standardised approach to determine the market risk capital requirements for trading desks that are out-of-scope for model approval. The positions in these out-of-scope trading desks are to be combined with all other positions that are subject to the standardised approach in order to determine the bank's standardised approach capital requirements.

(d) Trading desks that the bank does not nominate for model approval at the time of model approval will be ineligible to use the IMA for a period of at least one year from the date of the latest internal model approval.

(3) The bank must receive supervisory approval to use the IMA on individual trading desks. Following the identification of eligible trading desks, this step determines which trading desks will be in-scope to use the IMA and which risk factors within in-scope trading desks are eligible to be included in the bank's internal expected shortfall (ES) models to determine market risk capital requirements as set out in MAR33.

(a) Each trading desk must satisfy profit and loss (P&L) attribution (PLA) tests on an ongoing basis to be eligible to use the IMA to determine market risk capital requirements. In order to conduct the PLA test, the bank must identify the set of risk factors to be used to determine its market risk capital requirements.

(b) Each trading desk also must satisfy backtesting requirements on an ongoing basis to be eligible to use the IMA to determine market risk capital requirements as set out in MAR32.4 to MAR32.19.

(c) Banks must conduct PLA tests and backtesting on a quarterly basis to update the eligibility and trading desk classification in PLA for trading desks in-scope to use the IMA.

(d) The market risk capital requirements for risk factors that satisfy the risk factor eligibility test as set out in MAR31.12 to MAR31.24 must be determined using ES models as specified in MAR33.1 to MAR33.15.

(e) The market risk capital requirements for risk factors that do not satisfy the risk factor eligibility test must be determined using stressed expected shortfall (SES) models as specified in MAR33.16 to MAR33.17.

FAQ

FAQ1 *The model approval process requires an overall assessment of a bank's bank-wide internal risk capital model. Does the use of the term "bank-wide" include a group of trading desks to be nominated as in-scope for model approval?*

The term "bank-wide" is defined as pertaining to the group of trading desks that the bank nominates as in-scope in their application for the IMA.

FAQ2 *As securitisations are out of scope for the IMA (IMA), are banks required to segregate desks to ensure securitisation and non-securitisation products reside in different trading desks? If not, how should banks test model eligibility?*

Securitisation positions are out of scope for IMA regulatory capital treatment, and as a result they are not taken into account for the model eligibility tests. This implies that banks are not allowed to include securitisations in trading desks for which they determine market risk capital requirements using the IMA. Securitisations must be included in trading desks for which capital requirements are determined using the standardised approach. Banks are allowed to also include hedging instruments in trading desks which include securitisations and are capitalised using the standardised approach.

Qualitative standards

30.5 In order to use the IMA to determine market risk capital requirements, the bank must have market risk management systems that are conceptually sound and implemented with integrity. Accordingly, the bank must meet the qualitative criteria set out below on an ongoing basis. Supervisors will assess that the bank has met the criteria before the bank is permitted to use the IMA.

30.6 The bank must have an independent risk control unit that is responsible for the design and implementation of the bank's market risk management system. The risk control unit should produce and analyse daily reports on the output of the trading desk's risk management model, including an evaluation of the relationship between measures of risk exposure and trading limits. This risk control unit must be independent of business trading units and should report directly to senior management of the bank.

30.7 The bank's risk control unit must conduct regular backtesting and PLA assessments at the trading desk level. The bank must also conduct regular backtesting of its bank-wide internal models used for determining market risk capital requirements.

30.8 A distinct unit of the bank that is separate from the unit that designs and implements the internal models must conduct the initial and ongoing validation of all internal models used to determine market risk capital requirements. The model validation unit must validate all internal models used for purposes of the IMA on at least an annual basis.

30.9 The board of directors and senior management of the bank must be actively involved in the risk control process and must devote appropriate resources to risk control as an essential aspect of the business. In this regard, the daily reports prepared by the independent risk control unit must be reviewed by a level of management with sufficient seniority and authority to enforce both reductions of positions taken by individual traders and reductions in the bank's overall risk exposure.

30.10 Internal models used to determine market risk capital requirements are likely to differ from those used by a bank in its day-to-day internal risk management functions. Nevertheless, the core design elements of both the market risk capital requirement model and the internal risk management model should be the same.

(1) Valuation models that are a feature of both models should be similar. These valuation models must be an integral part of the internal identification, measurement, management and internal reporting of price risks within the bank's trading desks.

(2) Internal risk management models should, at a minimum, be used to assess the risk of the positions that are subject to market risk capital requirements, although they may assess a broader set of positions.

(3) The construction of a trading desk risk management model must be based on the methodologies used in the bank's internal risk management model with regard to risk factor identification, parameter estimation and proxy concepts and deviate only if this is appropriate due to regulatory requirements. A bank's market risk capital requirement model and its internal risk management model should address an identical set of risk factors.

30.11 A routine and rigorous programme of stress testing is required. The results of stress testing must be:

(1) reviewed at least monthly by senior management;

(2) used in the bank's internal assessment of capital adequacy; and

(3) reflected in the policies and limits set by the bank's management and its board of directors.

30.12 Where stress tests reveal particular vulnerability to a given set of circumstances, the bank must take prompt action to mitigate those risks appropriately (eg by hedging against that outcome, reducing the size of the bank's exposures or increasing capital).

30.13 The bank must maintain a protocol for compliance with a documented set of internal manuals, policies, controls and procedures concerning the operation of the internal market risk management model. The bank's risk management model must be well documented. Such documentation may include a comprehensive risk management manual that describes the basic principles of the risk management model and that provides a detailed explanation of the empirical techniques used to measure market risk.

30.14 The bank must receive approval from its supervisory authority prior to implementing any significant changes to its internal models used to determine market risk capital requirements.

30.15 The bank's internal models for determining market risk capital requirements must address the full set of positions that are in the scope of application of the model. All models' measurements of risk must be based on a sound theoretical basis, calculated correctly, and reported accurately.

30.16 The bank's internal audit and validation functions or external auditor must conduct an independent review of the market risk measurement system on at least an annual basis. The scope of the independent review must include both the activities of the business trading units and the activities of the independent risk control unit. The independent review must be sufficiently detailed to determine which trading desks are impacted by any failings. At a minimum, the scope of the independent review must include the following:

(1) the organisation of the risk control unit;

(2) the adequacy of the documentation of the risk management model and process;

(3) the accuracy and appropriateness of market risk management models (including any significant changes);

(4) the verification of the consistency, timeliness and reliability of data sources used to run internal models, including the independence of such data sources;

(5) the approval process for risk pricing models and valuation systems used by the bank's

front- and back-office personnel;

(6) the scope of market risks reflected in the trading desk risk management models;

(7) the integrity of the management information system;

(8) the accuracy and completeness of position data;

(9) the accuracy and appropriateness of volatility and correlation assumptions;

(10) the accuracy of valuation and risk transformation calculations;

(11) the verification of trading desk risk management model accuracy through frequent backtesting and PLA assessments; and

(12) the general alignment between the model to determine market risk capital requirements and the model the bank uses in its day-to-day internal management functions.

Model validation standards

30.17 Banks must maintain a process to ensure that their internal models have been adequately validated by suitably qualified parties independent of the model development process to ensure that each model is conceptually sound and adequately reflects all material risks. Model validation must be conducted both when the model is initially developed and when any significant changes are made to the model. The bank must revalidate its models periodically, particularly when there have been significant structural changes in the market or changes to the composition of the bank's portfolio that might lead to the models no longer being adequate. Model validation must include PLA and backtesting, and must, at a minimum, also include the following:

(1) Tests to demonstrate that any assumptions made within internal models are appropriate and do not underestimate risk. This may include reviewing the appropriateness of assumptions of normal distributions and any pricing models.

(2) Further to the regulatory backtesting programmes, model validation must assess the hypothetical P&L (HPL) calculation methodology.

(3) The bank must use hypothetical portfolios to ensure that internal models are able to account for particular structural features that may arise. For example, where the data history for a particular instrument does not meet the quantitative standards in MAR33.1 to MAR33.12 and the bank maps these positions to proxies, the bank must ensure that the proxies produce conservative results under relevant market scenarios, with sufficient consideration given to ensuring:

(a) that material basis risks are adequately reflected (including mismatches between long and short positions by maturity or by issuer); and

(b) that the models reflect concentration risk that may arise in an undiversified portfolio.

External validation

30.18 The model validation conducted by external auditors and/or supervisory authorities of a bank's internal model to determine market risk capital requirements should, at a minimum, include the following steps:

(1) Verification that the internal validation processes described in MAR30.17 are operating in a satisfactory manner;

(2) Confirmation that the formulae used in the calculation process, as well as for the pricing of options and other complex instruments, are validated by a qualified unit,

which in all cases should be independent from the bank's trading area;

(3) Confirmation that the structure of internal models is adequate with respect to the bank's activities and geographical coverage;

(4) Review of the results of both the bank's backtesting of its internal models (ie comparison of value-at-risk with actual P&L and HPL) and its PLA process to ensure that the models provide a reliable measure of potential losses over time. On request, a bank should make available to its supervisory authority and/or to its external auditors the results as well as the underlying inputs to ES calculations and details of the PLA exercise; and

(5) Confirmation that data flows and processes associated with the risk measurement system are transparent and accessible. On request and in accordance with procedures, the bank should provide its supervisory authority and its external auditors access to the models' specifications and parameters.

Stress testing

30.19 Banks that use the IMA for determining market risk capital requirements must have in place a rigorous and comprehensive stress testing programme both at the trading desk level and at the bank-wide level.

30.20 Banks' stress scenarios must cover a range of factors that (i) can create extraordinary losses or gains in trading portfolios, or (ii) make the control of risk in those portfolios very difficult. These factors include low-probability events in all major types of risk, including the various components of market, credit and operational risks. A bank must design stress scenarios to assess the impact of such factors on positions that feature both linear and non-linear price characteristics (ie options and instruments that have option-like characteristics).

30.21 Banks' stress tests should be of a quantitative and qualitative nature, incorporating both market risk and liquidity risk aspects of market disturbances.

(1) Quantitative elements should identify plausible stress scenarios to which banks could be exposed.

(2) Qualitatively, a bank's stress testing programme should evaluate the capacity of the bank's capital to absorb potential significant losses and identify steps the bank can take to reduce its risk and conserve capital.

30.22 Banks should routinely communicate results of stress testing to senior management and should periodically communicate those results to the bank's board of directors.

30.23 Banks should combine the use of supervisory stress scenarios with stress tests developed by the bank itself to reflect its specific risk characteristics. Stress scenarios may include the following:

(1) Supervisory scenarios requiring no simulations by the bank. A bank should have information on the largest losses experienced during the reporting period and may be required to make this available for supervisory review. Supervisors may compare this loss information to the level of capital requirements that would result from a bank's internal measurement system. For example, the bank may be required to provide supervisory authorities with an assessment of how many days of peak day losses would have been covered by a given ES estimate.

(2) Scenarios requiring a simulation by the bank. Banks should subject their portfolios to a series of simulated stress scenarios and provide supervisory authorities with the

results. These scenarios could include testing the current portfolio against past periods of significant disturbance (eg the 1987 equity crash, the Exchange Rate Mechanism crises of 1992 and 1993, the increase in interest rates in the first quarter of 1994, the 1998 Russian financial crisis, the 2000 bursting of the technology stock bubble, the 2007–08 subprime mortgage crisis, or the 2011–12 Euro zone crisis) incorporating both the significant price movements and the sharp reduction in liquidity associated with these events. A second type of scenario would evaluate the sensitivity of the bank's market risk exposure to changes in the assumptions about volatilities and correlations. Applying this test would require an evaluation of the historical range of variation for volatilities and correlations and evaluation of the bank's current positions against the extreme values of the historical range. Due consideration should be given to the sharp variation that at times has occurred in a matter of days in periods of significant market disturbance. For example, the above-mentioned situations involved correlations within risk factors approaching the extreme values of 1 or –1 for several days at the height of the disturbance.

(3) Bank-developed stress scenarios. In addition to the scenarios prescribed by supervisory authorities under MAR30.23(1), a bank should also develop its own stress tests that it identifies as most adverse based on the characteristics of its portfolio (eg problems in a key region of the world combined with a sharp move in oil prices). A bank should provide supervisory authorities with a description of the methodology used to identify and carry out the scenarios as well as with a description of the results derived from these scenarios.

MAR31

Internal models approach: model requirements

Updated to include the following FAQ: MAR31.13 FAQ2.

Version effective as of 01 Jan 2023

Updated to include the following FAQ: MAR31.13 FAQ2.

Specification of market risk factors

31.1 An important part of a bank's trading desk internal risk management model is the specification of an appropriate set of market risk factors. Risk factors are the market rates and prices that affect the value of the bank's trading positions. The risk factors contained in a trading desk risk management model must be sufficient to represent the risks inherent in the bank's portfolio of on- and off-balance sheet trading positions. Although banks will have some discretion in specifying the risk factors for their internal models, the following requirements must be fulfilled.

31.2 A bank's market risk capital requirement models should include all risk factors that are used for pricing. In the event a risk factor is incorporated in a pricing model but not in the trading desk risk management model, the bank must support this omission to the satisfaction of its supervisory authority.

31.3 A bank's market risk capital requirement model must include all risk factors that are specified in the standardised approach for the corresponding risk class, as set out in MAR20 to MAR22.

(1) In the event a standardised approach risk factor is not included in the market risk capital requirement model, the bank must support this omission to the satisfaction of its supervisory authority.

(2) For securitised products, banks are prohibited from using internal models to determine market risk capital requirements. Banks must use the standardised approach to determine the market risk capital requirements for securitised products as set out in MAR11.9. Accordingly, a bank's market risk capital requirement model should not specify risk factors for securitisations as defined in MAR21.10 to MAR21.11.

31.4 A bank's market risk capital requirement model and any stress scenarios calculated for non-modellable risk factors must address non-linearities for options and other relevant products (eg mortgage-backed securities), as well as correlation risk and relevant basis risks (eg basis risks between credit default swaps and bonds).

31.5 A bank may use proxies for which there is an appropriate track record for their representation of a position (eg an equity index used as a proxy for a position in an individual stock). In the event a bank uses proxies, the bank must support their use to the satisfaction of the bank's supervisory authority.

31.6 For general interest rate risk, a bank must use a set of risk factors that corresponds to the interest rates associated with each currency in which the bank has interest rate sensitive on- or off-balance sheet trading positions.

(1) The trading desk risk management model must model the yield curve using one of a number of generally accepted approaches (eg estimating forward rates of zero coupon yields).

(2) The yield curve must be divided into maturity segments in order to capture variation in the volatility of rates along the yield curve.

(3) For material exposures to interest rate movements in the major currencies and markets, banks must model the yield curve using a minimum of six risk factors.

(4) The number of risk factors used ultimately should be driven by the nature of the bank's trading strategies. A bank with a portfolio of various types of securities across many points of the yield curve and that engages in complex arbitrage strategies would require the use of a greater number of risk factors than a bank with less complex portfolios.

31.7 The trading desk risk management model must incorporate separate risk factors to capture credit spread risk (eg between bonds and swaps). A variety of approaches may be used to reflect the credit spread risk arising from less-than-perfectly correlated movements between government and other fixed income instruments, such as specifying a completely separate yield curve for non-government fixed income instruments (eg swaps or municipal securities) or estimating the spread over government rates at various points along the yield curve.

31.8 For exchange rate risk, the trading desk risk management model must incorporate risk factors that correspond to the individual foreign currencies in which the bank's positions are denominated. Because the output of a bank's risk measurement system will be expressed in the bank's reporting currency, any net position denominated in a foreign currency will introduce foreign exchange risk. A bank must utilise risk factors that correspond to the exchange rate between the bank's reporting currency and each foreign currency in which the bank has a significant exposure.

31.9 For equity risk, a bank must utilise risk factors that correspond to each of the equity markets in which the bank holds significant positions.

(1) At a minimum, a bank must utilise risk factors that reflect market-wide movements in equity prices (eg a market index). Positions in individual securities or in sector indices may be expressed in beta-equivalents relative to a market-wide index.

(2) A bank may utilise risk factors that correspond to various sectors of the overall equity market (eg industry sectors or cyclical and non-cyclical sectors). Positions in individual securities within each sector may be expressed in beta-equivalents relative to a sector index.

(3) A bank may also utilise risk factors that correspond to the volatility of individual equities.

(4) The sophistication and nature of the modelling technique for a given market should correspond to the bank's exposure to the overall market as well as the bank's concentration in individual equities in that market.

31.10 For commodity risk, bank must utilise risk factors that correspond to each of the commodity markets in which the bank holds significant positions.

(1) For banks with relatively limited positions in commodity-based instruments, the bank may utilise a straightforward specification of risk factors. Such a specification could entail utilising one risk factor for each commodity price to which the bank is exposed (including different risk factors for different geographies where relevant).

(2) For a bank with active trading in commodities, the bank's model must account for variation in the convenience yield[1] between derivatives positions such as forwards and swaps and cash positions in the commodity.

Footnotes

[1] *The convenience yield reflects the benefits from direct ownership of the physical commodity (eg the ability to profit from temporary market shortages). The convenience yield is affected both by market conditions and by factors such as physical storage costs.*

31.11 For the risks associated with equity investments in funds:

(1) For funds that meet the criterion set out in RBC25.8(5)(a) (ie funds with look-through possibility), banks must consider the risks of the fund, and of any associated hedges, as if the fund's positions were held directly by the bank (taking into account the bank's share of the equity of the fund, and any leverage in the fund structure). The bank must

assign these positions to the trading desk to which the fund is assigned.

(2) For funds that do not meet the criterion set out in RBC25.8(5)(a), but meet both the criteria set out in RBC25.8(5)(b) (ie daily prices and knowledge of the mandate of the fund), banks must use the standardised approach to calculate capital requirements for the fund.

Model eligibility of risk factors

31.12 A bank must determine which risk factors within its trading desks that have received approval to use the internal models approach as set out in MAR32 are eligible to be included in the bank's internal expected shortfall (ES) model for regulatory capital requirements as set out in MAR33. For a risk factor to be classified as modellable by a bank, a necessary condition is that it passes the risk factor eligibility test (RFET). This test requires identification of a sufficient number of real prices that are representative of the risk factor. Collateral reconciliations or valuations cannot be considered real prices to meet the RFET. A price will be considered real if it meets at least one of the following criteria:

(1) It is a price at which the institution has conducted a transaction;

(2) It is a verifiable price for an actual transaction between other arms-length parties;

(3) It is a price obtained from a committed quote made by (i) the bank itself or (ii) another party. The committed quote must be collected and verified through a third-party vendor, a trading platform or an exchange; or

(4) It is a price that is obtained from a third-party vendor, where:

(a) the transaction or committed quote has been processed through the vendor;

(b) the vendor agrees to provide evidence of the transaction or committed quote to supervisors upon request; or

(c) the price meets any of the three criteria immediately listed in MAR31.12(1) to MAR31.12(3).

FAQ

FAQ1 *What is the definition of a "committed quote" as referenced in MAR31.12?*

A committed quote is a price from an arm's length provider at which the provider of the quote must buy or sell the financial instrument.

FAQ2 *Are all transactions and eligible committed quotes valid as real price observations, regardless of size?*

Orderly transactions and eligible committed quotes with a non-negligible volume, as compared to usual transaction sizes for the bank, reflective of normal market conditions can be generally accepted as valid.

31.13 To pass the RFET, a risk factor that a bank uses in an internal model must meet either of the following criteria on a quarterly basis. Any real price that is observed for a transaction should be counted as an observation for all of the risk factors for which it is representative.

(1) The bank must identify for the risk factor at least 24 real price observations per year (measured over the period used to calibrate the current ES model, with no more than one real price observation per day to be included in this count).[23] Moreover, over the previous 12 months there must be no 90-day period in which fewer than four real price observations are identified for the risk factor (with no more than one real price observation per day to be included in this count). The above criteria must be

monitored on a monthly basis; or

(2) The bank must identify for the risk factor at least 100 real price observations over the previous 12 months (with no more than one real price observation per day to be included in this count).

Footnotes

[2] *When a bank uses data for real price observations from an external source, and those observations are provided with a time lag (eg data provided for a particular day is only made available a number of weeks later), the period used for the RFET may differ from the period used to calibrate the current ES model. The difference in periods used for the RFET and calibration of the ES model should not be greater than one month, ie the banks could use, for each risk factor, a one-year time period finishing up to one month before the RFET assessment instead of the period used to calibrate the current ES model.*

[3] *In particular, a bank may add modellable risk factors, and replace non-modellable risk factors by a basis between these additional modellable risk factors and these non-modellable risk factors. This basis will then be considered a non-modellable risk factor. A combination between modellable and non-modellable risk factors will be a non-modellable risk factor.*

FAQ

FAQ1 *When a bank uses external data to determine whether a risk factor passes the RFET, the period of observations used for the RFET may differ from the period of observations used to calibrate the bank's expected shortfall model. According to footnote 2 in MAR31.13, the difference in periods used for the RFET and calibration of the ES model should not be greater than one month. Does the requirement set out in footnote 2 of MAR31.13 apply when a bank uses internal data to determine whether a risk factor passes the RFET?*

Yes. Regardless of whether data is from internal or external sources, when a bank uses data for real price observations, the difference in periods used for the RFET and calibration of the ES model must not exceed one month.

FAQ2 *Regarding the reform of benchmark reference rates, what guidance can the Committee provide on the count of real price observations for the risk factor eligibility test (RFET)?*

Risk factors must have sufficient market liquidity, evidenced by records of trades, to be eligible for modelling. The replacement of risk factors due to benchmark rate reform could raise particular challenges for the count of real price observations for the risk factor eligibility test (RFET). Hence, when conducting the RFET for a new benchmark rate, banks can count both: (i) real price observations of the old benchmark rate (that has been replaced by the new benchmark rate) from before the discontinuation of the old benchmark rate; and (ii) real price observations of the new benchmark rate, until one year after the discontinuation of the old benchmark rate (eg in the UK, LIBOR discontinuation is expected to be 31 December 2021). In this context, discontinuation includes cessation of the old benchmark rate or an event whereby the old benchmark rate is deemed by its regulator to no longer be representative of the underlying market.

31.14 In order for a risk factor to pass the RFET, a bank may also count real price observations based on information collected from a third-party vendor provided all of the following criteria are met:

(1) The vendor communicates to the bank the number of corresponding real prices observed and the dates at which they have been observed.

(2) The vendor provides, individually, a minimum necessary set of identifier information to enable banks to map real prices observed to risk factors.

(3) The vendor is subject to an audit regarding the validity of its pricing information. The results and reports of this audit must be made available on request to the relevant supervisory authority and to banks as a precondition for the bank to be allowed to use real price observations collected by the third-party vendor. If the audit of a third-party vendor is not satisfactory to a supervisory authority, the supervisory authority may decide to prevent the bank from using data from this vendor.[4]

Footnotes

[4] *In this case, the bank may be permitted to use real price observations from this vendor for other risk factors.*

31.15 A real price is representative for a risk factor of a bank where the bank is able to extract the value of the risk factor from the value of the real price. The bank must have policies and procedures that describe its mapping of real price observations to risk factors. The bank must provide sufficient information to its supervisory authorities in order to determine if the methodologies the bank uses are appropriate.

Bucketing approach for the RFET

31.16 Where a risk factor is a point on a curve or a surface (and other higher dimensional objects such as cubes), in order to count real price observations for the RFET, banks may choose from the following bucketing approaches:

(1) The own bucketing approach. Under this approach, the bank must define the buckets it will use and meet the following requirements:

(a) Each bucket must include only one risk factor, and all risk factors must correspond to the risk factors that are part of the risk-theoretical profit and loss (RTPL) of the bank for the purpose of the profit and loss (P&L) attribution (PLA) test.[5]

(b) The buckets must be non-overlapping.

(2) The regulatory bucketing approach. Under this approach, the bank must use the following set of standard buckets as set out in Table 1.

(a) For interest rate, foreign exchange and commodity risk factors with one maturity dimension (excluding implied volatilities) (t, where t is measured in years), the buckets in row (A) below must be used.

(b) For interest rate, foreign exchange and commodity risk factors with several maturity dimensions (excluding implied volatilities) (t, where t is measured in years), the buckets in row (B) below must be used.

(c) Credit spread and equity risk factors with one or several maturity dimensions (excluding implied volatilities) (t, where t is measured in years), the buckets in row (C) below must be used.

(d) For any risk factors with one or several strike dimensions (delta, δ; ie the probability that an option is "in the money" at maturity), the buckets in row (D) below must be used.[6]

(e) For expiry and strike dimensions of implied volatility risk factors (excluding those of interest rate swaptions), only the buckets in rows (C) and (D) below must be used.

(f) For maturity, expiry and strike dimensions of implied volatility risk factors from interest rate swaptions, only the buckets in row (B), (C) and (D) below must be used.

Standard buckets for the regulatory bucketing approach								Table 1	
Row	Bucket								
	1	2	3	4	5	6	7	8	9
(A)	0≤t<0.75	0.75≤t<1.5	1.5≤t<4	4≤t<7	7≤t<12	12≤t<18	18≤t<25	25≤t<35	35≤t<∞
(B)	0≤t<0.75	0.75≤t<4	4≤t<10	10≤t<18	18≤t<30	30≤t<∞			
(C)	0≤t<1.5	1.5≤t<3.5	3.5≤t<7.5	7.5≤t<15	15≤t<∞				
(D)	0≤δ<0.05	0.05≤δ<0.3	0.3≤δ<0.7	0.7≤δ<0.95	0.95≤δ<1.00				

Footnotes

[5] *The requirement to use the same buckets or segmentation of risk factors for the PLA test and the RFET recognises that there is a trade-off in determining buckets for an ES model. The use of more granular buckets may facilitate a trading desk's success in meeting the requirements of the PLA test, but additional granularity may challenge a bank's ability to source a sufficient number of real observed prices per bucket to satisfy the RFET. Banks should consider this trade-off when designing their ES models.*

[6] *For options markets where alternative definitions of moneyness are standard, banks shall convert the regulatory delta buckets to the market-standard convention using their own approved pricing models.*

31.17 Banks may count all real price observations allocated to a bucket to assess whether it passes the RFET for any risk factors that belong to the bucket. A real price observation must be allocated to a bucket for which it is representative of any risk factors that belong to the bucket.

31.18 As debt instruments mature, real price observations for those products that have been identified within the prior 12 months are usually still counted in the maturity bucket to which they were initially allocated per MAR31.17. When banks no longer need to model a credit spread risk factor belonging to a given maturity bucket, banks are allowed to re-allocate the real price observations of this bucket to the adjacent (shorter) maturity bucket.[7] A real price observation may only be counted in a single maturity bucket for the purposes of the RFET.

Footnotes

[7] *For example, if a bond with an original maturity of four years, had a real price observation on its issuance date eight months ago, banks can opt to allocate the real price observation to the bucket associated with a maturity between 1.5 and 3.5 years instead of to the bucket associated with a maturity between 3.5 and 7.5 years to which it would normally be allocated.*

31.19 Where a bank uses a parametric function to represent a curve/surface and defines the function's parameters as the risk factors in its risk measurement system, the RFET must be passed at the level of the market data used to calibrate the function's parameters and not be passed directly at the level of these risk factor parameters (due to the fact that real price observations may not exist that are directly representative of these risk factors).

31.20 A bank may use systematic credit or equity risk factors within its models that are designed to capture market-wide movements for a given economy, region or sector, but not the idiosyncratic risk of a specific issuer (the idiosyncratic risk of a specific issuer would be a

non-modellable risk factor (NMRF) unless there are sufficient real price observations of that issuer). Real price observations of market indices or instruments of individual issuers may be considered representative for a systematic risk factor as long as they share the same attributes as the systematic risk factor.

31.21 In addition to the approach set out in MAR31.20, where systematic risk factors of credit or equity risk factors include a maturity dimension (eg a credit spread curve), one of the bucketing approaches set out above must be used for this maturity dimension to count "real" price observations for the RFET.

31.22 Once a risk factor has passed the RFET, the bank should choose the most appropriate data to calibrate its model. The data used for calibration of the model does not need to be the same data used to pass the RFET.

31.23 Once a risk factor has passed the RFET, the bank must demonstrate that the data used to calibrate its ES model are appropriate based on the principles contained in MAR31.25 to MAR31.26. Where a bank has not met these principles to the satisfaction of its supervisory authority for a particular risk factor, the supervisory authority may choose to deem the data unsuitable for use to calibrate the model and, in such case, the risk factor must be excluded from the ES model and subject to capital requirements as an NMRF.

31.24 There may, on very rare occasions, be a valid reason why a significant number of modellable risk factors across different banks may become non-modellable due to a widespread reduction in trading activities (for instance, during periods of significant cross-border financial market stress affecting several banks or when financial markets are subjected to a major regime shift). One possible supervisory response in this instance could be to consider as modellable a risk factor that no longer passes the RFET. However, such a response should not facilitate a decrease in capital requirements. Supervisory authorities should only pursue such a response under the most extraordinary, systemic circumstances.

Principles for the modellability of risk factors that pass the RFET

31.25 Banks use many different types of models to determine the risks resulting from trading positions. The data requirements for each model may be different. For any given model, banks may use different sources or types of data for the model's risk factors. Banks must not rely solely on the number of observations of real prices to determine whether a risk factor is modellable. The accuracy of the source of the risk factor real price observation must also be considered.

31.26 In addition to the requirements specified in MAR31.12 to MAR31.23, banks must apply the principles below to determine whether a risk factor that passed the RFET can be modelled using the ES model or should be subject to capital requirements as an NMRF. Banks are required to demonstrate to their supervisory authorities that these principles are being followed. Supervisory authorities may determine risk factors to be non-modellable in the event these principles are not applied.

(1) Principle one. The data used may include combinations of modellable risk factors. Banks often price instruments as a combination of risk factors. Generally, risk factors derived solely from a combination of modellable risk factors are modellable. For example, risk factors derived through multifactor beta models for which inputs and calibrations are based solely on modellable risk factors, can be classified as modellable and can be included within the ES model. A risk factor derived from a combination of modellable risk factors that are mapped to distinct buckets of a given curve/surface is

modellable only if this risk factor also passes the RFET.

(a) Interpolation based on combinations of modellable risk factors should be consistent with mappings used for PLA testing (to determine the RTPL) and should not be based on alternative, and potentially broader, bucketing approaches. Likewise, banks may compress risk factors into a smaller dimension of orthogonal risk factors (eg principal components) and/or derive parameters from observations of modellable risk factors, such as in models of stochastic implied volatility, without the parameters being directly observable in the market.

(b) Subject to the approval of the supervisor, banks may extrapolate up to a reasonable distance from the closest modellable risk factor. The extrapolation should not rely solely on the closest modellable risk factor but on more than one modellable risk factor. In the event that a bank uses extrapolation, the extrapolation must be considered in the determination of the RTPL.

(2) Principle two. The data used must allow the model to pick up both idiosyncratic and general market risk. General market risk is the tendency of an instrument's value to change with the change in the value of the broader market, as represented by an appropriate index or indices. Idiosyncratic risk is the risk associated with a particular issuance, including default provisions, maturity and seniority. The data must allow both components of market risk to be captured in any market risk model used to determine capital requirements. If the data used in the model do not reflect either idiosyncratic or general market risk, the bank must apply an NMRF charge for those aspects that are not adequately captured in its model.

(3) Principle three. The data used must allow the model to reflect volatility and correlation of the risk positions. Banks must ensure that they do not understate the volatility of an asset (eg by using inappropriate averaging of data or proxies). Further, banks must ensure that they accurately reflect the correlation of asset prices, rates across yield curves and/or volatilities within volatility surfaces. Different data sources can provide dramatically different volatility and correlation estimates for asset prices. The bank should choose data sources so as to ensure that (i) the data are representative of real price observations; (ii) price volatility is not understated by the choice of data; and (iii) correlations are reasonable approximations of correlations among real price observations. Furthermore, any transformations must not understate the volatility arising from risk factors and must accurately reflect the correlations arising from risk factors used in the bank's ES model.

(4) Principle four. The data used must be reflective of prices observed and/or quoted in the market. Where data used are not derived from real price observations, the bank must demonstrate that the data used are reasonably representative of real price observations. To that end, the bank must periodically reconcile price data used in a risk model with front office and back office prices. Just as the back office serves to check the validity of the front office price, risk model prices should be included in the comparison. The comparison of front or back office prices with risk prices should consist of comparisons of risk prices with real price observations, but front office and back office prices can be used where real price observations are not widely available. Banks must document their approaches to deriving risk factors from market prices.

(5) Principle five. The data used must be updated at a sufficient frequency. A market risk model may require large amounts of data, and it can be challenging to update such large data sets frequently. Banks should strive to update their model data as often as possible to account for frequent turnover of positions in the trading portfolio and changing market conditions. Banks should update data at a minimum on a monthly

basis, but preferably daily. Additionally, banks should have a workflow process for updating the sources of data. Furthermore, where the bank uses regressions to estimate risk factor parameters, these must be re-estimated on a regular basis, generally no less frequently than every two weeks. Calibration of pricing models to current market prices must also be sufficiently frequent, ideally no less frequent than the calibration of front office pricing models. Where appropriate, banks should have clear policies for backfilling and/or gap-filling missing data.

(6) Principle six. The data used to determine stressed expected shortfall (ESR,S) must be reflective of market prices observed and/or quoted in the period of stress. The data for the ESR,S model should be sourced directly from the historical period whenever possible. There are cases where the characteristics of current instruments in the market differ from those in the stress period. Nevertheless, banks must empirically justify any instances where the market prices used for the stress period are different from the market prices actually observed during that period. Further, in cases where instruments that are currently traded did not exist during a period of significant financial stress, banks must demonstrate that the prices used match changes in prices or spreads of similar instruments during the stress period.In cases where banks do not sufficiently justify the use of current market data for products whose characteristics have changed since the stress period, the bank must omit the risk factor for the stressed period and meet the requirement of MAR33.5(2)(b) that the reduced set of risk factors explain 75% of the fully specified ES model. Moreover, if name-specific risk factors are used to calculate the ES in the actual period and these names were not available in the stressed period, there is a presumption that the idiosyncratic part of these risk factors are not in the reduced set of risk factors. Exposures for risk factors that are included in the current set but not in the reduced set need to be mapped to the most suitable risk factor of the reduced set for the purposes of calculating ES measures in the stressed period.

(7) Principle seven. The use of proxies must be limited, and proxies must have sufficiently similar characteristics to the transactions they represent. Proxies must be appropriate for the region, quality and type of instrument they are intended to represent. Supervisors will assess whether methods for combining risk factors are conceptually and empirically sound.

(a) For example, the use of indices in a multifactor model must capture the correlated risk of the assets represented by the indices, and the remaining idiosyncratic risk must be demonstrably uncorrelated across different issuers. A multifactor model must have significant explanatory power for the price movements of assets and must provide an assessment of the uncertainty in the final outcome due to the use of a proxy. The coefficients (betas) of a multifactor model must be empirically based and must not be determined based on judgment. Instances where coefficients are set by judgment generally should be considered as NMRFs.

(b) If risk factors are represented by proxy data in the current period ES model, the proxy data representation of the risk factor – not the risk factor itself – must be used in the RTPL unless the bank has identified the basis between the proxy and the actual risk factor and properly capitalised the basis either by including the basis in the ES model (if the risk factor is a modellable) or capturing the basis as a NMRF. If the capital requirement for the basis is properly determined, then the bank can choose to include in the RTPL either:

(i) the proxy risk factor and the basis; or

(ii) the actual risk factor itself.

MAR32

Internal models approach: backtesting and P&L attribution test requirements

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

Version effective as of 01 Jan 2023

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

Introduction

32.1 As set out in MAR30.4, a bank that intends to use the internal models approach (IMA) to determine market risk capital requirements for a trading desk must conduct and successfully pass backtesting at the bank-wide level and both the backtesting and profit and loss (P&L) attribution (PLA) test at the trading desk level as identified in MAR30.4(2).

32.2 For a bank to remain eligible to use the IMA to determine market risk capital requirements, a minimum of 10% of the bank's aggregated market risk capital requirement must be based on positions held in trading desks that qualify for use of the bank's internal models for market risk capital requirements by satisfying the backtesting and PLA test as set out in this chapter. This 10% criterion must be assessed by the bank on a quarterly basis when calculating the aggregate capital requirement for market risk according to MAR33.43.

32.3 The implementation of the backtesting programme and the PLA test must begin on the date that the internal models capital requirement becomes effective.

(1) For supervisory approval of a model, the bank must provide a one-year backtesting and PLA test report to confirm the quality of the model.

(2) The bank's supervisory authority may require backtesting and PLA test results prior to that date.

(3) The bank's supervisory authority will determine any necessary supervisory response to backtesting results based on the number of exceptions over the course of 12 months (ie 250 trading days) generated by the bank's model.

(a) Based on the assessment on the significance of exceptions, the supervisory authority may initiate a dialogue with the bank to determine if there is a problem with a bank's model.

(b) In the most serious cases, the supervisory authority will impose an additional increase in a bank's capital requirement or disallow use of the model.

Backtesting requirements

32.4 Backtesting requirements compare the value-at-risk (VaR) measure calibrated to a one-day holding period against each of the actual P&L (APL) and hypothetical P&L (HPL) over the prior 12 months. Specific requirements to be applied at the bank-wide level and trading desk level are set out below.

32.5 Backtesting of the bank-wide risk model must be based on a VaR measure calibrated at a 99th percentile confidence level.

(1) An exception or an outlier occurs when either the actual loss or the hypothetical loss of the bank-wide trading book registered in a day of the backtesting period exceeds the corresponding daily VaR measure given by the model. As per MAR99.8, exceptions for actual losses are counted separately from exceptions for hypothetical losses; the overall number of exceptions is the greater of these two amounts.

(2) In the event either the P&L or the daily VaR measure is not available or impossible to compute, it will count as an outlier.

32.6 In the event an outlier can be shown by the bank to relate to a non-modellable risk factor, and the capital requirement for that non-modellable risk factor exceeds the actual or hypothetical loss for that day, it may be disregarded for the purpose of the overall backtesting process if the supervisory authority is notified accordingly and does not object to this treatment. In these cases, a bank must document the history of the movement of the

value of the relevant non-modellable risk factor and have supporting evidence that the non-modellable risk factor has caused the relevant loss.

FAQ

FAQ1 *Please confirm if this treatment applies to desk-level backtesting exceptions as well. Also, please confirm if the stressed capital add-on (SES) should be compared with the full loss amount or just the excess amount, ie the difference between APL/HPL and VaR.*

If the backtesting exception at a desk-level test is being driven by a non-modellable risk factor that receives an SES capital requirement that is in excess of the maximum of the APL loss or HPL loss for that day, it is permitted to be disregarded for the purposes of the desk-level backtesting. The bank must be able to calculate a non-modellable risk factor capital requirement for the specific desk and not only for the respective risk factor across all desks.

For example, if the P&L for a desk is EUR –1.5 million and VaR is EUR 1 million, a non-modellable risk factor capital requirement (at desk level) of EUR 0.8 million would not be sufficient to disregard an exception for the purpose of desk-level backtesting. The non-modellable risk factor capital requirement attributed to the standalone desk level (without VaR) must be greater than the loss of EUR 1.5 million in order to disregard an exception for the purpose of desk-level backtesting.

32.7 The scope of the portfolio subject to bank-wide backtesting should be updated quarterly based on the results of the latest trading desk-level backtesting, risk factor eligibility test and PLA tests.

32.8 The framework for the supervisory interpretation of backtesting results for the bank-wide capital model encompasses a range of possible responses, depending on the strength of the signal generated from the backtesting. These responses are classified into three backtesting zones, distinguished by colours into a hierarchy of responses.

(1) Green zone. This corresponds to results that do not themselves suggest a problem with the quality or accuracy of a bank's model.

(2) Amber zone. This encompasses results that do raise questions in this regard, for which such a conclusion is not definitive.

(3) Red zone. This indicates a result that almost certainly indicates a problem with a bank's risk model.

32.9 These zones are defined according to the number of exceptions generated in the backtesting programme considering statistical errors as explained in MAR99.9 to MAR99.21. Table 1 sets out boundaries for these zones and the presumptive supervisory response for each backtesting outcome, based on a sample of 250 observations.

Backtesting zones	Table 1	
Backtesting zone	Number of exceptions	Backtesting dependent multiplier (to be added to any qualitative add-on per MAR33.44)
Green	0 1 2 3 4	1.50 1.50 1.50 1.50 1.50
Amber	5 6 7 8 9	1.70 1.76 1.83 1.88 1.92
Red	10 or more	2.00

32.10 The backtesting green zone generally would not initiate a supervisory increase in capital requirements for backtesting (ie no backtesting add-on would apply).

32.11 Outcomes in the backtesting amber zone could result from either accurate or inaccurate models. However, they are generally deemed more likely for inaccurate models than for accurate models. Within the backtesting amber zone, the supervisory authority will impose a higher capital requirement in the form of a backtesting add-on. The number of exceptions should generally inform the size of any backtesting add-on, as set out in Table 1 of MAR32.9.

32.12 A bank must also document all of the exceptions generated from its ongoing backtesting programme, including an explanation for each exception.

32.13 A bank may also implement backtesting for confidence intervals other than the 99th percentile, or may perform other statistical tests not set out in this standard.

32.14 Besides a higher capital requirement for any outcomes that place the bank in the backtesting amber zone, in the case of severe problems with the basic integrity of the model, the supervisory authority may consider whether to disallow the bank's use of the model for market risk capital requirement purposes altogether.

32.15 If a bank's model falls into the backtesting red zone, the supervisor will automatically increase the multiplication factor applicable to the bank's model or may disallow use of the model.

Backtesting at the trading desk level

32.16 The performance of a trading desk's risk management model will be tested through daily backtesting.

32.17 The backtesting assessment is considered to be complementary to the PLA assessment when determining the eligibility of a trading desk for the IMA.

32.18 At the trading desk level, backtesting must compare each desk's one-day VaR measure (calibrated to the most recent 12 months' data, equally weighted) at both the 97.5th percentile and the 99th percentile, using at least one year of current observations of the desk's one-day P&L.

(1) An exception or an outlier occurs when either the actual or hypothetical loss of the trading desk registered in a day of the backtesting period exceeds the corresponding daily VaR measure determined by the bank's model. Exceptions for actual losses are counted separately from exceptions for hypothetical losses; the overall number of exceptions is the greater of these two amounts.

(2) In the event either the P&L or the risk measure is not available or impossible to compute, it will count as an outlier.

FAQ

FAQ1 *Are banks permitted to use volatility scaling of returns for the VaR calculation?*

Volatility scaling of returns for VaR calculation at the discretion of the bank that results in a shorter observation period being used is not allowed. A bank may scale up the volatility of all observations for a selected (group of) risk factor(s) to reflect a recent stress period. The bank may use this scaled data to calculate future VaR and expected shortfall estimates only after ex ante notification of such a scaling to the supervisor.

32.19 If any given trading desk experiences either more than 12 exceptions at the 99th percentile or 30 exceptions at the 97.5th percentile in the most recent 12-month period, the capital requirement for all of the positions in the trading desk must be determined using the standardised approach.[1]

Footnotes

[1] *Desks with exposure to issuer default risk must pass a two-stage approval process. First, the market risk model must pass backtesting and PLA. Conditional on approval of the market risk model, the desk may then apply for approval to model default risk. Desks that fail either test must be capitalised under the standardised approach.*

PLA test requirements

32.20 The PLA test compares daily risk-theoretical P&L (RTPL) with the daily HPL for each trading desk. It intends to:

(1) measure the materiality of simplifications in a banks' internal models used for determining market risk capital requirements driven by missing risk factors and differences in the way positions are valued compared with their front office systems; and

(2) prevent banks from using their internal models for the purposes of capital requirements when such simplifications are considered material.

32.21 The PLA test must be performed on a standalone basis for each trading desk in scope for use of the IMA.

Definition of profits and losses used for the PLA test and backtesting

32.22 The RTPL is the daily trading desk-level P&L that is produced by the valuation engine of the trading desk's risk management model.

(1) The trading desk's risk management model must include all risk factors that are included in the bank's expected shortfall (ES) model with supervisory parameters and any risk factors deemed not modellable by the supervisory authority, and which are therefore not included in the ES model for calculating the respective regulatory capital requirement, but are included in non-modellable risk factors.

(2) The RTPL must not take into account any risk factors that the bank does not include in its trading desk's risk management model.

32.23 Movements in all risk factors contained in the trading desk's risk management model should be included, even if the forecasting component of the internal model uses data that incorporates additional residual risk. For example, a bank using a multifactor beta-based index model to capture event risk might include alternative data in the calibration of the residual component to reflect potential events not observed in the name-specific historical time series. The fact that the name is a risk factor in the model, albeit modelled in a multifactor model environment, means that, for the purposes of the PLA test, the bank would include the actual return of the name in the RTPL (and in the HPL) and receive recognition for the risk factor coverage of the model.

32.24 The PLA test compares a trading desk's RTPL with its HPL. The HPL used for the PLA test should be identical to the HPL used for backtesting purposes. This comparison is performed to determine whether the risk factors included and the valuation engines used in the trading desk's risk management model capture the material drivers of the bank's P&L by determining if there is a significant degree of association between the two P&L

measures observed over a suitable time period. The RTPL can differ from the HPL for a number of reasons. However, a trading desk risk management model should provide a reasonably accurate assessment of the risks of a trading desk to be deemed eligible for the internal models-based approach.

32.25 The HPL must be calculated by revaluing the positions held at the end of the previous day using the market data of the present day (ie using static positions). As HPL measures changes in portfolio value that would occur when end-of-day positions remain unchanged, it must not take into account intraday trading nor new or modified deals, in contrast to the APL. Both APL and HPL include foreign denominated positions and commodities included in the banking book.

32.26 Fees and commissions must be excluded from both APL and HPL as well as valuation adjustments for which separate regulatory capital approaches have been otherwise specified as part of the rules (eg credit valuation adjustment and its associated eligible hedges) and valuation adjustments that are deducted from Common Equity Tier 1 (eg the impact on the debt valuation adjustment component of the fair value of financial instruments must be excluded from these P&Ls).

32.27 Any other market risk-related valuation adjustments, irrespective of the frequency by which they are updated, must be included in the APL while only valuation adjustments updated daily must be included in the HPL, unless the bank has received specific agreement to exclude them from its supervisory authority. Smoothing of valuation adjustments that are not calculated daily is not allowed. P&L due to the passage of time should be included in the APL and should be treated consistently in both HPL and RTPL.[2]

Footnotes

[2] *Time effects can include various elements such as: the sensitivity to time, or theta effect (ie using mathematical terminology, the first-order derivative of the price relative to the time) and carry or costs of funding.*

32.28 Valuation adjustments that the bank is unable to calculate at the trading desk level (eg because they are assessed in terms of the bank's overall positions/risks or because of other constraints around the assessment process) are not required to be included in the HPL and APL for backtesting at the trading desk level, but should be included for bank-wide backtesting. To the satisfaction of its supervisory authority, the bank must provide support for valuation adjustments that are not computed at a trading desk level.

32.29 Both APL and HPL must be computed based on the same pricing models (eg same pricing functions, pricing configurations, model parametrisation, market data and systems) as the ones used to produce the reported daily P&L.

PLA test data input alignment

32.30 For the sole purpose of the PLA assessment, banks are allowed to align RTPL input data for its risk factors with the data used in HPL if these alignments are documented, justified to the supervisory authority and the requirements set out below are fulfilled:

(1) Banks must demonstrate that HPL input data can be appropriately used for RTPL purposes, and that no risk factor differences or valuation engine differences are omitted when transforming HPL input data into a format which can be applied to the risk factors used in RTPL calculation.

(2) Any adjustment of RTPL input data must be properly documented, validated and justified to the supervisory authority.

(3) Banks must have procedures in place to identify changes with regard to the adjustments of RTPL input data. Banks must notify the supervisory authority of any such changes.

(4) Banks must provide assessments on the effect these input data alignments would have on the RTPL and the PLA test. To do so, banks must compare RTPL based on HPL-aligned market data with the RTPL based on market data without alignment. This comparison must be performed when designing or changing the input data alignment process and upon the request of the bank's supervisory authority.

32.31 Adjustments to RTPL input data will be allowed when the input data for a given risk factor that is included in both the RTPL and the HPL differs due to different providers of market data sources or time fixing of market data sources, or transformations of market data into input data suitable for the risk factors of the underlying pricing models. These adjustments can be done either:

(1) by direct replacement of the RTPL input data (eg par rate tenor x, provider a) with the HPL input data (eg par rate tenor x, provider b); or

(2) by using the HPL input data (eg par rate tenor x, provider b) as a basis to calculate the risk factor data needed in the RTPL/ES model (eg zero rate tenor x).

FAQ

FAQ1 *In the event trading desks of a bank operate in different time zones compared to the location of the bank's risk control department, data for risk modelling could be retrieved at different snapshot times compared to the data on which the desks' front office P&L is based. Are banks permitted to align RTPL and HPL in terms of data snapshot times for these desks?*

Banks are permitted to align the snapshot time used for the calculation of the RTPL of a desk to the snapshot time used for the derivation of its HPL.

32.32 If the HPL uses market data in a different manner to RTPL to calculate risk parameters that are essential to the valuation engine, these differences must be reflected in the PLA test and as a result in the calculation of HPL and RTPL. In this regard, HPL and RTPL are allowed to use the same market data only as a basis, but must use their respective methods (which can differ) to calculate the respective valuation engine parameters. This would be the case, for example, where market data are transformed as part of the valuation process used to calculate RTPL. In that instance, banks may align market data between RTPL and HPL pre-transformation but not post-transformation.

32.33 Banks are not permitted to align HPL input data for risk factors with input data used in RTPL. Adjustments to RTPL or HPL to address residual operational noise are not permitted. Residual operational noise arises from computing HPL and RTPL in two different systems at two different points in time. It may originate from transitioning large portions of data across systems, and potential data aggregations may result in minor reconciliation gaps below tolerance levels for intervention; or from small differences in static/reference data and configuration.

PLA test metrics

32.34 The PLA requirements are based on two test metrics:

(1) the Spearman correlation metric to assess the correlation between RTPL and HPL; and

(2) the Kolmogorov-Smirnov (KS) test metric to assess similarity of the distributions of RTPL and HPL.

32.35 To calculate each test metric for a trading desk, the bank must use the time series of the most recent 250 trading days of observations of RTPL and HPL.

Process for determining the Spearman correlation metric

32.36 For a time series of HPL, banks must produce a corresponding time series of ranks based on the size of the P&L . That is, the lowest value in the HPL time series receives a rank of 1, the next lowest value receives a rank of 2 and so on.
(R_{HPL})

32.37 Similarly, for a time series of RTPL, banks must produce a corresponding time series of ranks based on size .
(R_{RTPL})

32.38 Banks must calculate the Spearman correlation coefficient of the two time series of rank values of based on size using the following formula, where and are the standard deviations of .
$R_{RTPL}\ and\ R_{HPL}$
$\sigma_{R_{HPL}}$
$\sigma_{R_{RTPL}}$
$R_{RTPL}\ and\ R_{HPL}$

$$r_S = \frac{cov\left(R_{HPL}, R_{RTPL}\right)}{\sigma_{R_{HPL}} \times \sigma_{R_{RTPL}}}$$

Process for determining Kolmogorov-Smirnov test metrics

32.39 The bank must calculate the empirical cumulative distribution function of RTPL. For any value of RTPL, the empirical cumulative distribution is the product of 0.004 and the number of RTPL observations that are less than or equal to the specified RTPL.

32.40 The bank must calculate the empirical cumulative distribution function of HPL. For any value of HPL, the empirical cumulative distribution is the product of 0.004 and number of HPL observations that are less than or equal to the specified HPL.

32.41 The KS test metric is the largest absolute difference observed between these two empirical cumulative distribution functions at any P&L value.

PLA test metrics evaluation

32.42 Based on the outcome of the metrics, a trading desk is allocated to a PLA test red zone, an amber zone or a green zone as set out in Table 2.

(1) A trading desk is in the PLA test green zone if both

(a) the correlation metric is above 0.80; and

(b) the KS distributional test metric is below 0.09 (p-value = 0.264).

(2) A trading desk is in the PLA test red zone if the correlation metric is less than 0.7 or if the KS distributional test metric is above 0.12 (p-value = 0.055).

(3) A trading desk is in the PLA amber zone if it is allocated neither to the green zone nor to the red zone.

PLA test thresholds	Table 2

Zone	Spearman correlation	KS test
Amber zone thresholds	0.80	0.09 (p-value = 0.264)
Red zone thresholds	0.70	0.12 (p-value = 0.055)

32.43 If a trading desk is in the PLA test red zone, it is ineligible to use the IMA to determine market risk capital requirements and must be use the standardised approach.

(1) Risk exposures held by these ineligible trading desks must be included with the out-of-scope trading desks for purposes of determining capital requirement per the standardised approach.

(2) A trading desk deemed ineligible to use the IMA must remain out-of-scope to use the IMA until:

(a) the trading desk produces outcomes in the PLA test green zone; and

(b) the trading desk has satisfied the backtesting exceptions requirements over the past 12 months.

32.44 If a trading desk is in the PLA test amber zone, it is not considered an out-of-scope trading desk for use of the IMA.

(1) If a trading desk is in the PLA test amber zone, it cannot return to the PLA test green zone until:

(a) the trading desk produces outcomes in the PLA test green zone; and

(b) the trading desk has satisfied its backtesting exceptions requirements over the prior 12 months.

(2) Trading desks in the PLA test amber zone are subject to a capital surcharge as specified in MAR33.43.

Treatment for exceptional situations

32.45 There may, on very rare occasions, be a valid reason why a series of accurate trading desk level-models across different banks will produce many backtesting exceptions or inadequately track the P&L produced by the front office pricing model (for instance, during periods of significant cross-border financial market stress affecting several banks or when financial markets are subjected to a major regime shift). One possible supervisory response in this instance would be to permit the relevant trading desks to continue to use the IMA but require each trading desk's model to take account of the regime shift or significant market stress as quickly as practicable while maintaining the integrity of its procedures for updating the model. Supervisory authorities should only pursue such a response under the most extraordinary, systemic circumstances.

MAR33

Internal models approach: capital requirements calculation

Updated to include the following FAQ: MAR33.5 FAQ4.

Version effective as of 01 Jan 2023

Updated to include the following FAQ: MAR33.5 FAQ4.

Calculation of expected shortfall

33.1 Banks will have flexibility in devising the precise nature of their expected shortfall (ES) models, but the following minimum standards will apply for the purpose of calculating market risk capital requirements. Individual banks or their supervisory authorities will have discretion to apply stricter standards.
FAQ

FAQ1 *Does the internal models approach (IMA) require all products to be simulated on full revaluation? Can a parametric approach be used on simple products, such as a forward rate agreement?*

The IMA does not require all products to be simulated on full revaluation. Simplifications (eg sensitivities-based valuation) may be used provided the bank's supervisor agrees that the method used is adequate for the instruments covered.

33.2 ES must be computed on a daily basis for the bank-wide internal models to determine market risk capital requirements. ES must also be computed on a daily basis for each trading desk that uses the internal models approach (IMA).

33.3 In calculating ES, a bank must use a 97.5th percentile, one-tailed confidence level.

33.4 In calculating ES, the liquidity horizons described in MAR33.12 must be reflected by scaling an ES calculated on a base horizon. The ES for a liquidity horizon must be calculated from an ES at a base liquidity horizon of 10 days with scaling applied to this base horizon result as expressed below, where:

(1) ES is the regulatory liquidity-adjusted ES;

(2) T is the length of the base horizon, ie 10 days;

(3) EST(P) is the ES at horizon T of a portfolio with positions P = (pi) with respect to shocks to all risk factors that the positions P are exposed to;

(4) EST(P, j) is the ES at horizon T of a portfolio with positions P = (pi) with respect to shocks for each position pi in the subset of risk factors Q(pi , j), with all other risk factors held constant;

(5) the ES at horizon T, EST(P) must be calculated for changes in the risk factors, and EST(P, j) must be calculated for changes in the relevant subset Q(pi , j) of risk factors, over the time interval T without scaling from a shorter horizon;

(6) Q(pi , j) is the subset of risk factors for which liquidity horizons, as specified in MAR33.12, for the desk where pi is booked are at least as long as LHj according to the table below. For example, Q(pi,4) is the set of risk factors with a 60-day horizon and a 120-day liquidity horizon. Note that Q(pi , j) is a subset of Q(pi , j-1);

(7) the time series of changes in risk factors over the base time interval T may be determined by overlapping observations; and

(8) LHj is the liquidity horizon j, with lengths in the following table:

Liquidity horizons, j — Table 1

j	LH_j
1	10
2	20
3	40

4	60
5	120

$$ES = \sqrt{\left(ES_T\left(P\right)\right)^2 + \sum_{j\geq 2}\left(ES_T\left(P,j\right)\sqrt{\frac{\left(LH_j - LH_{j-1}\right)}{T}}\right)^2}$$

33.5 The ES measure must be calibrated to a period of stress.

(1) Specifically, the ES measure must replicate an ES outcome that would be generated on the bank's current portfolio if the relevant risk factors were experiencing a period of stress. This is a joint assessment across all relevant risk factors, which will capture stressed correlation measures.

(2) This calibration is to be based on an indirect approach using a reduced set of risk factors. Banks must specify a reduced set of risk factors that are relevant for their portfolio and for which there is a sufficiently long history of observations.

(a) This reduced set of risk factors is subject to supervisory approval and must meet the data quality requirements for a modellable risk factor as outlined in MAR31.12 to MAR31.24.

(b) The identified reduced set of risk factors must be able to explain a minimum of 75% of the variation of the full ES model (ie the ES of the reduced set of risk factors should be at least equal to 75% of the fully specified ES model on average measured over the preceding 12-week period).

FAQ

FAQ1 *What indicator must be maximised for the identification of the stressed period?*

The aggregate capital requirement for modellable risk factors (IMCC) as per MAR33.15 has to be maximised for the modellable risk factors.

FAQ2 *Is it correct that the reduced set of risk factors must explain a minimum of 75% of the variation of the full ES at the group level (ie top level) only and not at the desk level in order to be consistent with the stressed period selection performed at the group level?*

Yes, the reduced set of risk factors must be able to explain a minimum of 75% of the variation of the full ES model at the group level for the aggregate of all desks with IMA model approval.

FAQ3 *How should banks determine whether the ES measure calculated using a reduced set of risk factors explains at least 75% of the variation of the full ES model?*

The average of the measurements of the ratio (ES using reduced set of risk factors and current period (ESR,C) to ES using full set of risk factors and current period (ESF,C)) over the preceding 12-week period must be at least 75%.

FAQ4 *Regarding the reform of benchmark reference rates, what guidance can the Committee provide on the calculation of expected shortfall (ES) if the new benchmark rate was not available during a stress period for the purposes of MAR33?*

If the new benchmark rate is currently eligible for modelling according to MAR31 but was not available during the stress period, it may pose a challenge to banks calculating the expected shortfall (ES) for the current and stress period per MAR33. To address this, if the new benchmark rate is eligible for modelling according to MAR31

but was not available during the stress period, banks may use:

(i) for the current period, the new benchmark rate in the full set of risk factors (ESF,C) and in the reduced set of risk factors (ESR,C); and

(ii) for the stress period, the old benchmark rate in the reduced set of risk factors (ESR,S).

This interpretation does not annul the specification in MAR33.5(2) that the reduced set is subject to supervisory approval and must meet the data quality requirements.

33.6 The ES for market risk capital purposes is therefore expressed as follows, where:

(1) The ES for the portfolio using the above reduced set of risk factors (ESR,S), is calculated based on the most severe 12-month period of stress available over the observation horizon.

(2) ESR,S is then scaled up by the ratio of (i) the current ES using the full set of risk factors to (ii) the current ES measure using the reduced set of factors. For the purpose of this calculation, this ratio is floored at 1.

(a) ESF,C is the ES measure based on the current (most recent) 12-month observation period with the full set of risk factors; and

(b) ESR,C is the ES measure based on the current period with a reduced set of risk factors.

$$ES = ES_{R,S} \times \frac{ES_{F,C}}{ES_{R,C}}$$

33.7 For measures based on stressed observations (ES), banks must identify the 12-month period of stress over the observation horizon in which the portfolio experiences the largest loss. The observation horizon for determining the most stressful 12 months must, at a minimum, span back to and include 2007. Observations within this period must be equally weighted. Banks must update their 12-month stressed periods at least quarterly, or whenever there are material changes in the risk factors in the portfolio. Whenever a bank updates its 12-month stressed periods it must also update the reduced set of risk factors (as the basis for the calculations of E and E) accordingly.

33.8 For measures based on current observations (ES), banks must update their data sets no less frequently than once every three months and must also reassess data sets whenever market prices are subject to material changes.

(1) This updating process must be flexible enough to allow for more frequent updates.

(2) The supervisory authority may also require a bank to calculate its ES using a shorter observation period if, in the supervisor's judgement; this is justified by a significant upsurge in price volatility. In this case, however, the period should be no shorter than six months.

33.9 No particular type of ES model is prescribed. Provided that each model used captures all the material risks run by the bank, as confirmed through profit and loss (P&L) attribution (PLA) tests and backtesting, and conforms to each of the requirements set out above and below, supervisors may permit banks to use models based on either historical simulation, Monte Carlo simulation, or other appropriate analytical methods.

33.10 Banks will have discretion to recognise empirical correlations within broad regulatory risk factor classes (interest rate risk, equity risk, foreign exchange risk, commodity risk and

credit risk, including related options volatilities in each risk factor category). Empirical correlations across broad risk factor categories will be constrained by the supervisory aggregation scheme, as described in MAR33.14 to MAR33.15, and must be calculated and used in a manner consistent with the applicable liquidity horizons, clearly documented and able to be explained to supervisors on request.

33.11 Banks' models must accurately capture the risks associated with options within each of the broad risk categories. The following criteria apply to the measurement of options risk:

(1) Banks' models must capture the non-linear price characteristics of options positions.

(2) Banks' risk measurement systems must have a set of risk factors that captures the volatilities of the rates and prices underlying option positions, ie vega risk. Banks with relatively large and/or complex options portfolios must have detailed specifications of the relevant volatilities. Banks must model the volatility surface across both strike price and vertex (ie tenor).

33.12 As set out in MAR33.4, a scaled ES must be calculated based on the liquidity horizon n defined below. n is calculated per the following conditions:

(1) Banks must map each risk factor on to one of the risk factor categories shown below using consistent and clearly documented procedures.

(2) The mapping of risk factors must be:

(a) set out in writing;

(b) validated by the bank's risk management;

(c) made available to supervisors; and

(d) subject to internal audit.

(3) n is determined for each broad category of risk factor as set out in Table 2. However, on a desk-by-desk basis, n can be increased relative to the values in the table below (ie the liquidity horizon specified below can be treated as a floor). Where n is increased, the increased horizon must be 20, 40, 60 or 120 days and the rationale must be documented and be subject to supervisory approval. Furthermore, liquidity horizons should be capped at the maturity of the related instrument.

Liquidity horizon n by risk factor			Table 2
Risk factor category	*n*	Risk factor category	*n*
Interest rate: specified currencies - EUR, USD, GBP, AUD, JPY, SEK, CAD and domestic currency of a bank	10	Equity price (small cap): volatility	60
Interest rate: unspecified currencies	20	Equity: other types	60
Interest rate: volatility	60	Foreign exchange (FX) rate: specified currency pairs[1]	10
Interest rate: other types	60	FX rate: currency pairs	20
Credit spread: sovereign (investment grade, or IG)	20	FX: volatility	40
Credit spread: sovereign (high yield, or HY)	40	FX: other types	40
Credit spread: corporate (IG)	40	Energy and carbon emissions trading price	20
Credit spread: corporate (HY)	60	Precious metals and non-ferrous metals price	20

Credit spread: volatility	120		Other commodities price	60
Credit spread: other types	120		Energy and carbon emissions trading price: volatility	60
			Precious metals and non-ferrous metals price: volatility	60
Equity price (large cap)	10		Other commodities price: volatility	120
Equity price (small cap)	20		Commodity: other types	120
Equity price (large cap): volatility	20			

Footnotes

[1] *USD/EUR, USD/JPY, USD/GBP, USD/AUD, USD/CAD, USD/CHF, USD/MXN, USD/CNY, USD/NZD, USD/RUB, USD/HKD, USD/SGD, USD/TRY, USD/KRW, USD/SEK, USD/ZAR, USD/INR, USD/NOK, USD/BRL, EUR/JPY, EUR/GBP, EUR/CHF and JPY/AUD. Currency pairs forming first-order crosses across these specified currency pairs are also subject to the same liquidity horizon.*

FAQ

FAQ1 *Please clarify the liquidity horizon to be used for equity dividends and equity repo risk factors.*

The liquidity horizon for equity large cap repo and dividend risk factors is 20 days. All other equity repo and dividend risk factors are subject to a liquidity horizon of 60 days.

FAQ2 *For mono-currency and cross-currency basis risk, should liquidity horizons of 10 days and 20 days for interest rate-specified currencies and unspecified currencies, respectively, be applied?*

Yes.

FAQ3 *To which liquidity horizon should inflation risk factors be assigned? Should the liquidity horizon for inflation risk factors be treated consistently with interest rates?*

The liquidity horizon for inflation risk factors should be consistent with the liquidity horizons for interest rate risk factors for a given currency.

FAQ4 *How must a bank treat risk factors in instruments that mature before the liquidity horizon of the respective risk factor prescribed in MAR33.12?*

If the maturity of the instrument is shorter than the respective liquidity horizon of the risk factor as prescribed in MAR33.12, the next longer liquidity horizon length (out of the lengths of 10, 20, 40, 60 or 120 days as set out in the paragraph) compared with the maturity of the instrument itself must be used. For example, although the liquidity horizon for interest rate volatility is prescribed as 60 days, if an instrument matures in 30 days, a 40-day liquidity horizon would apply for the instrument's interest rate volatility.

FAQ5 *Which liquidity horizon should be mapped to multi-sector credit and equity indices (ie where different risk factor categories are involved)?*

To determine the liquidity horizon of multi-sector credit and equity indices, the respective liquidity horizons of the underlying instruments must be used. A weighted average of liquidity horizons of the instruments contained in the index must be determined by multiplying the liquidity horizon of each individual instrument by its weight in the index (ie the weight used to construct the index) and summing across all instruments. The liquidity horizon of the index is the shortest liquidity horizon (out of 10, 20, 40, 60 and

120 days) that is equal to or longer than the weighted average liquidity horizon. For example, if the weighted average liquidity horizon is 12 days, the liquidity horizon of the index would be 20 days.

Calculation of capital requirement for modellable risk factors

33.13 For those trading desks that are permitted to use the IMA, all risk factors that are deemed to be modellable must be included in the bank's internal, bank-wide ES model. The bank must calculate its internally modelled capital requirement at the bank-wide level using this model, with no supervisory constraints on cross-risk class correlations (IMCC(C)).

FAQ

FAQ1 *Are banks permitted to not capitalise certain risks or risk factors via ES or stressed expected shortfall (SES) (as appropriate) as long as those risks or risk factors are not included in the model eligibility tests?*

Banks design their own models for use under the IMA. As a result, they may exclude risk factors from IMA models as long as the bank's supervisor does not conclude that the risk factor must be capitalised by either ES or SES. Moreover, at a minimum, the risk factors defined in MAR31.1 to MAR31.11 need to be covered in the IMA. If a risk factor is capitalised by neither ES nor SES, it is to be excluded from the calculation of risk-theoretical P&L.

33.14 The bank must calculate a series of partial ES capital requirements (ie all other risk factors must be held constant) for the range of broad regulatory risk classes (interest rate risk, equity risk, foreign exchange risk, commodity risk and credit spread risk). These partial, non-diversifiable (constrained) ES values (IMCC(C)) will then be summed to provide an aggregated risk class ES capital requirement.

33.15 The aggregate capital requirement for modellable risk factors (IMCC) is based on the weighted average of the constrained and unconstrained ES capital requirements, where:

(1) The stress period used in the risk class level ESR,S,i should be the same as that used to calculate the portfolio-wide ESR,S.

(2) Rho (ρ) is the relative weight assigned to the firm's internal model. The value of ρ is 0.5.

(3) B stands for broad regulatory risk classes as set out in MAR33.14.

$$IMCC = \rho\left(IMCC\left(C\right)\right) + \left(1-\rho\right)\left(\sum_{i=1}^{B} IMCC\left(C_i\right)\right)$$

$$\text{where } IMCC\left(C\right) = ES_{R,S}\frac{ES_{F,C}}{ES_{R,C}} \text{ and } IMCC\left(C_i\right) = ES_{R,S,i}\frac{ES_{F,C,i}}{ES_{R,C,i}}$$

FAQ

FAQ1 *To calculate the aggregate capital requirement for modellable risk factors (internally modelled capital charge, IMCC) up to 63 daily ES calculations would be necessary if each ES measure were required to be calculated daily. Is it permissible to calculate some of the ES measures weekly or must all measures be calculated daily?*

The formula specified in MAR33.15, , can be rewritten as with . While ES , ES and ES must be calculated daily, it is generally acceptable that the ratio of undiversified IMCC(C) to diversified IMCC(C), , may be calculated on a weekly basis.

By defining as the formula for the calculation of IMCC can be rearranged, leading to the following expression of IMCC: . Hence, IMCC can be calculated as a multiple

of IMCC(C), where IMCC(C) is calculated daily and the multiplier is updated weekly.

Banks must have procedures and controls in place to ensure that the weekly calculation of the "undiversified IMCC(C) to diversified IMCC(C)" ratio does not lead to a systematic underestimation of risks relative to daily calculation. Banks must be in a position to switch to daily calculation upon supervisory direction.

Calculation of capital requirement for non-modellable risk factors

33.16 Capital requirements for each non-modellable risk factor (NMRF) are to be determined using a stress scenario that is calibrated to be at least as prudent as the ES calibration used for modelled risks (ie a loss calibrated to a 97.5% confidence threshold over a period of stress). In determining that period of stress, a bank must determine a common 12-month period of stress across all NMRFs in the same risk class. Subject to supervisory approval, a bank may be permitted to calculate stress scenario capital requirements at the bucket level (using the same buckets that the bank uses to disprove modellability, per MAR31.16) for risk factors that belong to curves, surfaces or cubes (ie a single stress scenario capital requirement for all the NMRFs that belong to the same bucket).

(1) For each NMRF, the liquidity horizon of the stress scenario must be the greater of the liquidity horizon assigned to the risk factor in MAR33.12 and 20 days. The bank's supervisory authority may require a higher liquidity horizon.

(2) For NMRFs arising from idiosyncratic credit spread risk, banks may apply a common 12-month stress period. Likewise, for NMRFs arising from idiosyncratic equity risk arising from spot, futures and forward prices, equity repo rates, dividends and volatilities, banks may apply a common 12-month stress scenario. Additionally, a zero correlation assumption may be used when aggregating gains and losses provided the bank conducts analysis to demonstrate to its supervisor that this is appropriate.[2] Correlation or diversification effects between other non-idiosyncratic NMRFs are recognised through the formula set out in MAR33.17.

(3) In the event that a bank cannot provide a stress scenario which is acceptable for the supervisor, the bank will have to use the maximum possible loss as the stress scenario.

Footnotes

[2] *The tests are generally done on the residuals of panel regressions where the dependent variable is the change in issuer spread while the independent variables can be either a change in a market factor or a dummy variable for sector and/or region. The assumption is that the data on the names used to estimate the model suitably proxies the names in the portfolio and the idiosyncratic residual component captures the multifactor-name basis. If the model is missing systematic explanatory factors or the data suffers from measurement error, then the residuals would exhibit heteroscedasticity (which can be tested via White, Breuche Pagan tests etc) and/or serial correlation (which can be tested with Durbin Watson, Lagrange multiplier (LM) tests etc) and/or cross-sectional correlation (clustering).*

33.17 The aggregate regulatory capital measure for I (non-modellable idiosyncratic credit spread risk factors that have been demonstrated to be appropriate to aggregate with zero correlation), J (non-modellable idiosyncratic equity risk factors that have been demonstrated to be appropriate to aggregate with zero correlation) and the remaining K (risk factors in model-eligible trading desks that are non-modellable (SES)) is calculated as follows, where:

(1) $I_{SESNM,i}$ is the stress scenario capital requirement for idiosyncratic credit spread non-modellable risk i from the I risk factors aggregated with zero correlation;

(2) ISESNM,j is the stress scenario capital requirement for idiosyncratic equity non-modellable risk j from the J risk factors aggregated with zero correlation;

(3) SESNM,k is the stress scenario capital requirement for non-modellable risk k from K risk factors; and

(4) Rho (ρ) is equal to 0.6.

$$SES = \sqrt{\sum_{i=1}^{I} ISES_{NM,i}^2} + \sqrt{\sum_{j=1}^{J} ISES_{NM,j}^2} + \sqrt{\left(\rho \times \sum_{k=1}^{K} SES_{NM,k}\right)^2 + \left(1-\rho^2\right) \times \sum_{k=1}^{K} SES_{NM,k}^2}$$

Calculation of default risk capital requirement

33.18 Banks must have a separate internal model to measure the default risk of trading book positions. The general criteria in MAR30.1 to MAR30.4 and the qualitative standards in MAR30.5 to MAR30.16 also apply to the default risk model.

33.19 Default risk is the risk of direct loss due to an obligor's default as well as the potential for indirect losses that may arise from a default event.

33.20 Default risk must be measured using a value-at-risk (VaR) model.

(1) Banks must use a default simulation model with two types of systematic risk factors.

(2) Default correlations must be based on credit spreads or on listed equity prices. Correlations must be based on data covering a period of 10 years that includes a period of stress as defined in MAR33.5 and based on a one-year liquidity horizon.

(3) Banks must have clear policies and procedures that describe the correlation calibration process, documenting in particular in which cases credit spreads or equity prices are used.

(4) Banks have the discretion to apply a minimum liquidity horizon of 60 days to the determination of default risk capital (DRC) requirement for equity sub-portfolios.

(5) The VaR calculation must be conducted weekly and be based on a one-year time horizon at a one-tail, 99.9 percentile confidence level.

FAQ

FAQ1 *MAR33.20 and MAR33.28 state that correlations must be measured over a liquidity horizon of one year in line with MAR33.23, which states that a bank must assume constant positions over the one-year capital horizon. However, according to MAR33.23, a minimum liquidity horizon of 60 days can be applied to equity sub-portfolios. Should the correlations for equity sub-portfolios be calibrated utilising a 60-day liquidity horizon for consistency?*

Banks are permitted to calibrate correlations to liquidity horizons of 60 days in the case that a separate calculation is performed for equity sub-portfolios and these desks deal predominately in equity exposures. In the case of a desk with both equity and bond exposures, for which a joint calculation for default risk of equities and bonds needs to be performed, the correlations need to be calibrated to a liquidity horizon of one year.

In this case, a bank is permitted to consistently use a 60-day probability of default (PD) for equities and a one-year PD for bonds.

FAQ2 *MAR33.20(2) states: "Default correlations must be based on credit spreads or on*

listed equity prices." Are banks permitted to also include additional data sources (eg rating time series) in addition to equity prices in order to correct for a correlation bias observed in equity data?

Only credit spreads or listed equity prices are permitted. No additional data sources (eg rating time series) are permitted.

FAQ3 *MAR33.20(1) specifies that banks must use a default simulation model with two types of systematic risk factors. To meet this condition, should the model always have two random variables that correspond to the systematic risk factors?*

Yes. Systematic risk in a DRC requirement model must be accounted for via multiple systematic factors of two different types. The random variable that determines whether an obligor defaults must be an obligor-specific function of the systematic factors of both types and of an idiosyncratic factor. For example, in a Merton-type model, obligor defaults when its asset return falls below an obligor-specific threshold that determines the obligor's probability of default. Systematic risk can be described via systematic regional factors ( ) and systematic industry factors ( ). For each obligor , region factor loadings and industry factor loadings that describe the sensitivity of the obligor's asset return to each systematic factor need to be chosen. There must be at least one non-zero factor loading for the region type and at least one non-zero factor loading for the industry type. The asset return of obligor can be represented as , where is the idiosyncratic risk factor and is the idiosyncratic factor loading.

FAQ4 *Is a 60-day liquidity horizon permitted to be used for all equity positions? Are banks permitted to use a longer liquidity horizon where appropriate, eg where equity is held to hedge hybrid positions (such as convertibles)?*

Yes, banks are permitted to use a 60-day liquidity horizon for all equity positions but are permitted to use a longer liquidity horizon where appropriate.

33.21 All positions subject to market risk capital requirements that have default risk as defined in MAR33.19, with the exception of those positions subject to the standardised approach, are subject to the DRC requirement model.

(1) Sovereign exposures (including those denominated in the sovereign's domestic currency), equity positions and defaulted debt positions must be included in the model.

(2) For equity positions, the default of an issuer must be modelled as resulting in the equity price dropping to zero.

33.22 The DRC requirement model capital requirement is the greater of:

(1) the average of the DRC requirement model measures over the previous 12 weeks; or

(2) the most recent DRC requirement model measure.

33.23 A bank must assume constant positions over the one-year horizon, or 60 days in the context of designated equity sub-portfolios.

FAQ

FAQ1 *MAR33.20 and MAR33.27 state that correlations must be measured over a liquidity horizon of one year in line with MAR33.23, which states that a bank must assume constant positions over the one-year capital horizon. However, according to MAR33.23, a minimum liquidity horizon of 60 days can be applied to equity sub-portfolios. Should the correlations for equity sub-portfolios be calibrated utilising a*

60-day liquidity horizon for consistency?

Banks are permitted to calibrate correlations to liquidity horizons of 60 days in the case that a separate calculation is performed for equity sub-portfolios and these desks deal predominately in equity exposures. In the case of a desk with both equity and bond exposures, for which a joint calculation for default risk of equities and bonds needs to be performed, the correlations need to be calibrated to a liquidity horizon of one year.

In this case, a bank is permitted to consistently use a 60-day probability of default (PD) for equities and a one-year PD for bonds.

FAQ2 *MAR33.23 states that a bank must have constant positions over the chosen liquidity horizon. However, MAR33.28 states that a bank must capture material mismatches between the position and its hedge. Please explain how these two paragraphs are to be consistently applied to securities with a maturity of less than one year.*

The concept of constant positions has changed in the market risk framework because the capital horizon is now meant to always be synonymous with the new definition of liquidity horizon and no new positions are added when positions expire during the capital horizon. For securities with a maturity under one year, a constant position can be maintained within the liquidity horizon but, much like under the Basel II.5 incremental risk charge, any maturity of a long or short position must be accounted for when the ability to maintain a constant position within the liquidity horizon cannot be contractually assured.

33.24 Default risk must be measured for each obligor.

(1) Probabilities of default (PDs) implied from market prices are not acceptable unless they are corrected to obtain an objective probability of default.[3]

(2) PDs are subject to a floor of 0.03%.

Footnotes

[3] *Market-implied PDs are not acceptable.*

33.25 A bank's model may reflect netting of long and short exposures to the same obligor. If such exposures span different instruments with exposure to the same obligor, the effect of the netting must account for different losses in the different instruments (eg differences in seniority).

33.26 The basis risk between long and short exposures of different obligors must be modelled explicitly. The potential for offsetting default risk among long and short exposures across different obligors must be included through the modelling of defaults. The pre-netting of positions before input into the model other than as described in MAR33.25 is not allowed.

33.27 The DRC requirement model must recognise the impact of correlations between defaults among obligors, including the effect on correlations of periods of stress as described below.

(1) These correlations must be based on objective data and not chosen in an opportunistic way where a higher correlation is used for portfolios with a mix of long and short positions and a low correlation used for portfolios with long only exposures.

(2) A bank must validate that its modelling approach for these correlations is appropriate for its portfolio, including the choice and weights of its systematic risk factors. A bank must document its modelling approach and the period of time used to calibrate the model.

(3) These correlations must be measured over a liquidity horizon of one year.

(4) These correlations must be calibrated over a period of at least 10 years.

(5) Banks must reflect all significant basis risks in recognising these correlations, including, for example, maturity mismatches, internal or external ratings, vintage etc.

FAQ

FAQ1 *MAR33.20 and MAR33.27 state that correlations must be measured over a liquidity horizon of one year in line with MAR33.23, which states that a bank must assume constant positions over the one-year capital horizon. However, according to MAR33.23, a minimum liquidity horizon of 60 days can be applied to equity sub-portfolios. Should the correlations for equity sub-portfolios be calibrated utilising a 60-day liquidity horizon for consistency?*

Banks are permitted to calibrate correlations to liquidity horizons of 60 days in the case that a separate calculation is performed for equity sub-portfolios and these desks deal predominately in equity exposures. In the case of a desk with both equity and bond exposures, for which a joint calculation for default risk of equities and bonds needs to be performed, the correlations need to be calibrated to a liquidity horizon of one year.

In this case, a bank is permitted to consistently use a 60-day PD for equities and a one-year PD for bonds.

FAQ2 *MAR33.23 states that a bank must have constant positions over the chosen liquidity horizon. However, MAR33.28 states that a bank must capture material mismatches between the position and its hedge. Please explain how these two paragraphs are to be consistently applied to securities with a maturity of less than one year.*

The concept of constant positions has changed in the market risk framework because the capital horizon is now meant to always be synonymous with the new definition of liquidity horizon and no new positions are added when positions expire during the capital horizon. For securities with a maturity under one year, a constant position can be maintained within the liquidity horizon but, much like under the Basel II.5 incremental risk charge, any maturity of a long or short position must be accounted for when the ability to maintain a constant position within the liquidity horizon cannot be contractually assured.

33.28 The bank's model must capture any material mismatch between a position and its hedge. With respect to default risk within the one-year capital horizon, the model must account for the risk in the timing of defaults to capture the relative risk from the maturity mismatch of long and short positions of less than one-year maturity.

33.29 The bank's model must reflect the effect of issuer and market concentrations, as well as concentrations that can arise within and across product classes during stressed conditions.

33.30 As part of this DRC requirement model, the bank must calculate, for each and every position subjected to the model, an incremental loss amount relative to the current valuation that the bank would incur in the event that the obligor of the position defaults.

33.31 Loss estimates must reflect the economic cycle; for example, the model must incorporate the dependence of the recovery on the systemic risk factors.

33.32 The bank's model must reflect the non-linear impact of options and other positions with material non-linear behaviour with respect to default. In the case of equity derivatives

positions with multiple underlyings, simplified modelling approaches (for example modelling approaches that rely solely on individual jump-to-default sensitivities to estimate losses when multiple underlyings default) may be applied (subject to supervisory approval).

FAQ

FAQ1 *MAR33.32 indicates that a bank may use a simplified modelling approach for equity derivative positions with multiple underlyings. May a similar simplified approach be used for non-correlation trading portfolio credit derivative positions with multiple underlyings?*

No. The simplified treatment applies only to equity derivatives.

33.33 Default risk must be assessed from the perspective of the incremental loss from default in excess of the mark-to-market losses already taken into account in the current valuation.

33.34 Owing to the high confidence standard and long capital horizon of the DRC requirement, robust direct validation of the DRC model through standard backtesting methods at the 99.9%/one-year soundness standard will not be possible.

(1) Accordingly, validation of a DRC model necessarily must rely more heavily on indirect methods including but not limited to stress tests, sensitivity analyses and scenario analyses, to assess its qualitative and quantitative reasonableness, particularly with regard to the model's treatment of concentrations.

(2) Given the nature of the DRC soundness standard, such tests must not be limited to the range of events experienced historically.

(3) The validation of a DRC model represents an ongoing process in which supervisors and firms jointly determine the exact set of validation procedures to be employed.

33.35 Banks should strive to develop relevant internal modelling benchmarks to assess the overall accuracy of their DRC models.

33.36 Due to the unique relationship between credit spread and default risk, banks must seek approval for each trading desk with exposure to these risks, both for credit spread risk and default risk. Trading desks which do not receive approval will be deemed ineligible for internal modelling standards and be subject to the standardised capital framework.

33.37 Where a bank has approved PD estimates as part of the internal ratings-based (IRB) approach, this data must be used. Where such estimates do not exist, or the bank's supervisor determines that they are not sufficiently robust, PDs must be computed using a methodology consistent with the IRB methodology and satisfy the following conditions.

(1) Risk-neutral PDs should not be used as estimates of observed (historical) PDs.

(2) PDs must be measured based on historical default data including both formal default events and price declines equivalent to default losses. Where possible, this data should be based on publicly traded securities over a complete economic cycle. The minimum historical observation period for calibration purposes is five years.

(3) PDs must be estimated based on historical data of default frequency over a one-year period. The PD may also be calculated on a theoretical basis (eg geometric scaling) provided that the bank is able to demonstrate that such theoretical derivations are in line with historical default experience.

(4) PDs provided by external sources may also be used by banks, provided they can be shown to be relevant for the bank's portfolio.

33.38 Where a bank has approved loss-given-default (LGD)[4] estimates as part of the IRB approach, this data must be used. Where such estimates do not exist, or the supervisor determines that they are not sufficiently robust, LGDs must be computed using a methodology consistent with the IRB methodology and satisfy the following conditions.

(1) LGDs must be determined from a market perspective, based on a position's current market value less the position's expected market value subsequent to default. The LGD should reflect the type and seniority of the position and cannot be less than zero.

(2) LGDs must be based on an amount of historical data that is sufficient to derive robust, accurate estimates.

(3) LGDs provided by external sources may also be used by institutions, provided they can be shown to be relevant for the bank's portfolio.

Footnotes

[4] *LGD should be interpreted in this context as 1 – recovery rate.*

33.39 Banks must establish a hierarchy ranking their preferred sources for PDs and LGDs, in order to avoid the cherry-picking of parameters.

Calculation of capital requirement for model-ineligible trading desks

33.40 The regulatory capital requirement associated with trading desks that are either out-of-scope for model approval or that have been deemed ineligible to use an internal model (Cu) is to be calculated by aggregating all such risks and applying the standardised approach.

Aggregation of capital requirement

33.41 The aggregate (non-DRC) capital requirement for those trading desks approved and eligible for the IMA (ie trading desks that pass the backtesting requirements and that have been assigned to the PLA test green zone or amber zone (C) in MAR32.43 to MAR32.45) is equal to the maximum of the most recent observation and a weighted average of the previous 60 days scaled by a multiplier and is calculated as follows where SES is the aggregate regulatory capital measure for the risk factors in model-eligible trading desks that are non-modellable.

$$C_A = max\left\{IMCC_{t-1} + SES_{t-1}; m_c \cdot IMCC_{avg} + SES_{avg}\right\}$$

33.42 The multiplication factor m is fixed at 1.5 unless it is set at a higher level by the supervisory authority to reflect the addition of a qualitative add on and/or a backtesting add-on per the following considerations.

(1) Banks must add to this factor a "plus" directly related to the ex-post performance of the model, thereby introducing a built-in positive incentive to maintain the predictive quality of the model.

(2) For the backtesting add-on, the plus will range from 0 to 0.5 based on the outcome of the backtesting of the bank's daily VaR at the 99th percentile based on current observations on the full set of risk factors (VaRFC).

(3) If the backtesting results are satisfactory and the bank meets all of the qualitative standards set out in MAR30.5 to MAR30.16, the plus factor could be zero. MAR32 presents in detail the approach to be applied for backtesting and the plus factor.

(4) The backtesting add-on factor is determined based on the maximum of the exceptions generated by the backtesting results against actual P&L (APL) and hypothetical P&L

(HPL) as described MAR32.

33.43 The aggregate capital requirement for market risk (ACR) is equal to the aggregate capital requirement for approved and eligible trading desks (IMA =C+ DRC) plus the standardised approach capital requirement for trading desks that are either out-of-scope for model approval or that have been deemed ineligible to use the internal models approach (C). If at least one eligible trading desk is in the PLA test amber zone, a capital surcharge is added. The impact of the capital surcharge is limited by the formula:

$$ACR_{total} = \min\left\{IMA_{G,A} + Capital\,surcharge + C_U\ ; SA_{all\ desk}\right\} + \max\left\{0; IMA_{G,A} - SA_{G,A}\right\}$$

33.44 For the purposes of calculating the capital requirement, the risk factor eligibility test, the PLA test and the trading desk-level backtesting are applied on a quarterly basis to update the modellability of risk factors and desk classification to the PLA test green zone, amber zone, or red zone. In addition, the stressed period and the reduced set of risk factors (E and E) must be updated on a quarterly basis. The reference dates to perform the tests and to update the stress period and selection of the reduced set of risk factors should be consistent. Banks must reflect updates to the stressed period and to the reduced set of risk factors as well as the test results in calculating capital requirements in a timely manner. The averages of the previous 60 days (IMCC, SES) and or respectively 12 weeks (DRC) have only to be calculated at the end of the quarter for the purpose of calculating the capital requirement.

33.45 The capital surcharge is calculated as the difference between the aggregated standardised capital charges (SA) and the aggregated internal models-based capital charges ( ) multiplied by a factor k. To determine the aggregated capital charges, positions in all of the trading desks in the PLA green zone or amber zone are taken into account. The capital surcharge is floored at zero. In the formula below:

$\mathrm{IMA}_{\mathrm{G,A}} = \mathrm{C}_{\mathrm{A}} + \mathrm{DRC}$

(1)

$$k = 0.5 \times \frac{\sum_{i \in A} SA_i}{\sum_{i \in G,A} SA_i}$$

;

(2)

SA_i

denotes the standardised capital requirement for all the positions of trading desk “i”;

(3)

$i \in A$

denotes the indices of all the approved trading desks in the amber zone; and

(4)

$i \in G, A$

denotes the indices of all the approved trading desks in the green zone or amber zone.

$$Capital\,surcharge = k \cdot \max\left\{0, SA_{G,A} - IMA_{G,A}\right\}$$

33.46 The risk-weighted assets for market risk under the IMA are determined by multiplying the capital requirements calculated as set out in this chapter by 12.5.

MAR40

Simplified standardised approach

Updated to take account of the revised implementation date announced on 27 March 2020. Cross references to the securitisation chapters updated to include a reference to the chapter on NPL securitisations (CRE45) published on 26 November 2020. Updated to incorporate the FAQ published on 5 July 2024.

Version effective as of 01 Jan 2023

Updated to take account of the revised implementation date announced on 27 March 2020. Cross references to the securitisation chapters updated to include a reference to the chapter on NPL securitisations (CRE45) published on 26 November 2020. Updated to incorporate the FAQ published on 5 July 2024.

Risk-weighted assets and capital requirements

40.1 The risk-weighted assets for market risk under the simplified standardised approach are determined by multiplying the capital requirements calculated as set out in this chapter by 12.5.

(1) MAR40.3 to MAR40.73 deal with interest rate, equity, foreign exchange (FX) and commodities risk.

(2) MAR40.74 to MAR40.86 set out a number of possible methods for measuring the price risk in options of all kinds.

(3) The capital requirement under the simplified standardised approach will be the measures of risk obtained from MAR40.2 to MAR40.86, summed arithmetically.

40.2 The capital requirement arising from the simplified standardised approach is the simple sum of the recalibrated capital requirements arising from each of the four risk classes – namely interest rate risk, equity risk, FX risk and commodity risk as detailed in the formula below, where:

(1)

CR_{IRR}

= capital requirement under MAR40.3 to MAR40.40 (interest rate risk), plus additional requirements for option risks from debt instruments (non-delta risks) under MAR40.74 to MAR40.86 (treatment of options);

(2)

CR_{EQ}

= capital requirement under MAR40.41 to MAR40.52 (equity risk), plus additional requirements for option risks from equity instruments (non-delta risks) under MAR40.74 to MAR40.86 (treatment of options);

(3)

CR_{FX}

= capital requirement under MAR40.53 to MAR40.62 (FX risk), plus additional requirements for option risks from foreign exchange instruments (non-delta risks) under MAR40.74 to MAR40.86 (treatment of options);

(4)

CR_{COMM}

= capital requirement under MAR40.63 to MAR40.73 (commodities risk), plus additional requirements for option risks from commodities instruments (non-delta risks) under MAR40.74 to MAR40.86 (treatment of options);

(5)

SF_{IRR}

= Scaling factor of 1.30;

(6)

SF_{EQ}

= Scaling factor of 3.50;

(7)

SF_{COMM}

= Scaling factor of 1.90; and

(8)

SF_{FX}
= Scaling factor of 1.20.

$$Capital\ requirement = CR_{IRR} * SF_{IRR} + CR_{EQ} * SF_{EQ} + CR_{FX} * SF_{FX} + CR_{COMM} * SF_{Comm}$$

FAQ

FAQ1 *Should the scaling factors under the simplified standardised approach be applied to capital requirements calculated for securitisation positions?*

The scaling factor for interest rate risk should be applied to all capital requirements calculated in MAR40.3 to MAR40.40.

Interest rate risk

40.3 This section sets out the simplified standard approach for measuring the risk of holding or taking positions in debt securities and other interest rate related instruments in the trading book. The instruments covered include all fixed-rate and floating-rate debt securities and instruments that behave like them, including non-convertible preference shares.[1] Convertible bonds, ie debt issues or preference shares that are convertible, at a stated price, into common shares of the issuer, will be treated as debt securities if they trade like debt securities and as equities if they trade like equities. The basis for dealing with derivative products is considered in MAR40.31 to MAR40.40.

Footnotes

[1] *Traded mortgage securities and mortgage derivative products possess unique characteristics because of the risk of prepayment. Accordingly, for the time being, no common treatment will apply to these securities, which will be dealt with at national discretion. A security that is the subject of a repurchase or securities lending agreement will be treated as if it were still owned by the lender of the security, ie it will be treated in the same manner as other securities positions.*

40.4 The minimum capital requirement is expressed in terms of two separately calculated amounts, one applying to the "specific risk" of each security, whether it is a short or a long position, and the other to the interest rate risk in the portfolio (termed "general market risk") where long and short positions in different securities or instruments can be offset.

Specific risk

40.5 The capital requirement for specific risk is designed to protect against an adverse movement in the price of an individual security owing to factors related to the individual issuer. In measuring the risk, offsetting will be restricted to matched positions in the identical issue (including positions in derivatives). Even if the issuer is the same, no offsetting will be permitted between different issues since differences in coupon rates, liquidity, call features, etc mean that prices may diverge in the short run.

FAQ

FAQ1 *What could be the conditions under which trading book positions that are subject to interest rate specific risk could be netted in order to derive either the net long position or the net short position? Are the rules considering a perfect hedge only? Is it allowed to net cash and synthetic securitisations for the purpose of the capital calculation for structured products under the simplified standardised approach for correlation trading?*

Netting is only allowed under limited circumstances for interest rate specific risk as explained in MAR40.5: "offsetting will be restricted to matched positions in the

identical issue (including positions in derivatives). Even if the issuer is the same, no offsetting will be permitted between different issues since differences in coupon rates, liquidity, call features, etc means that prices may diverge in the short run."

In addition, partial offsetting is allowed in two other sets of circumstances. One set of circumstances is described in MAR40.21 and concerns nth-to-default basked products. The other set of circumstances described in MAR40.16 to MAR40.18 pertains to offsetting between a credit derivative (whether total return swap or credit default swap) and the underlying exposure (ie cash position). Although this treatment applies generally in a one-for-one fashion, it is possible that multiple instruments could combine to create a hedge that would be eligible for consideration for partial offsetting. Supervisors should recognise that, in the case of multiple instruments comprising one side of the position, necessary conditions (ie the value of two legs moving in opposite directions, key contractual features of the credit derivative, identical reference obligations and currency/maturity mismatches) will be extremely difficult to meet, in practice.

40.6 The specific risk capital requirements for "government" and "other" categories will be as follows:

Specific risk capital requirements for issuer risk Government and "other" categories		Table 1
Categories	External credit assessment	Specific risk capital requirement
Government	AAA to AA–	0%
	A+ to BBB–	0.25% (residual term to final maturity 6 months or less) 1.00% (residual term to final maturity greater than 6 and up to and including 24 months) 1.60% (residual term to final maturity exceeding 24 months)
	BB+ to B–	8.00%
	Below B–	12.00%
	Unrated	8.00%
Qualifying		0.25% (residual term to final maturity 6 months or less) 1.00% (residual term to final maturity greater than 6 and up to and including 24 months) 1.60% (residual term to final maturity exceeding 24 months)
Other	BB+ to BB–	8.00%
	Below BB–	12.00%
	Unrated	8.00%

40.7 The government category will include all forms of government[2] paper including bonds, treasury bills and other short-term instruments, but national authorities reserve the right to apply a specific risk capital requirement to securities issued by certain foreign governments, especially to securities denominated in a currency other than that of the issuing government.

Footnotes

[2] *Including, at national discretion, local and regional governments subject to a zero credit risk weight in CRE20.*

40.8 When the government paper is denominated in the domestic currency and funded by the bank in the same currency, at national discretion a lower specific risk capital requirement

may be applied.

40.9 The qualifying category includes securities issued by public sector entities and multilateral development banks, plus other securities that are:

(1) rated investment grade (IG)[3] by at least two credit rating agencies specified by the national authority; or

(2) rated IG by one rating agency and not less than IG by any other rating agency specified by the national authority (subject to supervisory oversight); or

(3) subject to supervisory approval, unrated, but deemed to be of comparable investment quality by the reporting bank, and the issuer has securities listed on a recognised stock exchange.

Footnotes

[3] *For example, IG include rated Baa or higher by Moody's and BBB or higher by Standard and Poor's.*

40.10 Each supervisory authority will be responsible for monitoring the application of these qualifying criteria, particularly in relation to the last criterion where the initial classification is essentially left to the reporting banks. National authorities will also have discretion to include within the qualifying category debt securities issued by banks in countries which have implemented this framework, subject to the express understanding that supervisory authorities in such countries undertake prompt remedial action if a bank fails to meet the capital standards set forth in this framework. Similarly, national authorities will have discretion to include within the qualifying category debt securities issued by securities firms that are subject to equivalent rules.

40.11 Furthermore, the qualifying category shall include securities issued by institutions that are deemed to be equivalent to IG quality and subject to supervisory and regulatory arrangements comparable to those under this framework.

40.12 Unrated securities may be included in the qualifying category when they are subject to supervisory approval, unrated, but deemed to be of comparable investment quality by the reporting bank, and the issuer has securities listed on a recognised stock exchange. This will remain unchanged for banks using the simplified standardised approach. For banks using the internal ratings-based (IRB) approach for a portfolio, unrated securities can be included in the qualifying category if both of the following conditions are met:

(1) the securities are rated equivalent[4] to IG under the reporting bank's internal rating system, which the national supervisor has confirmed complies with the requirements for an IRB approach; and

(2) the issuer has securities listed on a recognised stock exchange.

Footnotes

[4] *Equivalent means the debt security has a one-year probability of default (PD) equal to or less than the one year PD implied by the long-run average one-year PD of a security rated IG or better by a qualifying rating agency.*

40.13 However, since this may in certain cases considerably underestimate the specific risk for debt instruments which have a high yield to redemption relative to government debt securities, each national supervisor will have the discretion:

(1) to apply a higher specific risk charge to such instruments; and/or

(2) to disallow offsetting for the purposes of defining the extent of general market risk

between such instruments and any other debt instruments.

40.14 The specific risk capital requirement of securitisation positions as defined in CRE40.1 to CRE40.6 that are held in the trading book is to be calculated according to the revised method for such positions in the banking book as set out in CRE40 to CRE45. A bank shall calculate the specific risk capital requirement applicable to each net securitisation position by dividing the risk weight calculated as if it were held in the banking book by 12.5.

40.15 Banks may limit the capital requirement for an individual position in a credit derivative or securitisation instrument to the maximum possible loss. For a short risk position this limit could be calculated as a change in value due to the underlying names immediately becoming default risk-free. For a long risk position, the maximum possible loss could be calculated as the change in value in the event that all the underlying names were to default with zero recoveries. The maximum possible loss must be calculated for each individual position.

FAQ

FAQ1 *When a bank buys credit protection for an asset-backed security (ABS) tranche and (due to netting rules) the bank is treated as having a net short position, the simplified standardised capital requirement for the net short position is often determined by the max potential loss. This is particularly true when the underlying ABS tranche has been severely downgraded and written down. In particular, banks note that if the underlying ABS continues to deteriorate, the overall capital requirement progressively increases and is dominated by the charge against the short side of the hedged position.*

Some examples (without and with off-set) illustrate how the Max Loss principle should apply.

Max loss without offset:

Suppose the bank has net long and net short positions that reference similar, but not the same, underlying assets. In other words the bank hedges an A-rated mezzanine residential mortgage-backed security (RMBS) tranche (notional = USD 100) with a credit default swap (CDS) on a similar but different A-rated mezzanine RMBS (also having notional = USD 100).

Suppose the RMBS tranche owned by the bank is now rated C, and has value of USD 15. Also assume that the value of the CDS on the different RMBS has a current value of USD 80. Further, suppose that the current value of the RMBS underlying this CDS is USD 20 and is also rated C. Finally, suppose that the CDS would be valued at USD –2 if the underlying RMBS tranche were to recover unexpectedly and become risk-free.

The correct treatment is as follows: min (USD 15, USD 15) (long leg) + min (USD 20, USD 82) (short leg) = USD 35.

No off-set would be permissible in this example, because the same underlying asset has not been hedged. The capital requirement should, therefore, be calculated by summing the charges against the long and short legs. The maximum loss principle would apply to each individual position.

Please note that the market value of the underlying has been applied in determining the exposure value of the CDS.

Max loss with offset:

Suppose the bank hedges an A-rated mezzanine RMBS tranche with a CDS

referencing the same RMBS having notional of USD 100. Suppose the RMBS tranche is now rated C, and has value USD 15, while the current value of the CDS is USD 85. Suppose that the value of the CDS would equal USD –2 if the RMBS tranche were to recover unexpectedly and become risk-free.

In this example, if the CDS exactly matched the RMBS in tenor, then offsetting could potentially apply. In that instance, the capital requirement should equal 20% of max{min(USD 15, USD 15), min(USD 15, USD 87)} = USD 3.

If the tenors were not matched (ie maturity mismatch), then the capital requirement should equal max{min(USD 15, USD 15), min(USD 15, USD 87)} = USD 15.

Please note that the maximum loss principle cannot be applied on a portfolio basis.

40.16 Full allowance will be recognised for positions hedged by credit derivatives when the values of two legs (ie long and short) always move in the opposite direction and broadly to the same extent. This would be the case in the following situations, in which cases no specific risk capital requirement applies to both sides of the position:

(1) the two legs consist of completely identical instruments; or

(2) a long cash position (or credit derivative) is hedged by a total rate of return swap (or vice versa) and there is an exact match between the reference obligation and the underlying exposure (ie the cash position).[5]

Footnotes

[5] *The maturity of the swap itself may be different from that of the underlying exposure.*

FAQ

FAQ1 *According to MAR40.16 to MAR40.18, the offsetting treatment is applied to a cash position that is hedged by a credit derivative or a credit derivative that is hedged by another credit derivative, assuming there is an exact match in terms of the reference obligations. Please illustrate the treatment.*

MAR40.16 to MAR40.18, are applicable not only when the underlying position being hedged is a cash position, but also when the position being hedged is a credit default swap (CDS) or other credit derivative. They also apply regardless of whether the cash positions or reference obligations of the credit derivative are single-name or securitisation exposures.

For example, when a long cash position is hedged using a CDS, the 80% offset treatment of MAR40.17 (the partial allowance treatment of MAR40.18) generally applies when the reference obligation of the CDS is the cash instrument being hedged and the currencies and remaining maturities of the two positions are (are not) identical. Similarly, when a purchased CDS is hedged with a sold CDS, the 80% offset treatment (the partial allowance treatment) generally applies when both the long and short CDSs have the same reference obligations and the currencies and remaining maturities of the long and short CDSs are (are not) identical. The full allowance (100% offset) treatment generally applies only when there is zero basis risk between the instrument being hedged and the hedging instrument, such as when a cash position is hedged with a total rate of return swap referencing the same cash instrument and there is no currency mismatch, or when a purchased CDS position is hedged by selling a CDS with identical terms in all respects, including reference obligation, currency, maturity, documentation clauses (eg credit payout events, methods for determining payouts for credit events, etc), and structure of fixed and variable payments over time.

As explained in FAQ1 to MAR40.5, it is worth noting that the conditions under which partial or full offsetting of risk positions that are subject to interest rate specific risk are narrowly defined. In practice, offsets between securitisation positions and credit derivatives are unlikely to be recognised in most cases due to the explicit requirements in MAR40.16 to MAR40.18 on reference names etc.

40.17 An 80% offset will be recognised when the value of two legs (ie long and short) always moves in the opposite direction but not broadly to the same extent. This would be the case when a long cash position (or credit derivative) is hedged by a credit default swap (CDS) or a credit-linked note (or vice versa) and there is an exact match in terms of the reference obligation, the maturity of both the reference obligation and the credit derivative, and the currency of the underlying exposure. In addition, key features of the credit derivative contract (eg credit event definitions, settlement mechanisms) should not cause the price movement of the credit derivative to materially deviate from the price movements of the cash position. To the extent that the transaction transfers risk (ie taking account of restrictive payout provisions such as fixed payouts and materiality thresholds), an 80% specific risk offset will be applied to the side of the transaction with the higher capital requirement, while the specific risk requirement on the other side will be zero.

FAQ

FAQ1 *According to MAR40.16 to MAR40.18, the offsetting treatment is applied to a cash position that is hedged by a credit derivative or a credit derivative that is hedged by another credit derivative, assuming there is an exact match in terms of the reference obligations. Please illustrate the treatment.*

MAR40.16 to MAR40.18 are applicable not only when the underlying position being hedged is a cash position, but also when the position being hedged is a CDS or other credit derivative. They also apply regardless of whether the cash positions or reference obligations of the credit derivative are single-name or securitisation exposures.

For example, when a long cash position is hedged using a CDS, the 80% offset treatment of MAR40.17 (the partial allowance treatment of MAR40.18) generally applies when the reference obligation of the CDS is the cash instrument being hedged and the currencies and remaining maturities of the two positions are (are not) identical. Similarly, when a purchased CDS is hedged with a sold CDS, the 80% offset treatment (the partial allowance treatment) generally applies when both the long and short CDSs have the same reference obligations and the currencies and remaining maturities of the long and short CDSs are (are not) identical. The full allowance (100% offset) treatment generally applies only when there is zero basis risk between the instrument being hedged and the hedging instrument, such as when a cash position is hedged with a total rate of return swap referencing the same cash instrument and there is no currency mismatch, or when a purchased CDS position is hedged by selling a CDS with identical terms in all respects, including reference obligation, currency, maturity, documentation clauses (eg credit payout events, methods for determining payouts for credit events, etc), and structure of fixed and variable payments over time.

As explained in FAQ1 to MAR40.5, it is worth noting that the conditions under which partial or full offsetting of risk positions that are subject to interest rate specific risk are narrowly defined. In practice, offsets between securitisation positions and credit derivatives are unlikely to be recognised in most cases due to the explicit requirements in MAR40.16 to MAR40.18 on reference names etc.

40.18 Partial allowance will be recognised when the value of the two legs (ie long and short) usually moves in the opposite direction. This would be the case in the following situations:

(1) The position is captured in MAR40.16(2), but there is an asset mismatch between the reference obligation and the underlying exposure. Nonetheless, the position meets the requirements in CRE22.74.

(2) The position is captured in MAR40.16(1) or MAR40.17 but there is a currency or maturity mismatch[6] between the credit protection and the underlying asset.

(3) The position is captured in MAR40.17 but there is an asset mismatch between the cash position (or credit derivative) and the credit derivative hedge. However, the underlying asset is included in the (deliverable) obligations in the credit derivative documentation.

Footnotes

[6] *Currency mismatches should feed into the normal reporting of FX risk.*

FAQ

FAQ1 *According to MAR40.16 to MAR40.18, the offsetting treatment is applied to a cash position that is hedged by a credit derivative or a credit derivative that is hedged by another credit derivative, assuming there is an exact match in terms of the reference obligations. Please illustrate the treatment.*

MAR40.16 to MAR40.18 are applicable not only when the underlying position being hedged is a cash position, but also when the position being hedged is a CDS or other credit derivative. They also apply regardless of whether the cash positions or reference obligations of the credit derivative are single-name or securitisation exposures.

For example, when a long cash position is hedged using a CDS, the 80% offset treatment of MAR40.17 (the partial allowance treatment of MAR40.18) generally applies when the reference obligation of the CDS is the cash instrument being hedged and the currencies and remaining maturities of the two positions are (are not) identical. Similarly, when a purchased CDS is hedged with a sold CDS, the 80% offset treatment (the partial allowance treatment) generally applies when both the long and short CDSs have the same reference obligations and the currencies and remaining maturities of the long and short CDSs are (are not) identical. The full allowance (100% offset) treatment generally applies only when there is zero basis risk between the instrument being hedged and the hedging instrument, such as when a cash position is hedged with a total rate of return swap referencing the same cash instrument and there is no currency mismatch, or when a purchased CDS position is hedged by selling a CDS with identical terms in all respects, including reference obligation, currency, maturity, documentation clauses (eg credit payout events, methods for determining payouts for credit events, etc), and structure of fixed and variable payments over time.

As explained in FAQ1 to MAR40.5, it is worth noting that the conditions under which partial or full offsetting of risk positions that are subject to interest rate specific risk are narrowly defined. In practice, offsets between securitisation positions and credit derivatives are unlikely to be recognised in most cases due to the explicit requirements in MAR40.16 to MAR40.18 on reference names etc.

40.19 In each of these cases in MAR40.16 to MAR40.18, the following rule applies. Rather than adding the specific risk capital requirements for each side of the transaction (ie the credit protection and the underlying asset) only the higher of the two capital requirements will apply.

40.20 In cases not captured in MAR40.16 to MAR40.18, a specific risk capital requirement will be assessed against both sides of the position.

40.21 An nth-to-default credit derivative is a contract where the payoff is based on the nth asset to default in a basket of underlying reference instruments. Once the nth default occurs the transaction terminates and is settled.

(1) The capital requirement for specific risk for a first-to-default credit derivative is the lesser of:

(a) the sum of the specific risk capital requirements for the individual reference credit instruments in the basket; and

(b) the maximum possible credit event payment under the contract.

(2) Where a bank has a risk position in one of the reference credit instruments underlying a first-to-default credit derivative and this credit derivative hedges the bank's risk position, the bank is allowed to reduce, with respect to the hedged amount, both the capital requirement for specific risk for the reference credit instrument and that part of the capital requirement for specific risk for the credit derivative that relates to this particular reference credit instrument. Where a bank has multiple risk positions in reference credit instruments underlying a first-to-default credit derivative, this offset is allowed only for that underlying reference credit instrument having the lowest specific risk capital requirement.

(3) The capital requirement for specific risk for an nth-to-default credit derivative with n greater than one is the lesser of:

(a) the sum of the specific risk capital requirements for the individual reference credit instruments in the basket but disregarding the (n-1) obligations with the lowest specific risk capital requirements; and

(b) the maximum possible credit event payment under the contract. For nth-to-default credit derivatives with n greater than 1, no offset of the capital requirement for specific risk with any underlying reference credit instrument is allowed.

(4) If a first or other nth-to-default credit derivative is externally rated, then the protection seller must calculate the specific risk capital requirement using the rating of the derivative and apply the respective securitisation risk weights as specified in MAR40.14, as applicable.

(5) The capital requirement against each net nth-to-default credit derivative position applies irrespective of whether the bank has a long or short position, ie obtains or provides protection.

FAQ

FAQ1 *The framework mentions only tranches and nth-to-default products explicitly, but not nth to n+m-th-to-default products (eg the value depends on the default of the 5th, 6th, 7th and 8th default in a pool; only in specific cases such as the same nominal for all underlyings can this product be represented by, for example, a 5% to 8% tranche). Are nth to n+m-th-to-default products covered in the framework?*

Yes. Such products are to be decomposed into individual nth-to-default products and the rules for nth-to-default products in MAR40.21 apply.

In the example cited above, the capital requirement for a basket default swap covering defaults five to eight would be calculated as the sum of the capital requirements for a 5th-to-default swap, a 6th-to-default swap, a 7th-to-default swap and an 8th-to-default swap.

40.22 A bank must determine the specific risk capital requirement for the correlation trading portfolio (CTP) as follows:

(1) The bank computes:

(a) the total specific risk capital requirements that would apply just to the net long positions from the net long correlation trading exposures combined; and

(b) the total specific risk capital requirements that would apply just to the net short positions from the net short correlation trading exposures combined.

(2) The larger of these total amounts is then the specific risk capital requirement for the CTP.

FAQ

FAQ1 *Can the approach of taking the larger of the specific risk capital requirements for net long positions and the specific risk capital requirement for net short positions be applied to leveraged securitisation positions or option products on securitisation positions?*

No. Leveraged securitisation positions and option products on securitisation positions are securitisation positions. They are not admissible for the CTP. The capital requirements for specific risk will be determined as the sum of the capital requirements for specific risk against net long and net short positions.

General market risk

40.23 The capital requirements for general market risk are designed to capture the risk of loss arising from changes in market interest rates. A choice between two principal methods of measuring the risk is permitted – a maturity method and a duration method. In each method, the capital requirement is the sum of four components:

(1) the net short or long position in the whole trading book;

(2) a small proportion of the matched positions in each time band (the "vertical disallowance");

(3) a larger proportion of the matched positions across different time bands (the "horizontal disallowance"); and

(4) a net charge for positions in options, where appropriate (see MAR40.84 and MAR40.85).

40.24 Separate maturity ladders should be used for each currency and capital requirements should be calculated for each currency separately and then summed with no offsetting between positions of the opposite sign. In the case of those currencies in which business is insignificant, separate maturity ladders for each currency are not required. Rather, the bank may construct a single maturity ladder and slot, within each appropriate time band, the net long or short position for each currency. However, these individual net positions are to be summed within each time band, irrespective of whether they are long or short positions, to produce a gross position figure.

40.25 In the maturity method (see MAR40.29 for the duration method), long or short positions in debt securities and other sources of interest rate exposures including derivative instruments, are slotted into a maturity ladder comprising 13 time bands (or 15 time bands in the case of low coupon instruments). Fixed rate instruments should be allocated according to the residual term to maturity and floating-rate instruments according to the residual term to the next repricing date. Opposite positions of the same amount in the same issues (but not different issues by the same issuer), whether actual or notional, can

be omitted from the interest rate maturity framework, as well as closely matched swaps, forwards, futures and forward rate agreements (FRAs) which meet the conditions set out in MAR40.35 and MAR40.36 below.

40.26 The first step in the calculation is to weight the positions in each time band by a factor designed to reflect the price sensitivity of those positions to assumed changes in interest rates. The weights for each time band are set out in Table 4. Zero-coupon bonds and deep-discount bonds (defined as bonds with a coupon of less than 3%) should be slotted according to the time bands set out in the second column of Table 4.

Maturity method: time bands and weights			Table 4
Coupon 3% or more	Coupon less than 3%	Risk weight	Assumed changes in yield
1 month or less	1 month or less	0.00%	1.00
1 to 3 months	1 to 3 months	0.20%	1.00
3 to 6 months	3 to 6 months	0.40%	1.00
6 to 12 months	6 to 12 months	0.70%	1.00
1 to 2 years	1.0 to 1.9 years	1.25%	0.90
2 to 3 years	1.9 to 2.8 years	1.75%	0.80
3 to 4 years	2.8 to 3.6 years	2.25%	0.75
4 to 5 years	3.6 to 4.3 years	2.75%	0.75
5 to 7 years	4.3 to 5.7 years	3.25%	0.70
7 to 10 years	5.7 to 7.3 years	3.75%	0.65
10 to 15 years	7.3 to 9.3 years	4.50%	0.60
15 to 20 years	9.3 to 10.6 years	5.25%	0.60
Over 20 years	10.6 to 12 years	6.00%	0.60
	12 to 20 years	8.00%	0.60
	Over 20 years	12.50%	0.60

40.27 The next step in the calculation is to offset the weighted longs and shorts in each time band, resulting in a single short or long position for each band. Since, however, each band would include different instruments and different maturities, a 10% capital requirement to reflect basis risk and gap risk will be levied on the smaller of the offsetting positions, be it long or short. Thus, if the sum of the weighted longs in a time band is USD 100 million and the sum of the weighted shorts USD 90 million, the so-called vertical disallowance for that time band would be 10% of USD 90 million (ie USD 9 million).

40.28 The result of the above calculations is to produce two sets of weighted positions, the net long or short positions in each time band (USD 10 million long in the example above) and the vertical disallowances, which have no sign.

(1) In addition, however, banks will be allowed to conduct two rounds of horizontal offsetting:

(a) first between the net positions in each of three zones, where zone 1 is set as zero to one year, zone 2 is set as one year to four years, and zone 3 is set as four years and over (however, for coupons less than 3%, zone 2 is set as one year to 3.6 years and zone 3 is set as 3.6 years and over); and

(b) subsequently between the net positions in the three different zones.

(2) The offsetting will be subject to a scale of disallowances expressed as a fraction of the matched positions, as set out in Table 5. The weighted long and short positions in each of three zones may be offset, subject to the matched portion attracting a disallowance factor that is part of the capital requirement. The residual net position in each zone may be carried over and offset against opposite positions in other zones, subject to a second set of disallowance factors.

Horizontal disallowances				Table 5
Zones[7]	Time band[7]	Within the zone	Between adjacent zones	Between zones 1 and 3
	0-1 month			
Zone 1	1-3 months	40%		
	3-6 months			
	6-12 months		40%	
	1-2 years			
Zone 2	2-3 years	30%		100%
	3-4 years			
	4-5 years		40%	
	5-7 years			
Zone 3	7-10 years			
	10-15 years	30%		
	15-20 years			
	Over 20 years			

Footnotes

[7] *The zones for coupons less than 3% are 0 to 1 year, 1 to 3.6 years, and 3.6 years and over.*

40.29 Under the alternative duration method, banks with the necessary capability may, with their supervisors' consent, use a more accurate method of measuring all of their general market risk by calculating the price sensitivity of each position separately. Banks must elect and use the method on a continuous basis (unless a change in method is approved by the national authority) and will be subject to supervisory monitoring of the systems used. The mechanics of this method are as follows:

(1) First calculate the price sensitivity of each instrument in terms of a change in interest rates of between 0.6 and 1.0 percentage points depending on the maturity of the instrument (see Table 6);

(2) Slot the resulting sensitivity measures into a duration-based ladder with the 15 time bands set out in Table 6;

(3) Subject long and short positions in each time band to a 5% vertical disallowance designed to capture basis risk; and

(4) Carry forward the net positions in each time band for horizontal offsetting subject to the disallowances set out in Table 5 above.

Duration method: time bands and assumed changes in yield	Table 6

	Assumed change in yield		Assumed change in yield
Zone 1:		Zone 3:	
1 month or less	1.00	3.6 to 4.3 years	0.75
1 to 3 months	1.00	4.3 to 5.7 years	0.70
3 to 6 months	1.00	5.7 to 7.3 years	0.65
6 to 12 months	1.00	7.3 to 9.3 years	0.60
Zone 2:		9.3 to 10.6 years	0.60
1.0 to 1.9 years	0.90	10.6 to 12 years	0.60
1.9 to 2.8 years	0.80	12 to 20 years	0.60
2.8 to 3.6 years	0.75	Over 20 years	0.60

40.30 In the case of residual currencies (see MAR40.24 above) the gross positions in each time band will be subject to either the risk weightings set out in MAR40.26, if positions are reported using the maturity method, or the assumed change in yield set out in MAR40.29, if positions are reported using the duration method, with no further offsets.

Interest rate derivatives

40.31 The measurement system should include all interest-rate derivatives and off-balance sheet instruments in the trading book which react to changes in interest rates (eg FRAs, other forward contracts, bond futures, interest rate and cross-currency swaps and forward foreign exchange positions). Options can be treated in a variety of ways as described in MAR40.74 to MAR40.86. A summary of the rules for dealing with interest rate derivatives is set out in MAR40.40.

40.32 The derivatives should be converted into positions in the relevant underlying and become subject to specific and general market risk charges as described above. In order to calculate the standard formula described above, the amounts reported should be the market value of the principal amount of the underlying or of the notional underlying resulting from the prudent valuation guidance set out in CAP50.[8]

Footnotes

[8] *For instruments where the apparent notional amount differs from the effective notional amount, banks must use the effective notional amount.*

40.33 Futures and forward contracts (including FRAs) are treated as a combination of a long and a short position in a notional government security. The maturity of a future or an FRA will be the period until delivery or exercise of the contract, plus – where applicable – the life of the underlying instrument. For example, a long position in a June three-month interest rate future (taken in April) is to be reported as a long position in a government security with a five-month maturity and a short position in a government security with a two-month maturity. Where a range of deliverable instruments may be delivered to fulfil the contract, the bank has flexibility to elect which deliverable security goes into the maturity or duration ladder but should take account of any conversion factor defined by the exchange. In the case of a future on a corporate bond index, positions will be included at the market value of the notional underlying portfolio of securities.

40.34 Swaps will be treated as two notional positions in government securities with relevant

maturities. For example, an interest rate swap under which a bank is receiving floating rate interest and paying fixed will be treated as a long position in a floating rate instrument of maturity equivalent to the period until the next interest fixing and a short position in a fixed-rate instrument of maturity equivalent to the residual life of the swap. For swaps that pay or receive a fixed or floating interest rate against some other reference price, eg a stock index, the interest rate component should be slotted into the appropriate repricing maturity category, with the equity component being included in the equity framework. The separate legs of cross-currency swaps are to be reported in the relevant maturity ladders for the currencies concerned.

40.35 Banks may exclude from the interest rate maturity framework altogether (for both specific and general market risk) long and short positions (both actual and notional) in identical instruments with exactly the same issuer, coupon, currency and maturity. A matched position in a future or forward and its corresponding underlying may also be fully offset[10] and thus excluded from the calculation. When the future or the forward comprises a range of deliverable instruments offsetting of positions in the future or forward contract and its underlying is only permissible in cases where there is a readily identifiable underlying security that is most profitable for the trader with a short position to deliver. The price of this security, sometimes called the "cheapest-to-deliver", and the price of the future or forward contract should, in such cases, move in close alignment. No offsetting will be allowed between positions in different currencies; the separate legs of cross-currency swaps or forward FX deals are to be treated as notional positions in the relevant instruments and included in the appropriate calculation for each currency.

Footnotes

10 *The leg representing the time to expiry of the future should, however, be reported.*

40.36 In addition, opposite positions in the same category of instruments[10] can in certain circumstances be regarded as matched and allowed to offset fully. To qualify for this treatment, the positions must relate to the same underlying instruments, be of the same nominal value and be denominated in the same currency.[11] In addition:

(1) for futures: offsetting positions in the notional or underlying instruments to which the futures contract relates must be for identical products and mature within seven days of each other;

(2) for swaps and FRAs: the reference rate (for floating rate positions) must be identical and the coupon closely matched (ie within 15 basis points); and

(3) for swaps, FRAs and forwards: the next interest fixing date or, for fixed coupon positions or forwards, the residual maturity must correspond within the following limits:

  (a) less than one month hence: same day;

  (b) between one month and one year hence: within seven days; and

  (c) over one year hence: within 30 days.

Footnotes

10 *This includes the delta-equivalent value of options. The delta equivalent of the legs arising out of the treatment of caps and floors as set out in MAR40.78 can also be offset against each other under the rules laid down in this paragraph.*

11 *The separate legs of different swaps may also be matched subject to the same conditions.*

40.37 Banks with large swap books may use alternative formulae for these swaps to calculate the positions to be included in the maturity or duration ladder. One method would be to first convert the payments required by the swap into their present values. For that purpose, each payment should be discounted using zero coupon yields, and a single net figure for the present value of the cash flows entered into the appropriate time band using procedures that apply to zero- (or low-) coupon bonds; these figures should be slotted into the general market risk framework as set out above. An alternative method would be to calculate the sensitivity of the net present value implied by the change in yield used in the maturity or duration method and allocate these sensitivities into the time bands set out in MAR40.26 or MAR40.29. Other methods which produce similar results could also be used. Such alternative treatments will, however, only be allowed if:

(1) the supervisory authority is fully satisfied with the accuracy of the systems being used;

(2) the positions calculated fully reflect the sensitivity of the cash flows to interest rate changes and are entered into the appropriate time bands; and

(3) the positions are denominated in the same currency.

40.38 Interest rate and currency swaps, FRAs, forward FX contracts and interest rate futures will not be subject to a specific risk charge. This exemption also applies to futures on an interest rate index (eg London Interbank Offer Rate, or LIBOR). However, in the case of futures contracts where the underlying is a debt security, or an index representing a basket of debt securities, a specific risk charge will apply according to the credit risk of the issuer as set out in MAR40.5 to MAR40.21.

40.39 General market risk applies to positions in all derivative products in the same manner as for cash positions, subject only to an exemption for fully or very closely matched positions in identical instruments as defined in MAR40.35 and MAR40.36. The various categories of instruments should be slotted into the maturity ladder and treated according to the rules identified earlier.

40.40 Table 7 presents a summary of the regulatory treatment for interest rate derivatives, for market risk purposes.

Summary of treatment of interest rate derivatives		Table 7
Instrument	Specific risk charge[12]	General market risk charge
Exchanged-traded future		
Government debt security	Yes[13]	Yes, as two positions
Corporate debt security	Yes	Yes, as two positions
Index on interest rates (eg LIBOR)	No	Yes, as two positions
Over-the-counter (OTC) forward		
Government debt security	Yes[13]	Yes, as two positions
Corporate debt security	Yes	Yes, as two positions
Index on interest rates	No	Yes, as two positions
FRAs, swaps	No	Yes, as two positions
Forward FX	No	Yes, as one position in each currency
Options		Either
Government debt security	Yes[13]	(a) carve out together with

		the associated hedging positions: simplified approach; scenario analysis; internal models
Corporate debt security	Yes	(b) general market risk charge according to the delta-plus method (gamma and vega should receive separate capital requirements)
Index on interest rates	No	
FRAs, swaps	No	

Footnotes

12 *This is the specific risk charge relating to the issuer of the instrument. Under the credit risk rules, a separate capital requirement for the counterparty credit risk applies.*

13 *The specific risk capital requirement only applies to government debt securities that are rated below AA– (see MAR40.6 and MAR40.7).*

Equity risk

40.41 This section sets out a minimum capital standard to cover the risk of holding or taking positions in equities in the trading book. It applies to long and short positions in all instruments that exhibit market behaviour similar to equities, but not to non-convertible preference shares (which are covered by the interest rate risk requirements described in MAR40.3 to MAR40.40). Long and short positions in the same issue may be reported on a net basis. The instruments covered include common stocks (whether voting or non-voting), convertible securities that behave like equities, and commitments to buy or sell equity securities. The treatment of derivative products, stock indices and index arbitrage is described in MAR40.44 to MAR40.52 below.

Specific and general market risks

40.42 As with debt securities, the minimum capital standard for equities is expressed in terms of two separately calculated capital requirements for the specific risk of holding a long or short position in an individual equity and for the general market risk of holding a long or short position in the market as a whole. Specific risk is defined as the bank's gross equity positions (ie the sum of all long equity positions and of all short equity positions) and general market risk as the difference between the sum of the longs and the sum of the shorts (ie the overall net position in an equity market).The long or short position in the market must be calculated on a market-by-market basis, ie a separate calculation has to be carried out for each national market in which the bank holds equities.

40.43 The capital requirement for specific risk and for general market risk will each be 8%.

Equity derivatives

40.44 Except for options, which are dealt with in MAR40.74 to MAR40.86, equity derivatives and off-balance sheet positions that are affected by changes in equity prices should be included in the measurement system.[14] This includes futures and swaps on both individual equities and on stock indices. The derivatives are to be converted into positions in the relevant underlying. The treatment of equity derivatives is summarised in MAR40.52 below.

Footnotes

[14] *Where equities are part of a forward contract, a future or an option (quantity of equities to be received or to be delivered), any interest rate or foreign currency exposure from the other leg of the contract should be reported as set out in MAR40.3 to MAR40.40 and MAR40.53 to MAR40.62.*

40.45 In order to calculate the standard formula for specific and general market risk, positions in derivatives should be converted into notional equity positions:

(1) Futures and forward contracts relating to individual equities should in principle be reported at current market prices.

(2) Futures relating to stock indices should be reported as the marked-to-market value of the notional underlying equity portfolio.

(3) Equity swaps are to be treated as two notional positions.[15]

(4) Equity options and stock index options should be either carved out together with the associated underlyings or be incorporated in the measure of general market risk described in this section according to the delta-plus method.

Footnotes

[15] *For example, an equity swap in which a bank is receiving an amount based on the change in value of one particular equity or stock index and paying a different index will be treated as a long position in the former and a short position in the latter. Where one of the legs involves receiving/paying a fixed or floating interest rate, that exposure should be slotted into the appropriate repricing time band for interest rate related instruments as set out in MAR40.3 to MAR40.40. The stock index should be covered by the equity treatment.*

40.46 Matched positions in each identical equity or stock index in each market may be fully offset, resulting in a single net short or long position to which the specific and general market risk charges will apply. For example, a future in a given equity may be offset against an opposite cash position in the same equity.[16]

Footnotes

[16] *The interest rate risk arising out of the future, however, should be reported as set out in MAR40.3 to MAR40.40.*

40.47 Besides general market risk, a further capital requirement of 2% will apply to the net long or short position in an index contract comprising a diversified portfolio of equities. This capital requirement is intended to cover factors such as execution risk. National supervisory authorities will take care to ensure that this 2% risk weight applies only to well-diversified indices and not, for example, to sectoral indices.

40.48 In the case of the futures-related arbitrage strategies described below, the additional 2% capital requirement described above (set out in MAR40.47) may be applied to only one index with the opposite position exempt from a capital requirement. The strategies are:

(1) when the bank takes an opposite position in exactly the same index at different dates or in different market centres; and

(2) when the bank has an opposite position in contracts at the same date in different but similar indices, subject to supervisory oversight that the two indices contain sufficient common components to justify offsetting.

40.49 Where a bank engages in a deliberate arbitrage strategy, in which a futures contract on a broadly based index matches a basket of stocks, it will be allowed to carve out both

positions from the simplified standardised approach on condition that:

(1) the trade has been deliberately entered into and separately controlled; and

(2) the composition of the basket of stocks represents at least 90% of the index when broken down into its notional components.

40.50 In such a case as set out in MAR40.49 the minimum capital requirement will be 4% (ie 2% of the gross value of the positions on each side) to reflect divergence and execution risks. This applies even if all of the stocks comprising the index are held in identical proportions. Any excess value of the stocks comprising the basket over the value of the futures contract or excess value of the futures contract over the value of the basket is to be treated as an open long or short position.

40.51 If a bank takes a position in depository receipts against an opposite position in the underlying equity or identical equities in different markets, it may offset the position (ie bear no capital requirement) but only on condition that any costs on conversion are fully taken into account.[17]

Footnotes

17 *Any FX risk arising out of these positions has to be reported as set out in MAR40.53 to MAR40.67.*

40.52 Table 8 summarises the regulatory treatment of equity derivatives for market risk purposes.

Summary of treatment of equity derivatives		Table 8
Instrument	Specific risk 18	General market risk
Exchanged-traded or OTC future		
Individual equity	Yes	Yes, as underlying
Index	2%	Yes, as underlying
Options		Either
Individual equity	Yes	(a) carve out together with the associated hedging positions: simplified approach; scenario analysis; internal models
Index	2%	(b) general market risk charge according to the delta-plus method (gamma and vega should receive separate capital requirements)

Footnotes

18 *This is the specific risk charge relating to the issuer of the instrument. Under the credit risk rules], a separate capital requirement for the counterparty credit risk applies.*

Foreign exchange risk

40.53 This section sets out the simplified standardised approach for measuring the risk of holding or taking positions in foreign currencies, including gold.[19]

Footnotes

19 *Gold is to be dealt with as an FX position rather than a commodity because its volatility is more in line with foreign currencies and banks manage it in a similar*

manner to foreign currencies.

40.54 Two processes are needed to calculate the capital requirement for FX risk.

(1) The first is to measure the exposure in a single currency position as set out in MAR40.55 to MAR40.58.

(2) The second is to measure the risks inherent in a bank's mix of long and short positions in different currencies as set out in MAR40.59 to MAR40.62.

Measuring the exposure in a single currency

40.55 The bank's net open position in each currency should be calculated by summing:

(1) the net spot position (ie all asset items less all liability items, including accrued interest, denominated in the currency in question);

(2) the net forward position (ie all amounts to be received less all amounts to be paid under forward FX transactions, including currency futures and the principal on currency swaps not included in the spot position);

(3) guarantees (and similar instruments) that are certain to be called and are likely to be irrecoverable;

(4) net future income/expenses not yet accrued but already fully hedged (at the discretion of the reporting bank);

(5) any other item representing a profit or loss in foreign currencies (depending on particular accounting conventions in different countries); and

(6) the net delta-based equivalent of the total book of foreign currency options.[20]

Footnotes

[20] *Subject to a separately calculated capital requirement for gamma and vega as described in MAR40.77 to MAR40.80; alternatively, options and their associated underlyings are subject to one of the other methods described in MAR40.74 to MAR40.86.*

40.56 Positions in composite currencies need to be separately reported but, for measuring banks' open positions, may be either treated as a currency in their own right or split into their component parts on a consistent basis. Positions in gold should be measured in the same manner as described in MAR40.68.[21]

Footnotes

[21] *Where gold is part of a forward contract (quantity of gold to be received or to be delivered), any interest rate or foreign currency exposure from the other leg of the contract should be reported as set out in MAR40.3 to MAR40.40 and MAR40.55 above.*

40.57 Interest, other income and expenses should be treated as follows. Interest accrued (ie earned but not yet received) should be included as a position. Accrued expenses should also be included. Unearned but expected future interest and anticipated expenses may be excluded unless the amounts are certain and banks have taken the opportunity to hedge them. If banks include future income/expenses they should do so on a consistent basis, and not be permitted to select only those expected future flows which reduce their position.

40.58 Forward currency and gold positions should be measured as follows: Forward currency

and gold positions will normally be valued at current spot market exchange rates. Using forward exchange rates would be inappropriate since it would result in the measured positions reflecting current interest rate differentials to some extent. However, banks that base their normal management accounting on net present values are expected to use the net present values of each position, discounted using current interest rates and valued at current spot rates, for measuring their forward currency and gold positions.

Measuring the foreign exchange risk in a portfolio of foreign currency positions and gold

40.59 For measuring the FX risk in a portfolio of foreign currency positions and gold as set out in MAR40.54(2), a bank that is not approved to use internal models by its supervisory authority must use a shorthand method which treats all currencies equally.

40.60 Under the shorthand method, the nominal amount (or net present value) of the net position in each foreign currency and in gold is converted at spot rates into the reporting currency.[22] The overall net open position is measured by aggregating:

(1) the sum of the net short positions or the sum of the net long positions, whichever is the greater;[23] plus

(2) the net position (short or long) in gold, regardless of sign.

Footnotes

22 *Where the bank is assessing its FX risk on a consolidated basis, it may be technically impractical in the case of some marginal operations to include the currency positions of a foreign branch or subsidiary of the bank. In such cases, the internal limit in each currency may be used as a proxy for the positions. Provided there is adequate ex post monitoring of actual positions against such limits, the limits should be added, without regard to sign, to the net open position in each currency.*

23 *An alternative calculation, which produces an identical result, is to include the reporting currency as a residual and to take the sum of all the short (or long) positions.*

40.61 The capital requirement will be 8% of the overall net open position (see example in Table 9). In particular, the capital requirement would be 8% of the higher of either the net long currency positions or the net short currency positions (ie 300) and of the net position in gold (35) = 335 x 8% = 26.8.

Example of the shorthand measure of FX risk						Table 9
	JPY	EUR	GBP	CAD	USD	Gold
Net position per currency	+50	+100	+150	-20	-180	-35
Net open position		+300		-200		35

40.62 A bank of which business in foreign currency is insignificant and which does not take FX positions for its own account may, at the discretion of its national authority, be exempted from capital requirements on these positions provided that:

(1) its foreign currency business, defined as the greater of the sum of its gross long positions and the sum of its gross short positions in all foreign currencies, does not exceed 100% of eligible capital as defined in CAP10.1; and

(2) its overall net open position as defined in MAR40.60 above does not exceed 2% of its eligible capital as defined in CAP10.1.

Commodities risk

40.63 This section sets out the simplified standardised approach for measuring the risk of holding or taking positions in commodities, including precious metals, but excluding gold (which is treated as a foreign currency according to the methodology set out in MAR40.53 to MAR40.62 above). A commodity is defined as a physical product which is or can be traded on a secondary market, eg agricultural products, minerals (including oil) and precious metals.

40.64 The price risk in commodities is often more complex and volatile than that associated with currencies and interest rates. Commodity markets may also be less liquid than those for interest rates and currencies and, as a result, changes in supply and demand can have a more dramatic effect on price and volatility.[24] These market characteristics can make price transparency and the effective hedging of commodities risk more difficult.

Footnotes

[24] *Banks need also to guard against the risk that arises when the short position falls due before the long position. Owing to a shortage of liquidity in some markets, it might be difficult to close the short position and the bank might be squeezed by the market.*

40.65 The risks associated with commodities include the following risks:

(1) For spot or physical trading, the directional risk arising from a change in the spot price is the most important risk.

(2) However, banks using portfolio strategies involving forward and derivative contracts are exposed to a variety of additional risks, which may well be larger than the risk of a change in spot prices. These include:

(a) basis risk (the risk that the relationship between the prices of similar commodities alters through time);

(b) interest rate risk (the risk of a change in the cost of carry for forward positions and options); and

(c) forward gap risk (the risk that the forward price may change for reasons other than a change in interest rates).

(3) In addition, banks may face counterparty credit risk on over-the-counter derivatives, but this is captured by the methods set out in CRE51 to CRE55 and MAR50.

(4) The funding of commodities positions may well open a bank to interest rate or FX exposure and if that is so the relevant positions should be included in the measures of interest rate and FX risk described in MAR40.3 to MAR40.40 and MAR40.53 to MAR40.62, respectively.[25]

Footnotes

[25] *Where a commodity is part of a forward contract (quantity of commodities to be received or to be delivered), any interest rate or foreign currency exposure from the other leg of the contract should be reported as set out in MAR40.3 to MAR40.40 and MAR40.53 to MAR40.62. Positions which are purely stock financing (ie a physical stock has been sold forward and the cost of funding has been locked in until the date of the forward sale) may be omitted from the commodities risk calculation although they will be subject to interest rate and counterparty risk requirements.*

40.66 There are two alternatives for measuring commodities position risk under the simplified standardised approach that are described in MAR40.68 to MAR40.73 below. Commodities

risk can also be measured, using either (i) the maturity ladder approach, which is a measurement system that captures forward gap and interest rate risk separately by basing the methodology on seven time bands as set out in MAR40.68 to MAR40.71 below or (ii) the simplified approach, which is a very simple framework as set out in MAR40.72 and MAR40.73 below. Both the maturity ladder approach and the simplified approach are appropriate only for banks that, in relative terms, conduct only a limited amount of commodities business.

40.67 For the maturity ladder approach and the simplified approach, long and short positions in each commodity may be reported on a net basis for the purposes of calculating open positions. However, positions in different commodities will, as a general rule, not be offsettable in this fashion. Nevertheless, national authorities will have discretion to permit netting between different subcategories[26] of the same commodity in cases where the subcategories are deliverable against each other. They can also be considered as offsettable if they are close substitutes against each other and a minimum correlation of 0.9 between the price movements can be clearly established over a minimum period of one year. However, a bank wishing to base its calculation of capital requirements for commodities on correlations would have to satisfy the relevant supervisory authority of the accuracy of the method that has been chosen and obtain its prior approval.

Footnotes

26 *Commodities can be grouped into clans, families, subgroups and individual commodities. For example, a clan might be Energy Commodities, within which Hydro-Carbons are a family with Crude Oil being a subgroup and West Texas Intermediate, Arabian Light and Brent being individual commodities.*

Maturity ladder approach

40.68 In calculating the capital requirements under the maturity ladder approach, banks will first have to express each commodity position (spot plus forward) in terms of the standard unit of measurement (barrels, kilos, grams etc). The net position in each commodity will then be converted at current spot rates into the national currency.

40.69 Secondly, in order to capture forward gap and interest rate risk within a time band (which, together, are sometimes referred to as curvature/spread risk), matched long and short positions in each time band will carry a capital requirement. The methodology is similar to that used for interest rate related instruments as set out in MAR40.3 to MAR40.40. Positions in the separate commodities (expressed in terms of the standard unit of measurement) will first be entered into a maturity ladder while physical stocks should be allocated to the first time band. A separate maturity ladder will be used for each commodity as defined in MAR40.67 above.[27] For each time band as set out in Table 10, the sum of short and long positions that are matched will be multiplied first by the spot price for the commodity, and then by the spread rate of 1.5%.

Time bands and spread rates	Table 10
Time band	Spread rate
0-1 month	1.5%
1-3 months	1.5%
3-6 months	1.5%
6-12 months	1.5%
1-2 years	1.5%

2-3 years	1.5%
over 3 years	1.5%

Footnotes

27 *For markets that have daily delivery dates, any contracts maturing within 10 days of one another may be offset.*

40.70 The residual net positions from nearer time bands may then be carried forward to offset exposures in time bands that are further out. However, recognising that such hedging of positions among different time bands is imprecise, a surcharge equal to 0.6% of the net position carried forward will be added in respect of each time band that the net position is carried forward. The capital requirement for each matched amount created by carrying net positions forward will be calculated as in MAR40.69 above. At the end of this process, a bank will have either only long or only short positions, to which a capital requirement of 15% will apply.

40.71 All commodity derivatives and off-balance sheet positions that are affected by changes in commodity prices should be included in this measurement framework. This includes commodity futures, commodity swaps, and options where the "delta-plus" method[28] is used (see MAR40.77 to MAR40.80 below). In order to calculate the risk, commodity derivatives should be converted into notional commodities positions and assigned to maturities as follows:

(1) Futures and forward contracts relating to individual commodities should be incorporated as notional amounts of the standard unit of measurement (barrels, kilos, grams etc) and should be assigned a maturity with reference to expiry date.

(2) Commodity swaps where one leg is a fixed price and the other the current market price should be incorporated as a series of positions equal to the notional amount of the contract, with one position corresponding with each payment on the swap and slotted into the maturity ladder accordingly. The positions would be long positions if the bank is paying fixed and receiving floating, and short positions if the bank is receiving fixed and paying floating.[29]

(3) Commodity swaps where the legs are in different commodities are to be incorporated in the relevant maturity ladder. No offsetting will be allowed in this regard except where the commodities belong to the same subcategory as defined in MAR40.67 above.

Footnotes

28 *For banks using other approaches to measure options risk, all options and the associated underlyings should be excluded from both the maturity ladder approach and the simplified approach.*

29 *If one of the legs involves receiving/paying a fixed or floating interest rate, that exposure should be slotted into the appropriate repricing maturity band in the maturity ladder covering interest rate related instruments.*

Simplified approach

40.72 In calculating the capital requirement for directional risk under the simplified approach, the same procedure will be adopted as in the maturity ladder approach described above (see MAR40.68 and MAR40.71. Once again, all commodity derivatives and off-balance

sheet positions that are affected by changes in commodity prices should be included. The capital requirement will equal 15% of the net position, long or short, in each commodity.

40.73 In order to protect the bank against basis risk, interest rate risk and forward gap risk under the simplified approach, the capital requirement for each commodity as described in MAR40.68 and MAR40.71 above will be subject to an additional capital requirement equivalent to 3% of the bank's gross positions, long plus short, in that particular commodity. In valuing the gross positions in commodity derivatives for this purpose, banks should use the current spot price.

Treatment of options

40.74 In recognition of the wide diversity of banks' activities in options and the difficulties of measuring price risk for options, two alternative approaches will be permissible at the discretion of the national authority under the simplified standardised approach.

(1) Those banks which solely use purchased options[30] can use the simplified approach described in MAR40.76 below;

(2) Those banks which also write options are expected to use the delta-plus method or scenario approach which are the intermediate approaches as set out in MAR40.77 to MAR40.86. The more significant its trading activity is, the more the bank will be expected to use a sophisticated approach, and a bank with highly significant trading activity is expected to use the standardised approach or the internal models approach as set out in MAR20 to MAR23 or MAR30 to MAR33.

Footnotes

[30] *Unless all their written option positions are hedged by perfectly matched long positions in exactly the same options, in which case no capital requirement for market risk is required.*

40.75 In the simplified approach for options, the positions for the options and the associated underlying, cash or forward, are not subject to the standardised methodology but rather are carved-out and subject to separately calculated capital requirements that incorporate both general market risk and specific risk. The risk numbers thus generated are then added to the capital requirements for the relevant category, ie interest rate related instruments, equities, FX and commodities as described in MAR40.3 to MAR40.73. The delta-plus method uses the sensitivity parameters or Greek letters associated with options to measure their market risk and capital requirements. Under this method, the delta-equivalent position of each option becomes part of the simplified standardised approach set out in MAR40.3 to MAR40.73 with the delta-equivalent amount subject to the applicable general market risk charges. Separate capital requirements are then applied to the gamma and vega risks of the option positions. The scenario approach uses simulation techniques to calculate changes in the value of an options portfolio for changes in the level and volatility of its associated underlyings. Under this approach, the general market risk charge is determined by the scenario grid (ie the specified combination of underlying and volatility changes) that produces the largest loss. For the delta-plus method and the scenario approach, the specific risk capital requirements are determined separately by multiplying the delta-equivalent of each option by the specific risk weights set out in MAR40.3 to MAR40.52.

Simplified approach

40.76 Banks that handle a limited range of purchased options can use the simplified approach set out in Table 11 for particular trades. As an example of how the calculation would work,

if a holder of 100 shares currently valued at USD 10 each holds an equivalent put option with a strike price of USD 11, the capital requirement would be: USD 1,000 x 16% (ie 8% specific plus 8% general market risk) = USD 160, less the amount the option is in the money (USD 11 - USD 10) x 100 = USD 100, ie the capital requirement would be USD 60. A similar methodology applies for options whose underlying is a foreign currency, an interest rate related instrument or a commodity.

Simplified approach: capital requirements	Table 11
Position	Treatment
Long cash and long put or short cash and long call	The capital requirement will be the market value of the underlying security[31] multiplied by the sum of specific and general market risk charges[32] for the underlying less the amount the option is in the money (if any) bounded at zero[33]
Long call or long put	The capital requirement will be the lesser of: (i) the market value of the underlying security multiplied by the sum of specific and general market risk charges[32] for the underlying and (ii) the market value of the option[34]

Footnotes

[31] *In some cases such as FX, it may be unclear which side is the underlying security; this should be taken to be the asset that would be received if the option were exercised. In addition, the nominal value should be used for items where the market value of the underlying instrument could be zero, eg caps and floors, swaptions etc.*

[32] *Some options (eg where the underlying is an interest rate, a currency or a commodity) bear no specific risk but specific risk will be present in the case of options on certain interest rate related instruments (eg options on a corporate debt security or corporate bond index; see MAR40.3 to MAR40.40 for the relevant capital requirements) and for options on equities and stock indices (see MAR40.41 to MAR40.52). The charge under this measure for currency options will be 8% and for options on commodities 15%.*

[33] *For options with a residual maturity of more than six months, the strike price should be compared with the forward, not current, price. A bank unable to do this must take the in the money amount to be zero.*

[34] *Where the position does not fall within the trading book (ie options on certain FX or commodities positions not belonging to the trading book), it may be acceptable to use the book value instead.*

Delta-plus method

40.77 Banks that write options will be allowed to include delta-weighted options positions within the simplified standardised approach set out in MAR40.3 to MAR40.73. Such options should be reported as a position equal to the market value of the underlying multiplied by the delta. However, since delta does not sufficiently cover the risks associated with options positions, banks will also be required to measure gamma (which measures the rate of change of delta) and vega (which measures the sensitivity of the value of an option with respect to a change in volatility) sensitivities in order to calculate the total capital requirement. These sensitivities will be calculated according to an approved exchange model or to the bank's proprietary options pricing model subject to oversight by the national authority.[35]

Footnotes

35 *National authorities may wish to require banks doing business in certain classes of exotic options (eg barriers, digitals) or in options at the money that are close to expiry to use either the scenario approach or the internal models alternative, both of which can accommodate more detailed revaluation approaches.*

40.78 Delta-weighted positions with debt securities or interest rates as the underlying will be slotted into the interest rate time bands, as set out in MAR40.3 to MAR40.40, under the following procedure. A two-legged approach should be used as for other derivatives, requiring one entry at the time the underlying contract takes effect and a second at the time the underlying contract matures. For instance, a bought call option on a June three-month interest-rate future will in April be considered, on the basis of its delta-equivalent value, to be a long position with a five-month maturity and a short position with a two-month maturity.[36] The written option will be similarly slotted as a long position with a two-month maturity and a short position with a five-month maturity. Floating rate instruments with caps or floors will be treated as a combination of floating rate securities and a series of European-style options. For example, the holder of a three-year floating rate bond indexed to six month LIBOR with a cap of 15% will treat it as:

(1) a debt security that reprices in six months; and

(2) a series of five written call options on an FRA with a reference rate of 15%, each with a negative sign at the time the underlying FRA takes effect and a positive sign at the time the underlying FRA matures.[37]

Footnotes

36 *A two-month call option on a bond future where delivery of the bond takes place in September would be considered in April as being long the bond and short a five-month deposit, both positions being delta-weighted.*

37 *The rules applying to closely matched positions set out in MAR40.36 will also apply in this respect.*

40.79 The capital requirement for options with equities as the underlying will also be based on the delta-weighted positions that will be incorporated in the measure of equity risk described in MAR40.41 to MAR40.52. For purposes of this calculation each national market is to be treated as a separate underlying. The capital requirement for options on FX and gold positions will be based on the method for FX rate risk as set out in MAR40.53 to MAR40.62. For delta risk, the net delta-based equivalent of the foreign currency and gold options will be incorporated into the measurement of the exposure for the respective currency (or gold) position. The capital requirement for options on commodities will be based on the simplified or the maturity ladder approach for commodities risk as set out in MAR40.63 to MAR40.73. The delta-weighted positions will be incorporated in one of the measures described in that section.

40.80 In addition to the above capital requirements arising from delta risk, there are further capital requirements for gamma and vega risk. Banks using the delta-plus method will be required to calculate the gamma and vega for each option position (including hedge positions) separately. The capital requirements should be calculated in the following way:

(1) For each individual option a gamma impact should be calculated according to a Taylor series expansion as follows, where VU is the variation of the underlying of the option.

$$Gamma\,impact = \frac{1}{2} \times Gamma \times VU^2$$

(2) VU is calculated as follows:

(a) For interest rate options if the underlying is a bond, the market value of the underlying should be multiplied by the risk weights set out in MAR40.26. An equivalent calculation should be carried out where the underlying is an interest rate, again based on the assumed changes in the corresponding yield in MAR40.26.

(b) For options on equities and equity indices: the market value of the underlying should be multiplied by 8%.[38]

(c) For FX and gold options: the market value of the underlying should be multiplied by 8%.

(d) For options on commodities: the market value of the underlying should be multiplied by 15%.

(3) For the purpose of this calculation the following positions should be treated as the same underlying:

(a) for interest rates,[39] each time band as set out in MAR40.26;[40]

(b) for equities and stock indices, each national market;

(c) for foreign currencies and gold, each currency pair and gold; and

(d) for commodities, each individual commodity as defined in MAR40.67.

(4) Each option on the same underlying will have a gamma impact that is either positive or negative. These individual gamma impacts will be summed, resulting in a net gamma impact for each underlying that is either positive or negative. Only those net gamma impacts that are negative will be included in the capital requirement calculation.

(5) The total gamma risk capital requirement will be the sum of the absolute value of the net negative gamma impacts as calculated above.

(6) For volatility risk, banks will be required to calculate the capital requirements by multiplying the sum of the vega risks for all options on the same underlying, as defined above, by a proportional shift in volatility of ± 25%.

(7) The total capital requirement for vega risk will be the sum of the absolute value of the individual capital requirements that have been calculated for vega risk.

Footnotes

38 *The basic rules set out here for interest rate and equity options do not attempt to capture specific risk when calculating gamma capital requirements. However, national authorities may wish to require specific banks to do so.*

39 *Positions have to be slotted into separate maturity ladders by currency.*

40 *Banks using the duration method should use the time bands as set out in MAR40.29.*

Scenario approach

40.81 More sophisticated banks may opt to base the market risk capital requirement for options portfolios and associated hedging positions on scenario matrix analysis. This will be accomplished by specifying a fixed range of changes in the option portfolio's risk factors and calculating changes in the value of the option portfolio at various points along this grid. For the purpose of calculating the capital requirement, the bank will revalue the option portfolio using matrices for simultaneous changes in the option's underlying rate or price and in the volatility of that rate or price. A different matrix will be set up for each individual underlying as defined in MAR40.80 above. As an alternative, at the discretion of each national authority, banks that are significant traders in options will for interest rate options be permitted to base the calculation on a minimum of six sets of time bands.

When using this method, not more than three of the time bands as defined in MAR40.26 and MAR40.29 should be combined into any one set.

40.82 The options and related hedging positions will be evaluated over a specified range above and below the current value of the underlying. The range for interest rates is consistent with the assumed changes in yield in MAR40.26. Those banks using the alternative method for interest rate options set out in MAR40.81 above should use, for each set of time bands, the highest of the assumed changes in yield applicable to the group to which the time bands belong.[41] The other ranges are ± 8% for equities,[42] ± 8% for FX and gold, and ± 15% for commodities. For all risk categories, at least seven observations (including the current observation) should be used to divide the range into equally spaced intervals.

Footnotes

41 *If, for example, the time bands 3 to 4 years, 4 to 5 years and 5 to 7 years are combined the highest assumed change in yield of these three bands would be 0.75.*

42 *The basic rules set out here for interest rate and equity options do not attempt to capture specific risk when calculating gamma capital requirements. However, national authorities may wish to require specific banks to do so.*

40.83 The second dimension of the matrix entails a change in the volatility of the underlying rate or price. A single change in the volatility of the underlying rate or price equal to a shift in volatility of + 25% and - 25% is expected to be sufficient in most cases. As circumstances warrant, however, the supervisory authority may choose to require that a different change in volatility be used and/or that intermediate points on the grid be calculated.

40.84 After calculating the matrix, each cell contains the net profit or loss of the option and the underlying hedge instrument. The capital requirement for each underlying will then be calculated as the largest loss contained in the matrix.

40.85 The application of the scenario analysis by any specific bank will be subject to supervisory consent, particularly as regards the precise way that the analysis is constructed. Banks' use of scenario analysis as part of the simplified standardised approach will also be subject to validation by the national authority, and to those of the qualitative standards for internal models as set out in MAR30.

40.86 Besides the options risks mentioned above, the Committee is conscious of the other risks also associated with options, eg rho (rate of change of the value of the option with respect to the interest rate) and theta (rate of change of the value of the option with respect to time). While not proposing a measurement system for those risks at present, it expects banks undertaking significant options business at the very least to monitor such risks closely. Additionally, banks will be permitted to incorporate rho into their capital calculations for interest rate risk, if they wish to do so.

MAR50

Credit valuation adjustment framework

Changes to give effect to the target revisions to the CVA framework published on the 8 July 2020.

Version effective as of 01 Jan 2023

Changes to give effect to the target revisions to the CVA framework published on the 8 July 2020.

Definitions and application

50.1 The risk-weighted assets for credit value adjustment risk are determined by multiplying the capital requirements calculated as set out in this chapter by 12.5.

50.2 In the context of this document, CVA stands for credit valuation adjustment specified at a counterparty level. CVA reflects the adjustment of default risk-free prices of derivatives and securities financing transactions (SFTs) due to a potential default of the counterparty.

50.3 Unless explicitly specified otherwise, the term CVA in this document means regulatory CVA. Regulatory CVA may differ from CVA used for accounting purposes as follows:

(1) regulatory CVA excludes the effect of the bank's own default; and

(2) several constraints reflecting best practice in accounting CVA are imposed on calculations of regulatory CVA. .

50.4 CVA risk is defined as the risk of losses arising from changing CVA values in response to changes in counterparty credit spreads and market risk factors that drive prices of derivative transactions and SFTs.

50.5 The capital requirements for CVA risk must be calculated by all banks involved in covered transactions in both banking book and trading book. Covered transactions include:

(1) all derivatives except those transacted directly with a qualified central counterparty and except those transactions meeting the conditions of CRE54.14 to CRE54.16; and

(2) SFTs that are fair-valued by a bank for accounting purposes, if their supervisor determines that the bank's CVA loss exposures arising from SFT transactions are material. In case the bank deems the exposures immaterial, the bank must justify its assessment to its supervisory by providing relevant supporting documentation.

FAQ

FAQ1 *Are SFTs for which the accounting amount of CVA reserves is determined to be zero included in the scope of "SFTs that are fair-valued by a bank for accounting purposes"?*

For the purpose of CVA capital requirement, SFTs that are fair-valued for accounting purposes and for which a bank records zero for CVA reserves for accounting purposes are included in the scope of covered transactions if the CVA risk of those SFTs is deemed material as described in MAR50.5 (2).

50.6 The CVA risk capital requirements are calculated for a bank's "CVA portfolio" on a standalone basis. The CVA portfolio includes CVA for a bank's entire portfolio of covered transactions and eligible CVA hedges.

50.7 Two approaches are available for calculating CVA capital requirements: the standardised approach (SA-CVA) and the basic approach (BA-CVA). Banks must use the BA-CVA unless they receive approval from their relevant supervisory authority to use the SA-CVA.[1]

Footnotes

[1] *Note that this is in contrast to the application of the market risk approaches set out in MAR11, where banks do not need supervisory approval to use the standardised approach.*

50.8 Banks that have received approval of their supervisory authority to use the SA-CVA may carve out from the SA-CVA calculations any number of netting sets. CVA capital

requirements for all carved-out netting sets must be calculated using the BA-CVA. When applying the carve-out, a legal netting set may also be split into two synthetic netting sets, one containing the carved-out transactions subject to the BA-CVA and the other subject to the SA-CVA, subject to one or both of the following conditions:

(1) the split is consistent with the treatment of the legal netting set used by the bank for calculating accounting CVA (eg where certain transactions are not processed by the front office/accounting exposure model); or

(2) supervisory approval to use the SA-CVA is limited and does not cover all transactions within a legal netting set.

50.9 Banks that are below the materiality threshold specified in MAR50.9(1) may opt not to calculate its CVA capital requirements using the SA-CVA or BA-CVA and instead choose an alternative treatment.

(1) Any bank whose aggregate notional amount of non-centrally cleared derivatives is less than or equal to 100 billion euro is deemed as being below the materiality threshold.

(2) Any bank below the materiality threshold may choose to set its CVA capital requirement equal to 100% of the bank's capital requirement for counterparty credit risk (CCR).

(3) CVA hedges are not recognised under this treatment.

(4) If chosen, this treatment must be applied to the bank's entire portfolio instead of the BA-CVA or the SA-CVA.

(5) A bank's relevant supervisory authority, however, can remove this option if it determines that CVA risk resulting from the bank's derivative positions materially contributes to the bank's overall risk.

50.10 Eligibility criteria for CVA hedges are specified in MAR50.17 to MAR50.19 for the BA-CVA and in MAR50.37 to MAR50.39 for the SA-CVA.

50.11 CVA hedging instruments can be external (ie with an external counterparty) or internal (ie with one of the bank's trading desks).

(1) All external CVA hedges (including both eligible and ineligible external CVA hedges) that are covered transactions must be included in the CVA calculation of the counterparty providing to the hedge.

(2) All eligible external CVA hedges must be excluded from a bank's market risk capital requirement calculations under MAR10 through MAR40.

(3) Ineligible external CVA hedges are treated as trading book instruments and are capitalised under MAR10 through MAR40.

(4) An internal CVA hedge involves two perfectly offsetting positions: one of the CVA desk and the opposite position of the trading desk:

(a) If an internal CVA hedge is ineligible, both positions belong to the trading book where they cancel each other, so there is no impact on either the CVA portfolio or the trading book.

(b) If an internal CVA hedge is eligible, the CVA desk's position is part of the CVA portfolio where it is capitalised as set out in this chapter, while the trading desk's position is part of the trading book where it is capitalised as set out in MAR10 through MAR40.

(5) If an internal CVA hedge involves an instrument that is subject to curvature risk, default risk charge or the residual risk add-on under the standardised approach as set out in

MAR20 to MAR23, it can be eligible only if the trading desk that is the CVA desk's internal counterparty executes a transaction with an external counterparty that exactly offsets the trading desk's position with the CVA desk.

50.12 Banks that use the BA-CVA or the SA-CVA for calculating CVA capital requirements may cap the maturity adjustment factor at 1 for all netting sets contributing to CVA capital requirements when they calculate CCR capital requirements under the Internal Ratings Based (IRB) approach.

Basic approach for credit valuation adjustment risk

50.13 The BA-CVA calculations may be performed either via the reduced version or the full version. A bank under the BA-CVA approach can choose whether to implement the full version or the reduced version at its discretion. However, all banks using the BA-CVA must calculate the reduced version of BA-CVA capital requirements as the reduced BA-CVA is also part of the full BA-CVA capital calculations as a conservative means to limit hedging recognition.

(1) The full version recognises counterparty credit spread hedges and is intended for banks that hedge CVA risk.

(2) The reduced version eliminates the element of hedging recognition from the full version. The reduced version is designed to simplify BA-CVA implementation for less sophisticated banks that do not hedge CVA.

Reduced version of the BA-CVA (hedges are not recognised)

50.14 The capital requirements for CVA risk under the reduced version of the BA-CVA (DS × K, where the discount scalar DS = 0.65) are calculated as follows (where the summations are taken over all counterparties that are within scope of the CVA charge), where:

(1) SCVAc is the CVA capital requirement that counterparty c would receive if considered on a stand-alone basis (referred to as "stand-alone CVA capital" below). See MAR50.15 for its calculation;

(2) ρ = 50%. It is the supervisory correlation parameter. Its square, ρ2=25%, represents the correlation between credit spreads of any two counterparties.[2] In the formula below, the effect of ρ is to recognise the fact that the CVA risk to which a bank is exposed is less than the sum of the CVA risk for each counterparty, given that the credit spreads of counterparties are typically not perfectly correlated; and

(3) The first term under the square root in the formula below aggregates the systematic components of CVA risk, and the second term under the square root aggregates the idiosyncratic components of CVA risk.

$$K_{reduced} = \sqrt{\left(\rho \cdot \sum_C SCVA_C\right)^2 + \left(1-\rho^2\right) \cdot \sum_C SCVA_C^2}$$

Footnotes

[2] *One of the basic assumptions underlying the BA-CVA is that systematic credit spread risk is driven by a single factor. Under this assumption, ρ can be interpreted as the correlation between the credit spread of a counterparty and the single credit spread systematic factor.*

50.15 The stand-alone CVA capital requirements for counterparty c that are used in the formula in MAR50.14 (SCVA) are calculated as follows (where the summation is across all netting

sets with the counterparty), where:

(1) RWc is the risk weight for counterparty c that reflects the volatility of its credit spread. These risk weights are based on a combination of sector and credit quality of the counterparty as prescribed in MAR50.16.

(2) MNS is the effective maturity for the netting set NS. For banks that have supervisory approval to use the IMM, MNS is calculated as per CRE53.20 and CRE53.21, with the exception that the five year cap in CRE53.20 is not applied. For banks that do not have supervisory approval to use the IMM, MNS is calculated according to CRE32.46 to CRE32.54, with the exception that the five-year cap in CRE32.46 is not applied.

(3) EADNS is the exposure at default (EAD) of the netting set NS, calculated in the same way as the bank calculates it for minimum capital requirements for CCR.

(4) DFNS is a supervisory discount factor. It is 1 for banks using the IMM to calculate EAD, and is

$$\frac{1-e^{-0.05\cdot M_{NS}}}{0.05\cdot M_{NS}}$$

for banks not using the IMM.

(5) α = 1.4.[4]

$$SCVA_C = \frac{1}{\alpha}\cdot RW_C \cdot \sum_{NS} M_{NS}\cdot EAD_{NS}\cdot DF_{NS}$$

Footnotes

[3] *DF is the supervisory discount factor averaged over time between today and the netting set's effective maturity date. The interest rate used for discounting is set at 5%, hence 0.05 in the formula. The product of EAD and effective maturity in the BA-CVA formula is a proxy for the area under the discounted expected exposure profile of the netting set. The IMM definition of effective maturity already includes this discount factor, hence DF is set to 1 for IMM banks. Outside IMM, the netting set's effective maturity is defined as an average of actual trade maturities. This definition lacks discounting, so the supervisory discount factor is added to compensate for this.*

[4] *α is the multiplier used to convert Effective expected positive exposure (EEPE) to EAD in both SA-CCR and IMM. Its role in the calculation, therefore, is to convert the EAD of the netting set (EAD) back to EEPE.*

50.16 The supervisory risk weights (RW) are given in Table 1. Credit quality is specified as either investment grade (IG), high yield (HY), or not rated (NR). Where there are no external ratings or where external ratings are not recognised within a jurisdiction, banks may, subject to supervisory approval, map the internal rating to an external rating and assign a risk weight corresponding to either IG or HY. Otherwise, the risk weights corresponding to NR is to be applied.

Supervisory risk weights, RW_C	Table 1	
Sector of counterparty	Credit quality of counterparty	
	IG	HY and NR
Sovereigns including central banks and multilateral development banks	0.5%	2.0%
Local government, government-backed non-financials, education and public administration	1.0%	4.0%

Financials including government-backed financials	5.0%	12.0%
Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying	3.0%	7.0%
Consumer goods and services, transportation and storage, administrative and support service activities	3.0%	8.5%
Technology, telecommunications	2.0%	5.5%
Health care, utilities, professional and technical activities	1.5%	5.0%
Other sector	5.0%	12.0%

Full version of the BA-CVA (hedges are recognised)

50.17 As set out in MAR50.13(1) the full version of the BA-CVA recognises the effect of counterparty credit spread hedges. Only transactions used for the purpose of mitigating the counterparty credit spread component of CVA risk, and managed as such, can be eligible hedges.

50.18 Only single-name credit default swaps (CDS), single-name contingent CDS and index CDS can be eligible CVA hedges.

50.19 Eligible single-name credit instruments must:

(1) reference the counterparty directly; or

(2) reference an entity legally related to the counterparty, where legally related refers to cases where the reference name and the counterparty are either a parent and its subsidiary or two subsidiaries of a common parent; or

(3) reference an entity that belongs to the same sector and region as the counterparty.

50.20 Banks that intend to use the full version of BA-CVA must calculate the reduced version (K) as well. Under the full version, capital requirements for CVA risk DS × K is calculated as follows, where DS =0.65, and β=0.25 is the supervisory parameter that is used to provide a floor that limits the extent to which hedging can reduce the capital requirements for CVA risk:

$$K_{full} = \beta \cdot K_{reduced} + \left(1 - \beta\right) \cdot K_{hedged}$$

50.21 The part of capital requirements that recognises eligible hedges (K) is calculated as follows (where the summations are taken over all counterparties c that are within scope of the CVA charge), where:

(1) Both the stand-alone CVA capital (SCVAC) and the correlation parameter (ρ) are defined in exactly the same way as for the reduced version calculation of the BA-CVA.

(2) SNHC is a quantity that gives recognition to the reduction in CVA risk of the counterparty c arising from the bank's use of single-name hedges of credit spread risk. See MAR50.23 for its calculation.

(3) IH is a quantity that gives recognition to the reduction in CVA risk across all counterparties arising from the bank's use of index hedges. See MAR50.24 for its calculation.

(4) HMAC is a quantity characterising hedging misalignment, which is designed to limit the extent to which indirect hedges can reduce capital requirements given that they will

not fully offset movements in a counterparty's credit spread. That is, with indirect hedges present, Khedged cannot reach zero. See MAR50.25 for its calculation.

$$K_{hedged} = \sqrt{\left(\rho \cdot \sum_C \left(SCVA_C - SNH_C\right) - IH\right)^2 + \left(1-\rho^2\right) \cdot \sum_C \left(SCVA_C - SNH_C\right)^2 + \sum_C HMA_C}$$

50.22 The formula for K in MAR50.21 comprises three main terms as below:

(1) The first term

$(\rho \cdot \sum_c (SCVA_c - SNH_c) - IH)^2$

aggregates the systematic components of CVA risk arising from the bank's counterparties, the single-name hedges and the index hedges.

(2) The second term

$(1-\rho^2) \cdot \sum_c (SCVA_c - SNH_c)^2$

aggregates the idiosyncratic components of CVA risk arising from the bank's counterparties and the single-name hedges.

(3) The third term

$\sum_c HMA_c$

aggregates the components of indirect hedges that are not aligned with counterparties' credit spreads.

50.23 The quantity SNH is calculated as follows (where the summation is across all single name hedges h that the bank has taken out to hedge the CVA risk of counterparty c), where:

(1) rhc is the supervisory prescribed correlation between the credit spread of counterparty c and the credit spread of a single-name hedge h of counterparty c. The value of rhc is set out in the Table 2 of MAR50.26. It is set at 100% if the hedge directly references the counterparty c, and set at lower values if it does not.

(2)

M_h^{SN}

is the remaining maturity of single-name hedge h.

(3)

B_h^{SN}

is the notional of single-name hedge h. For single-name contingent CDS, the notional is determined by the current market value of the reference portfolio or instrument.

(4)

DF_h^{SN}

is the supervisory discount factor calculated as .

$$\frac{1-e^{-0.05 \cdot M_h^{SN}}}{0.05 \cdot M_h^{SN}}$$

(5) RWh is the supervisory risk weight of single-name hedge h that reflects the volatility of the credit spread of the reference name of the hedging instrument. These risk weights are based on a combination of the sector and the credit quality of the reference name of the hedging instrument as prescribed in Table 1 of MAR50.16.

$$SNH_c = \sum_{h \in c} r_{hc} \cdot RW_h \cdot M_h^{SN} \cdot B_h^{SN} \cdot DF_h^{SN}$$

50.24 The quantity IH is calculated as follows (where the summation is across all index hedges i that the bank has taken out to hedge CVA risk), where:

(1)

M_i^{ind}

is the remaining maturity of index hedge i.

(2)

B_i^{ind}

is the notional of the index hedge i.

(3)

DF_i^{ind}

is the supervisory discount factor calculated as

$$\frac{1-e^{-0.05\cdot M_i^{ind}}}{0.05\cdot M_i^{ind}}$$

(4) RWi is the supervisory risk weight of the index hedge i. RWi is taken from the Table 1 of MAR50.16 based on the sector and the credit quality of the index constituents and adjusted as follows:

(a) For an index where all index constituents belong to the same sector and are of the same credit quality, the relevant value in the Table 1 of MAR50.16 is multiplied by 0.7 to account for diversification of idiosyncratic risk within the index.

(b) For an index spanning multiple sectors or with a mixture of investment grade constituents and other grade constituents, the name-weighted average of the risk weights from the Table 1 of MAR50.16 should be calculated and then multiplied by 0.7.

$$IH = \sum_i RW_i \cdot M_i^{ind} \cdot B_i^{ind} \cdot DF_i^{ind}$$

50.25 The quantity HMA is calculated as follows (where the summation is across all single name hedges h that have been taken out to hedge the CVA risk of counterparty c), where and RW have the same definitions as set out in MAR50.23.

r_{hc}, M_h^{SN}, B_h^{SN}, DF_h^{SN}

$$HMA_c = \sum_{h\in c}\left(1-r_{hc}^2\right)\cdot\left(RW_h \cdot M_h^{SN} \cdot B_h^{SN} \cdot DF_h^{SN}\right)^2$$

50.26 The supervisory prescribed correlations between the credit spread of counterparty and the credit spread of its single-name hedge are set in Table 2 as follows:

Correlations between credit spread of counterparty and single-name hedge	Table 2
Single-name hedge *h* of counterparty *c*	Value of r_{hc}
references counterparty *c* directly	100%
has legal relation with counterparty *c*	80%
shares sector and region with counterparty *c*	50%

Standardised approach for credit valuation adjustment risk

50.27 The SA-CVA is an adaptation of the standardised approach for market risk set out in MAR20 to MAR23. The primary differences of the SA-CVA from the standardised approach for market risk are:

(1) the SA-CVA features a reduced granularity of market risk factors; and

(2) the SA-CVA does not include default risk and curvature risk.

50.28 Under the SA-CVA, capital requirements must be calculated and reported to supervisors at the same monthly frequency as for the market risk standardised approach. In addition, banks using the SA-CVA must have the ability to produce SA-CVA capital requirement calculations at the request of their supervisors and must accordingly provide the calculations.

50.29 The SA-CVA uses as inputs the sensitivities of regulatory CVA to counterparty credit spreads and market risk factors driving the values of covered transactions. Sensitivities must be computed by banks in accordance with the prudent valuation standards set out in CAP50.

50.30 For a bank to be considered eligible for the use of SA-CVA by its relevant supervisor as set out in MAR50.7, the bank must meet the following criteria at the minimum.

(1) A bank must be able to model exposure and calculate, on at least a monthly basis, CVA and CVA sensitivities to the market risk factors specified in MAR50.54 to MAR50.77.

(2) A bank must have a CVA desk (or a similar dedicated function) responsible for risk management and hedging of CVA.

Regulatory CVA calculations

50.31 A bank must calculate regulatory CVA for each counterparty with which it has at least one covered position for the purpose of the CVA risk capital requirements.

50.32 Regulatory CVA at a counterparty level must be calculated according to the following principles. A bank must demonstrate its compliance to the principles to its relevant supervisor.

(1) Regulatory CVA must be calculated as the expectation of future losses resulting from default of the counterparty under the assumption that the bank itself is free from the default risk. In expressing the regulatory CVA, non-zero losses must have a positive sign. This is reflected in MAR50.52 where

WS_k^{Hdg} must be subtracted from WS_k^{CVA}.

(2) The calculation must be based on at least the following three sets of inputs:

(a) term structure of market-implied probability of default (PD);

(b) market-consensus expected loss-given-default (ELGD);

(c) simulated paths of discounted future exposure.

(3) The term structure of market-implied PD must be estimated from credit spreads observed in the markets. For counterparties whose credit is not actively traded (ie illiquid counterparties), the market-implied PD must be estimated from proxy credit spreads estimated for these counterparties according to the following requirements:

(a) A bank must estimate the credit spread curves of illiquid counterparties from credit spreads observed in the markets of the counterparty's liquid peers via an algorithm

that discriminates on at least the following three variables: a measure of credit quality (eg rating), industry, and region.

(b) In certain cases, mapping an illiquid counterparty to a single liquid reference name can be allowed. A typical example would be mapping a municipality to its home country (ie setting the municipality credit spread equal to the sovereign credit spread plus a premium). A bank must justify to its supervisor each case of mapping an illiquid counterparty to a single liquid reference name.

(c) When no credit spreads of any of the counterparty's peers is available due to the counterparty's specific type (eg project finance, funds), a bank is allowed to use a more fundamental analysis of credit risk to proxy the spread of an illiquid counterparty. However, where historical PDs are used as part of this assessment, the resulting spread cannot be based on historical PD only - it must relate to credit markets.

(4) The market-consensus ELGD value must be the same as the one used to calculate the risk-neutral PD from credit spreads unless the bank can demonstrate that the seniority of the exposure resulting from covered positions differs from the seniority of senior unsecured bonds. Collateral provided by the counterparty does not change the seniority of the exposure.

(5) The simulated paths of discounted future exposure are produced by pricing all derivative transactions with the counterparty along simulated paths of relevant market risk factors and discounting the prices to today using risk-free interest rates along the path.

(6) All market risk factors material for the transactions with a counterparty must be simulated as stochastic processes for an appropriate number of paths defined on an appropriate set of future time points extending to the maturity of the longest transaction.

(7) For transactions with a significant level of dependence between exposure and the counterparty's credit quality, this dependence should be taken into account.

(8) For margined counterparties, collateral is permitted to be recognised as a risk mitigant under the following conditions:

(a) Collateral management requirements outlined in CRE53.39 and CRE53.40 are satisfied.

(b) All documentation used in collateralised transactions must be binding on all parties and legally enforceable in all relevant jurisdictions. Banks must have conducted sufficient legal review to verify this and have a well founded legal basis to reach this conclusion, and undertake such further review as necessary to ensure continuing enforceability.

(9) For margined counterparties, the simulated paths of discounted future exposure must capture the effects of margining collateral that is recognised as a risk mitigant along each exposure path. All the relevant contractual features such as the nature of the margin agreement (unilateral vs bilateral), the frequency of margin calls, the type of collateral, thresholds, independent amounts, initial margins and minimum transfer amounts must be appropriately captured by the exposure model. To determine collateral available to a bank at a given exposure measurement time point, the exposure model must assume that the counterparty will not post or return any collateral within a certain time period immediately prior to that time point. The assumed value of this time period, known as the margin period of risk (MPoR), cannot be less than a supervisory floor. For SFTs and client cleared transactions as specified in

CRE54.12, the supervisory floor for the MPoR is equal to 4+N business days, where N is the re-margining period specified in the margin agreement (in particular, for margin agreements with daily or intra-daily exchange of margin, the minimum MPoR is 5 business days). For all other transactions, the supervisory floor for the MPoR is equal to 9+N business days.

50.33 The simulated paths of discounted future exposure are obtained via the exposure models used by a bank for calculating front office/accounting CVA, adjusted (if needed) to meet the requirements imposed for regulatory CVA calculation. Model calibration process (with the exception of the MPoR), market and transaction data used for regulatory CVA calculation must be the same as the ones used for accounting CVA calculation.

50.34 The generation of market risk factor paths underlying the exposure models must satisfy and a bank must demonstrate to its relevant supervisors its compliance to the following requirements:

(1) Drifts of risk factors must be consistent with a risk-neutral probability measure. Historical calibration of drifts is not allowed.

(2) The volatilities and correlations of market risk factors must be calibrated to market data whenever sufficient data exist in a given market. Otherwise, historical calibration is permissible.

(3) The distribution of modelled risk factors must account for the possible non-normality of the distribution of exposures, including the existence of leptokurtosis (“fat tails”), where appropriate.

50.35 Netting recognition is the same as in the accounting CVA calculations used by the bank. In particular, netting uncertainty can be modelled.

50.36 A bank must satisfy and demonstrate to its relevant supervisors its compliance to the following requirements:

(1) Exposure models used for calculating regulatory CVA must be part of a CVA risk management framework that includes the identification, measurement, management, approval and internal reporting of CVA risk. A bank must have a credible track record in using these exposure models for calculating CVA and CVA sensitivities to market risk factors.

(2) Senior management should be actively involved in the risk control process and must regard CVA risk control as an essential aspect of the business to which significant resources need to be devoted.

(3) A bank must have a process in place for ensuring compliance with a documented set of internal policies, controls and procedures concerning the operation of the exposure system used for accounting CVA calculations.

(4) A bank must have an independent control unit that is responsible for the effective initial and ongoing validation of the exposure models. This unit must be independent from business credit and trading units (including the CVA desk), must be adequately staffed and must report directly to senior management of the bank.

(5) A bank must document the process for initial and ongoing validation of its exposure models to a level of detail that would enable a third party to understand how the models operate, their limitations, and their key assumptions; and recreate the analysis. This documentation must set out the minimum frequency with which ongoing validation will be conducted as well as other circumstances (such as a sudden change in market behaviour) under which additional validation should be conducted. In

addition, the documentation must describe how the validation is conducted with respect to data flows and portfolios, what analyses are used and how representative counterparty portfolios are constructed.

(6) The pricing models used to calculate exposure for a given path of market risk factors must be tested against appropriate independent benchmarks for a wide range of market states as part of the initial and ongoing model validation process. Pricing models for options must account for the non-linearity of option value with respect to market risk factors.

(7) An independent review of the overall CVA risk management process should be carried out regularly in the bank's own internal auditing process. This review should include both the activities of the CVA desk and of the independent risk control unit.

(8) A bank must define criteria on which to assess the exposure models and their inputs and have a written policy in place to describe the process to assess the performance of exposure models and remedy unacceptable performance.

(9) Exposure models must capture transaction-specific information in order to aggregate exposures at the level of the netting set. A bank must verify that transactions are assigned to the appropriate netting set within the model.

(10) Exposure models must reflect transaction terms and specifications in a timely, complete, and conservative fashion. The terms and specifications must reside in a secure database that is subject to formal and periodic audit. The transmission of transaction terms and specifications data to the exposure model must also be subject to internal audit, and formal reconciliation processes must be in place between the internal model and source data systems to verify on an ongoing basis that transaction terms and specifications are being reflected in the exposure system correctly or at least conservatively.

(11) The current and historical market data must be acquired independently of the lines of business and be compliant with accounting. They must be fed into the exposure models in a timely and complete fashion, and maintained in a secure database subject to formal and periodic audit. A bank must also have a well-developed data integrity process to handle the data of erroneous and/or anomalous observations. In the case where an exposure model relies on proxy market data, a bank must set internal policies to identify suitable proxies and the bank must demonstrate empirically on an ongoing basis that the proxy provides a conservative representation of the underlying risk under adverse market conditions.

Eligible hedges

50.37 Only whole transactions that are used for the purpose of mitigating CVA risk, and managed as such, can be eligible hedges. Transactions cannot be split into several effective transactions.

50.38 Eligible hedges can include:

(1) instruments that hedge variability of the counterparty credit spread; and

(2) instruments that hedge variability of the exposure component of CVA risk.

50.39 Instruments that are not eligible for the internal models approach for market risk under MAR30 to MAR33 (eg tranched credit derivatives) cannot be eligible CVA hedges.

Multiplier

50.40 Aggregated capital requirements can be scaled up by the multiplier m.

50.41 The multiplier m is set at 1. A bank's relevant supervisor may require a bank to use a higher value of m if the supervisor determines that the bank's CVA model risk warrants it (eg if the level of model risk for the calculation of CVA sensitivities is too high or the dependence between the bank's exposure to a counterparty and the counterparty's credit quality is not appropriately taken into account in its CVA calculations).

Calculations

50.42 The SA-CVA capital requirements are calculated as the sum of the capital requirements for delta and vega risks calculated for the entire CVA portfolio (including eligible hedges).

50.43 The capital requirements for delta risk are calculated as the simple sum of delta capital requirements calculated independently for the following six risk classes:

(1) interest rate risk;

(2) foreign exchange (FX) risk;

(3) counterparty credit spread risk;

(4) reference credit spread risk (ie credit spreads that drive the CVA exposure component);

(5) equity risk; and

(6) commodity risk.

50.44 If an instrument is deemed as an eligible hedge for credit spread delta risk, it must be assigned in its entirety (see MAR50.37) either to the counterparty credit spread or to the reference credit spread risk class. Instruments must not be split between the two risk classes.

50.45 The capital requirements for vega risk are calculated as the simple sum of vega capital requirements calculated independently for the following five risk classes. There is no vega capital requirements for counterparty credit spread risk.

(1) interest rate risk;

(2) FX risk;

(3) reference credit spread risk;

(4) equity risk; and

(5) commodity risk.

50.46 Delta and vega capital requirements are calculated in the same manner using the same procedures set out in MAR50.47 to MAR50.53.

50.47 For each risk class, (i) the sensitivity of the aggregate CVA, , and (ii) the sensitivity of the market value of all eligible hedging instruments in the CVA portfolio, , to each risk factor k in the risk class are calculated. The sensitivities are defined as the ratio of the change of the value in question (ie (i) aggregate CVA or (ii) market value of all CVA hedges) caused by a small change of the risk factor's current value to the size of the change. Specific definitions for each risk class are set out in MAR50.54 to MAR50.77. These definitions include specific values of changes or shifts in risk factors. However, a bank may use smaller values of risk factor shifts if doing so is consistent with internal risk management calculations.

S_k^{CVA}

S_k^{Hdg}

FAQ

FAQ1 *Are banks permitted under the SA-CVA to calculate CVA sensitivities via algorithmic techniques such as adjoint algorithmic differentiation (AAD)?*

Yes. A bank may use AAD and similar computational techniques to calculate CVA sensitivities under the SA-CVA if doing so is consistent with the bank's internal risk management calculations and the relevant validation standards described in the SA-CVA framework.

50.48 CVA sensitivities for vega risk are always material and must be calculated regardless of whether or not the portfolio includes options. When CVA sensitivities for vega risk are calculated, the volatility shift must apply to both types of volatilities that appear in exposure models:

(1) volatilities used for generating risk factor paths; and

(2) volatilities used for pricing options.

50.49 If a hedging instrument is an index, its sensitivities to all risk factors upon which the value of the index depends must be calculated. The index sensitivity to risk factor k must be calculated by applying the shift of risk factor k to all index constituents that depend on this risk factor and recalculating the changed value of the index. For example, to calculate delta sensitivity of S&P500 to large financial companies, a bank must apply the relevant shift to equity prices of all large financial companies that are constituents of S&P500 and re-compute the index.

50.50 For the following risk classes, a bank may choose to introduce a set of additional risk factors that directly correspond to qualified credit and equity indices. For delta risks, a credit or equity index is qualified if it satisfies liquidity and diversification conditions specified in MAR21.31; for vega risks, any credit or equity index is qualified. Under this option, a bank must calculate sensitivities of CVA and the eligible CVA hedges to the qualified index risk factors in addition to sensitivities to the non-index risk factors. Under this option, for a covered transaction or an eligible hedging instrument whose underlying is a qualified index, its contribution to sensitivities to the index constituents is replaced with its contribution to a single sensitivity to the underlying index. For example, for a portfolio consisting only of equity derivatives referencing only qualified equity indices, no calculation of CVA sensitivities to non-index equity risk factors is necessary. If more than 75% of constituents of a qualified index (taking into account the weightings of the constituents) are mapped to the same sector, the entire index must be mapped to that sector and treated as a single-name sensitivity in that bucket. In all other cases, the sensitivity must be mapped to the applicable index bucket.

(1) counterparty credit spread risk;

(2) reference credit spread risk; and

(3) equity risk.

50.51 The weighted sensitivities and for each risk factor k are calculated by multiplying the net sensitivities and , respectively, by the corresponding risk weight RW (the risk weights applicable to each risk class are specified in MAR50.54 to MAR50.77).

WS_k^{CVA}

WS_k^{Hdg}

S_k^{CVA}

S_k^{Hdg}

$WS_k^{CVA} = RW_k s_k^{CVA}$

$WS_k^{Hdg} = RW_k s_k^{Hdg}$

50.52 The net weighted sensitivity of the CVA portfolio to risk factor k is obtained by:[5]

$WS_k = WS_k^{CVA} - WS_k^{Hdg}$

Footnotes

[5] *Note that the formula in MAR50.52 is set out under the convention that the CVA is positive as specified in MAR50.32 (1). It intends to recognise the risk reducing effect of hedging. For example, when hedging the counterparty credit spread component of CVA risk for a specific counterparty by buying credit protection on the counterparty: if the counterparty's credit spread widens, the CVA (expressed as a positive value) increases resulting in the positive CVA sensitivity to the counterparty credit spread. At the same time, as the value of the hedge from the bank's perspective increases as well (as credit protection becomes more valuable), the sensitivity of the hedge is also positive. The positive weighted sensitivities of the CVA and its hedge offset each other using the formula with the minus sign. If CVA loss had been expressed as a negative value, the minus sign in MAR50.52 would have been replaced by a plus sign.*

50.53 For each risk class, the net sensitivities are aggregated as follows:

(1) The weighted sensitivities must be aggregated into a capital requirement Kb within each bucket b (the buckets and correlation parameters ρkl applicable to each risk class are specified in MAR50.54 to MAR50.77), where R is the hedging disallowance parameter, set at 0.01, that prevents the possibility of recognising perfect hedging of CVA risk.

$$K_b = \sqrt{\left(\sum_{k\in b} WS_k^2 + \sum_{k\in b}\sum_{l\in b, l\neq k} \rho_{kl} WS_k WS_l\right) + R\cdot\sum_{k\in b}\left(\left(WS_k^{Hdg}\right)^2\right)}$$

(2) Bucket-level capital requirements must then be aggregated across buckets within each risk class (the correlation parameters γbc applicable to each risk class are specified in MAR50.54 to MAR50.77). Note that this equation differs from the corresponding aggregation equation for market risk capital requirements in MAR21.4, including the multiplier mCVA.

$$K = m_{CVA}\sqrt{\sum_b K_b^2 + \sum_b\sum_{b\neq c}\gamma_{bc} S_b S_c}$$

(3) In calculating K in above (2), Sb is defined as the sum of the weighted sensitivities WSk for all risk factors k within bucket b, floored by -Kb and capped by Kb, and the Sc is defined in the same way for all risk factors k in bucket c:

$$S_b = \max\left\{-K_b; \min\left(\sum_{k\in b} WS_k; K_b\right)\right\}$$

$$S_c = \max\left\{-K_c; \min\left(\sum_{k \in c} WS_k; K_c\right)\right\}$$

Interest rate buckets, risk factors, sensitivities, risk weights and correlations

50.54 For interest rate delta and vega risks, buckets must be set per individual currencies.

50.55 For interest rate delta and vega risks, cross-bucket correlation γ is set at 0.5 for all currency pairs.

50.56 The interest rate delta risk factors for a bank's reporting currency and for the following currencies USD, EUR, GBP, AUD, CAD, SEK or JPY:

(1) The interest rate delta risk factors are the absolute changes of the inflation rate and of the risk-free yields for the following five tenors: 1 year, 2 years, 5 years, 10 years and 30 years.

(2) The sensitivities to the abovementioned risk-free yields are measured by changing the risk-free yield for a given tenor for all curves in a given currency by 1 basis point (0.0001 in absolute terms) and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.0001. The sensitivity to the inflation rate is obtained by changing the inflation rate by 1 basis point (0.0001 in absolute terms) and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.0001.

(3) The risk weights RWk are set as follows:

Risk weight for interest rate risk (specified currencies)						Table 3
Risk factor	1 year	2 years	5 years	10 years	30 years	Inflation
Risk weight	1.11%	0.93%	0.74%	0.74%	0.74%	1.11%

(4) The correlations between pairs of risk factors ρkl are set as follows:

Correlations for interest rate risk factors (specified currencies)						Table 4
	1 year	2 years	5 years	10 years	30 years	Inflation
1 year	100%	91%	72%	55%	31%	40%
2 years		100%	87%	72%	45%	40%
5 years			100%	91%	68%	40%
10 years				100%	83%	40%
30 years					100%	40%
Inflation						100%

50.57 The interest rate delta risk factors for other currencies not specified in MAR50.56:

(1) The interest rate risk factors are the absolute change of the inflation rate and the parallel shift of the entire risk-free yield curve for a given currency.

(2) The sensitivity to the yield curve is measured by applying a parallel shift to all risk-free yield curves in a given currency by 1 basis point (0.0001 in absolute terms) and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.0001. The sensitivity to the inflation rate is obtained by changing the inflation rate by 1 basis point (0.0001 in absolute terms) and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.0001.

(3) The risk weights for both the risk-free yield curve and the inflation rate RWk are set at 1.58%.

(4) The correlations between the risk-free yield curve and the inflation rate ρkl are set at 40%.

50.58 The interest rate vega risk factors for all currencies:

(1) The interest rate vega risk factors are a simultaneous relative change of all volatilities for the inflation rate and a simultaneous relative change of all interest rate volatilities for a given currency.

(2) The sensitivity to (i) the interest rate volatilities or (ii) inflation rate volatilities is measured by respectively applying a simultaneous shift to (i) all interest rate volatilities or (ii) inflation rate volatilities by 1% relative to their current values and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.01.

(3) The risk weights for both the interest rate volatilities and the inflation rate volatilities RWk are set to 100%.

(4) Correlations between the interest rate volatilities and the inflation rate volatilities ρkl are set at 40%.

Foreign exchange buckets, risk factors, sensitivities, risk weights and correlations

50.59 For FX delta and vega risks, buckets must be set per individual currencies except for a bank's own reporting currency.

50.60 For FX delta and vega risks, the cross-bucket correlation γ is set at 0.6 for all currency pairs.

50.61 The FX delta risk factors for all currencies:

(1) The single FX delta risk factor is defined as the relative change of the FX spot rate between a given currency and a bank's reporting currency, where the FX spot rate is the current market price of one unit of another currency expressed in the units of the bank's reporting currency.

(2) Sensitivities to FX spot rates are measured by shifting the exchange rate between the bank's reporting currency and another currency (ie the value of one unit of another currency expressed in units of the reporting currency) by 1% relative to its current value and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.01. For transactions that reference an exchange rate between a pair of non-reporting currencies, the sensitivities to the FX spot rates between the bank's reporting currency and each of the referenced non-reporting currencies must be measured.[6]

(3) The risk weights for all exchange rates between the bank's reporting currency and another currency are set at 11%.

Footnotes

[6] *For example, if a EUR-reporting bank holds an instrument that references the USD-GBP exchange rate, the bank must measure CVA sensitivity both to the EUR-GBP exchange rate and to the EUR-USD exchange rate.*

50.62 The FX vega risk factors for all currencies:

(1) The single FX vega risk factor is a simultaneous relative change of all volatilities for an exchange rate between a bank's reporting currency and another given currency.

(2) The sensitivities to the FX volatilities are measured by simultaneously shifting all

volatilities for a given exchange rate between the bank's reporting currency and another currency by 1% relative to their current values and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.01. For transactions that reference an exchange rate between a pair of non-reporting currencies, the volatilities of the FX spot rates between the bank's reporting currency and each of the referenced non-reporting currencies must be measured.

(3) The risk weights for FX volatilities RWk are set to 100%.

Counterparty credit spread buckets, risk factors, sensitivities, risk weights and correlations

50.63 Counterparty credit spread risk is not subject to vega risk capital requirements. Buckets for delta risk are set as follows:

(1) Buckets 1 to 7 are defined for factors that are not qualified indices as set out in MAR50.50;

(2) Bucket 8 is set for the optional treatment of qualified indices. Under the optional treatment, only instruments that reference qualified indices can be assigned to bucket 8, while all single-name and all non-qualified index hedges must be assigned to buckets 1 to 7 for calculations of CVA sensitivities and sensitivities. For any instrument referencing an index assigned to buckets 1 to 7, the look-through approach must be used (ie, sensitivity of the hedge to each index constituent must be calculated).

Buckets for counterparty credit spread delta risk	Table 5
Bucket number	Sector
1	a) Sovereigns including central banks, multilateral development banks
	b) Local government, government-backed non-financials, education and public administration
2	Financials including government-backed financials
3	Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying
4	Consumer goods and services, transportation and storage, administrative and support service activities
5	Technology, telecommunications
6	Health care, utilities, professional and technical activities
7	Other sector
8	Qualified Indices

50.64 For counterparty credit spread delta risk, the cross-bucket correlations γ are set as follows:

Cross-bucket correlations for counterparty credit spread delta risk								Table 6
Bucket	1	2	3	4	5	6	7	8
1	100%	10%	20%	25%	20%	15%	0%	45%
2		100%	5%	15%	20%	5%	0%	45%
3			100%	20%	25%	5%	0%	45%
4				100%	25%	5%	0%	45%
5					100%	5%	0%	45%

6						100%	0%	45%
7							100%	0%
8								100%

50.65 The counterparty credit spread delta risk factors for a given bucket:

(1) The counterparty credit spread delta risk factors are absolute shifts of credit spreads of individual entities (counterparties and reference names for counterparty credit spread hedges) and qualified indices (if the optional treatment is chosen) for the following tenors: 0.5 years, 1 year, 3 years, 5 years and 10 years.

(2) For each entity and each tenor point, the sensitivities are measured by shifting the relevant credit spread by 1 basis point (0.0001 in absolute terms) and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.0001.

(3) The risk weights RWk are set as follows depending on the entity's bucket, where IG, HY and NR represent "investment grade", "high yield" and "not rated" as specified for the BA-CVA in MAR50.16. The same risk weight for a given bucket and given credit quality applies to all tenors.

Risk weights for counterparty credit spread delta risk									Table 7
Bucket	1 a)	1 b)	2	3	4	5	6	7	8
IG names	0.5%	1.0%	5.0%	3.0%	3.0%	2.0%	1.5%	5.0%	1.5%
HY and NR names	2.0%	4.0%	12.0%	7.0%	8.5%	5.5%	5.0%	12.0%	5.0%

(4) For buckets 1 to 7, the correlation parameter ρkl between two weighted sensitivities WSk and WSl is calculated as follows, where:

(a) ρtenor is equal to 100% if the two tenors are the same and 90% otherwise;

(b) ρname is equal to 100% if the two names are the same, 90% if the two names are distinct, but legally related and 50% otherwise;

(c) ρquality is equal to 100% if the credit quality of the two names is the same (ie IG and IG or HY/NR and HY/NR) and 80% otherwise.

$$\rho_{kl} = \rho_{tenor} \cdot \rho_{name} \cdot \rho_{quality}$$

(5) For bucket 8, the correlation parameter ρkl between two weighted sensitivities WSk and WSl is calculated as follows, where:

(a) ρtenor is equal to 100% if the two tenors are the same and 90% otherwise;

(b) ρname is equal to 100% if the two indices are the same and of the same series, 90% if the two indices are the same, but of distinct series, and 80% otherwise;

(c) ρquality is equal to 100% if the credit quality of the two indices is the same (ie IG and IG or HY and HY) and 80% otherwise.

$$\rho_{kl} = \rho_{tenor} \cdot \rho_{name} \cdot \rho_{quality}$$

Reference credit spread buckets, risk factors, sensitivities, risk weights and correlations

50.66 Reference credit spread risk is subject to both delta and vega risk capital requirements.

Buckets for delta and vega risks are set as follows, where IG, HY and NR represent "investment grade", "high yield" and "not rated" as specified for the BA-CVA in MAR50.16:

Buckets for reference credit spread risk		Table 8
Bucket number	Credit quality	Sector
1	IG	Sovereigns including central banks, multilateral development banks
2		Local government, government-backed non-financials, education and public administration
3		Financials including government-backed financials
4		Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying
5		Consumer goods and services, transportation and storage, administrative and support service activities
6		Technology, telecommunications
7		Health care, utilities, professional and technical activities
8	HY and NR	Sovereigns including central banks, multilateral development banks
9		Local government, government-backed non-financials, education and public administration
10		Financials including government-backed financials
11		Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying
12		Consumer goods and services, transportation and storage, administrative and support service activities
13		Technology, telecommunications
14		Health care, utilities, professional and technical activities
15	(Not applicable)	Other sector
16	IG	Qualified Indices
17	HY	Qualified Indices

50.67 For reference credit spread delta and vega risks, the cross-bucket correlations γ are set as follows:

(1) The cross-bucket correlations γ_{bc} between buckets of the same credit quality (ie either IG or HY/NR) are set as follows:

Cross-bucket correlations for reference credit spread risk										Table 9
Bucket	1/8	2/9	3/10	4/11	5/12	6/13	7/14	15	16	17
1/8	100%	75%	10%	20%	25%	20%	15%	0%	45%	45%
2/9		100%	5%	15%	20%	15%	10%	0%	45%	45%
3/10			100%	5%	15%	20%	5%	0%	45%	45%
4/11				100%	20%	25%	5%	0%	45%	45%
5/12					100%	25%	5%	0%	45%	45%
6/13						100%	5%	0%	45%	45%

7/14												100%	0%	45%	45%
15													100%	0%	0%
16														100%	75%
17															100%

(2) For cross-bucket correlations ybc between buckets 1 to 14 of different credit quality (ie IG and HY/NR), the correlations ybc specified in MAR50.67(1) are divided by 2.

50.68 Reference credit spread delta risk factors for a given bucket:

(1) The single reference credit spread delta risk factor is a simultaneous absolute shift of the credit spreads of all tenors for all reference names in the bucket.

(2) The sensitivity to reference credit spread delta risk is measured by simultaneously shifting the credit spreads of all tenors for all reference names in the bucket by 1 basis point (0.0001 in absolute terms) and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.0001.

(3) The risk weights RWk are set as follows depending on the reference name's bucket:

Risk weights for reference credit spread delta risk								Table 10	
IG bucket	1	2	3	4	5	6	7	8	9
Risk weight	0.5%	1.0%	5.0%	3.0%	3.0%	2.0%	1.5%	2.0%	4.0%

HY/NR bucket	10	11	12	13	14	15	16	17	
Risk weight	12.0%	7.0%	8.5%	5.5%	5.0%	12.0%	1.5%	5.0%	

50.69 Reference credit spread vega risk factors for a given bucket:

(1) The single reference credit spread vega risk factor is a simultaneous relative shift of the volatilities of credit spreads of all tenors for all reference names in the bucket.

(2) The sensitivity to the reference credit spread vega risk factor is measured by simultaneously shifting the volatilities of credit spreads of all tenors for all reference names in the bucket by 1% relative to their current values and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.01.

(3) Risk weights for reference credit spread volatilities RWk are set to 100%.

Equity buckets, risk factors, sensitivities, risk weights and correlations

50.70 For equity delta and vega risks, buckets are set as follows, where:

(1) Market capitalisation ("market cap") is defined as the sum of the market capitalisations of the same legal entity or group of legal entities across all stock markets globally. The reference to "group of legal entities" covers cases where the listed entity is a parent company of a group of legal entities. Under no circumstances should the sum of the market capitalisations of multiple related listed entities be used to determine whether a listed entity is "large market cap" or "small market cap".

(2) "Large market cap" is defined as a market capitalisation equal to or greater than USD 2 billion and "small market cap" is defined as a market capitalisation of less than USD 2 billion.

(3) The advanced economies are Canada, the United States, Mexico, the euro area, the non-euro area western European countries (the United Kingdom, Norway, Sweden, Denmark and Switzerland), Japan, Oceania (Australia and New Zealand), Singapore and Hong Kong SAR.

(4) To assign a risk exposure to a sector, banks must rely on a classification that is commonly used in the market for grouping issuers by industry sector. The bank must assign each issuer to one of the sector buckets in the table above and it must assign all issuers from the same industry to the same sector. Risk positions from any issuer that a bank cannot assign to a sector in this fashion must be assigned to the "other sector" (ie bucket 11). For multinational multi-sector equity issuers, the allocation to a particular bucket must be done according to the most material region and sector in which the issuer operates.

Buckets for equity risk — Table 11

Bucket number	Size	Region	Sector
1	Large	Emerging market economies	Consumer goods and services, transportation and storage, administrative and support service activities, healthcare, utilities
2			Telecommunications, industrials
3			Basic materials, energy, agriculture, manufacturing, mining and quarrying
4			Financials including government-backed financials, real estate activities, technology
5		Advanced economies	Consumer goods and services, transportation and storage, administrative and support service activities, healthcare, utilities
6			Telecommunications, industrials
7			Basic materials, energy, agriculture, manufacturing, mining and quarrying
8			Financials including government-backed financials, real estate activities, technology
9	Small	Emerging market economies	All sectors described under bucket numbers 1, 2, 3, and 4
10		Advanced economies	All sectors described under bucket numbers 5, 6, 7, and 8
11	(Not applicable)		Other sector
12	Large cap, advanced economies		Qualified Indices
13	Other		Qualified Indices

50.71 For equity delta and vega risks, cross-bucket correlation γ is set at 15% for all cross-bucket pairs that fall within bucket numbers 1 to 10. The cross-bucket correlation between buckets 12 and 13 is set at 75% and the cross bucket correlation between buckets 12 or 13 and any of the buckets 1 to 10 is 45%. γ is set at 0% for all cross-bucket pairs that include bucket 11.

50.72 Equity delta risk factors for a given bucket:

(1) The single equity delta risk factor is a simultaneous relative shift of equity spot prices for all reference names in the bucket.

(2) The sensitivity to the equity delta risk factor is measured by simultaneously shifting the equity spot prices for all reference names in the bucket by 1% relative to their current

values and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.01.

(3) Risk weights RWk are set as follows depending on the reference name's bucket:

Risk weights for equity delta risk	Table 12
Bucket number	Risk weight
1	55%
2	60%
3	45%
4	55%
5	30%
6	35%
7	40%
8	50%
9	70%
10	50%
11	70%
12	15%
13	25%

50.73 Equity vega risk factors for a given bucket:

(1) The single equity vega risk factor is a simultaneous relative shift of the volatilities for all reference names in the bucket.

(2) The sensitivity to the equity vega risk factor is measured by simultaneously shifting the volatilities for all reference names in the bucket by 1% relative to their current values and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.01.

(3) The risk weights for equity volatilities RWk are set to 78% for large market capitalisation buckets and to 100% for the other buckets.

Commodity buckets, risk factors, sensitivities, risk weights and correlations

50.74 For commodity delta and vega risks, buckets are set as follows:

Buckets for commodity risk		Table 13
Bucket number	Commodity group	Examples
1	Energy – Solid combustibles	coal, charcoal, wood pellets, nuclear fuel (such as uranium)
2	Energy – Liquid combustibles	crude oil (such as Light-sweet, heavy, West Texas Intermediate and Brent); biofuels (such as bioethanol and biodiesel); petrochemicals (such as propane, ethane, gasoline, methanol and butane); refined fuels (such as jet fuel, kerosene, gasoil, fuel oil, naphtha, heating oil and diesel)
3	Energy – Electricity and carbon trading	electricity (such as spot, day-ahead, peak and off-peak); carbon emissions trading (such as certified emissions reductions, in-delivery month EU allowance, Regional Greenhouse Gas Initiative

		CO2 allowance and renewable energy certificates)
4	Freight	dry-bulk route (such as Capesize, Panamax, Handysize and Supramax); liquid-bulk/gas shipping route (such as Suezmax, Aframax and very large crude carriers)
5	Metals – non-precious	base metal (such as aluminium, copper, lead, nickel, tin and zinc); steel raw materials (such as steel billet, steel wire, steel coil, steel scrap and steel rebar, iron ore, tungsten, vanadium, titanium and tantalum); minor metals (such as cobalt, manganese, molybdenum)
6	Gaseous combustibles	natural gas; liquefied natural gas
7	Precious metals (including gold)	gold; silver; platinum; palladium
8	Grains & oilseed	corn; wheat; soybean (such as soybean seed, soybean oil and soybean meal); oats; palm oil; canola; barley; rapeseed (such as rapeseed seed, rapeseed oil, and rapeseed meal); red bean, sorghum; coconut oil; olive oil; peanut oil; sunflower oil; rice
9	Livestock & dairy	cattle (such live and feeder); hog; poultry; lamb; fish; shrimp; dairy (such as milk, whey, eggs, butter and cheese)
10	Softs and other agriculturals	cocoa; coffee (such as arabica and robusta); tea; citrus and orange juice; potatoes; sugar; cotton; wool; lumber and pulp; rubber
11	Other commodity	industrial minerals (such as potash, fertiliser and phosphate rocks), rare earths; terephthalic acid; flat glass

50.75 For commodity delta and vega risks, cross-bucket correlation γ is set at 20% for all cross-bucket pairs that fall within bucket numbers 1 to 10. γ is set at 0% for all cross-bucket pairs that include bucket 11.

50.76 Commodity delta risk factors for a given bucket:

(1) The single commodity delta risk factor is a simultaneous relative shift of the commodity spot prices fo all commodities in the bucket.

(2) The sensitivities to commodity delta risk factors are measured by simultaneously shifting the spot prices of all commodities in the bucket by 1% relative to their current values and dividing the resulting change in the aggregate CVA (or the value of CVA hedges) by 0.01.

(3) The risk weights RWk are set as follows depending on the reference name's bucket:

Risk weights for commodity delta risk										Table 14	
Bucket number	1	2	3	4	5	6	7	8	9	10	11
RW	30%	35%	60%	80%	40%	45%	20%	35%	25%	35%	50%

50.77 Commodity vega risk factors for a given bucket:

(1) The single commodity vega risk factor is a simultaneous relative shift of the volatilities for all commodities in the bucket.

(2) The sensitivity to the commodity vega risk factor is measured by simultaneously shifting the volatilities for all commodities in the bucket by 1% relative to their current values and dividing the resulting change in the aggregate CVA (or the value of CVA

hedges) by 0.01.

(3) The risk weights for commodity volatilities RWk are set to 100%.

MAR90

Transitional arrangements

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

Version effective as of 01 Jan 2023

First version in the format of the consolidated framework, updated to take account of the revised implementation date announced on 27 March 2020.

90.1 Banks are required to conduct the profit and loss (P&L) attribution (PLA) test beginning 1 January 2023 as set out in MAR32.3. The outcomes of the PLA test will be used for Pillar 2 purposes beginning 1 January 2023. The Pillar 1 capital requirement consequences of assignment to the PLA test amber zone or PLA test red zone, as set out in MAR32.43, MAR32.44 and MAR33.43, will apply beginning 1 January 2024.

MAR99

Guidance on use of the internal models approach

Reflects revisions to internal models approach and updated to take account of the revised implementation date announced on 27 March 2020.

Version effective as of 01 Jan 2023

Reflects revisions to internal models approach and updated to take account of the revised implementation date announced on 27 March 2020.

Trading desk-level backtesting

99.1 An additional consideration in specifying the appropriate risk measures and trading outcomes for profit and loss (P&L) attribution test and backtesting arises because the internally modelled risk measurement is generally based on the sensitivity of a static portfolio to instantaneous price shocks. That is, end-of-day trading positions are input into the risk measurement model, which assesses the possible change in the value of this static portfolio due to price and rate movements over the assumed holding period.

99.2 While this is straightforward in theory, in practice it complicates the issue of backtesting. For instance, it is often argued that neither expected shortfall nor value-at-risk measures can be compared against actual trading outcomes, since the actual outcomes will reflect changes in portfolio composition during the holding period. According to this view, the inclusion of fee income together with trading gains and losses resulting from changes in the composition of the portfolio should not be included in the definition of the trading outcome because they do not relate to the risk inherent in the static portfolio that was assumed in constructing the value-at-risk measure.

99.3 This argument is persuasive with regard to the use of risk measures based on price shocks calibrated to longer holding periods. That is, comparing the liquidity-adjusted time horizon 99th percentile risk measures from the internal models capital requirement with actual liquidity-adjusted time horizon trading outcomes would probably not be a meaningful exercise. In particular, in any given multi-day period, significant changes in portfolio composition relative to the initial positions are common at major trading institutions. For this reason, the backtesting framework described here involves the use of risk measures calibrated to a one-day holding period. Other than the restrictions mentioned in this paper, the test would be based on how banks model risk internally.

99.4 Given the use of one-day risk measures, it is appropriate to employ one-day trading outcomes as the benchmark to use in the backtesting programme. The same concerns about “contamination” of the trading outcomes discussed above continue to be relevant, however, even for one-day trading outcomes. That is, there is a concern that the overall one-day trading outcome is not a suitable point of comparison, because it reflects the effects of intraday trading, possibly including fee income that is booked in connection with the sale of new products.

99.5 On the one hand, intraday trading will tend to increase the volatility of trading outcomes and may result in cases where the overall trading outcome exceeds the risk measure. This event clearly does not imply a problem with the methods used to calculate the risk measure; rather, it is simply outside the scope of what the measure is intended to capture. On the other hand, including fee income may similarly distort the backtest, but in the other direction, since fee income often has annuity-like characteristics. Since this fee income is not typically included in the calculation of the risk measure, problems with the risk measurement model could be masked by including fee income in the definition of the trading outcome used for backtesting purposes.

99.6 To the extent that backtesting programmes are viewed purely as a statistical test of the integrity of the calculation of the risk measures, it is appropriate to employ a definition of daily trading outcome that allows for an uncontaminated test. To meet this standard, banks must have the capability to perform the tests based on the hypothetical changes in portfolio value that would occur were end-of-day positions to remain unchanged.

99.7 Backtesting using actual daily P&Ls is also a useful exercise since it can uncover cases where

the risk measures are not accurately capturing trading volatility in spite of being calculated with integrity.

99.8 For these reasons, the Committee requires banks to develop the capability to perform these tests using both hypothetical and actual trading outcomes. In combination, the two approaches are likely to provide a strong understanding of the relation between calculated risk measures and trading outcomes. The total number of backtesting exceptions for the purpose of the thresholds in MAR32.9 must be calculated as the maximum of the exceptions generated under hypothetical or actual trading outcomes.

Bank-wide backtesting

99.9 To place the definitions of three zones of the bank-wide backtesting in proper perspective, however, it is useful to examine the probabilities of obtaining various numbers of exceptions under different assumptions about the accuracy of a bank's risk measurement model.

99.10 Three zones have been delineated and their boundaries chosen in order to balance two types of statistical error:

(1) the possibility that an accurate risk model would be classified as inaccurate on the basis of its backtesting result, and

(2) the possibility that an inaccurate model would not be classified that way based on its backtesting result.

99.11 Table 1 reports the probabilities of obtaining a particular number of exceptions from a sample of 250 independent observations under several assumptions about the actual percentage of outcomes that the model captures (ie these are binomial probabilities). For example, the left-hand portion of Table 1 sets ou probabilities associated with an accurate model (that is, a true coverage level of 99%). Under these assumptions, the column labelled "exact" reports that exactly five exceptions can be expected in 6.7% of the samples.

Probabilities of exceptions from 250 independent observations										Table 1
Model is accurate			Model is inaccurate: possible alternative levels of coverage							
	Coverage = 99%		Coverage = 98%		Coverage = 97%		Coverage = 96%		Coverage = 95%	
	Exact	Type 1	Exact	Type 2	Exact	Type 2	Exact	Type 2	Exact	Type 2
0	8.1%	100.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
1	20.5%	91.9%	3.3%	0.6%	0.4%	0.0%	0.0%	0.0%	0.0%	0.0%
2	25.7%	71.4%	8.3%	3.9%	1.5%	0.4%	0.2%	0.0%	0.0%	0.0%
3	21.5%	45.7%	14.0%	12.2%	3.8%	1.9%	0.7%	0.2%	0.1%	0.0%
4	13.4%	24.2%	17.7%	26.2%	7.2%	5.7%	1.8%	0.9%	0.3%	0.1%
5	6.7%	10.8%	17.7%	43.9%	10.9%	12.8%	3.6%	2.7%	0.9%	0.5%
6	2.7%	4.1%	14.8%	61.6%	13.8%	23.7%	6.2%	6.3%	1.8%	1.3%
7	1.0%	1.4%	10.5%	76.4%	14.9%	37.5%	9.0%	12.5%	3.4%	3.1%
8	0.3%	0.4%	6.5%	86.9%	14.0%	52.4%	11.3%	21.5%	5.4%	6.5%
9	0.1%	0.1%	3.6%	93.4%	11.6%	66.3%	12.7%	32.8%	7.6%	11.9%
10	0.0%	0.0%	1.8%	97.0%	8.6%	77.9%	12.8%	45.5%	9.6%	19.5%
11	0.0%	0.0%	0.8%	98.7%	5.8%	86.6%	11.6%	58.3%	11.1%	29.1%

12	0.0%	0.0%	0.3%	99.5%	3.6%	92.4%	9.6%	69.9%	11.6%	40.2%
13	0.0%	0.0%	0.1%	99.8%	2.0%	96.0%	7.3%	79.5%	11.2%	51.8%
14	0.0%	0.0%	0.0%	99.9%	1.1%	98.0%	5.2%	86.9%	10.0%	62.9%
15	0.0%	0.0%	0.0%	100.0%	0.5%	99.1%	3.4%	92.1%	8.2%	72.9%

Notes to Table 1: The table reports both exact probabilities of obtaining a certain number of exceptions from a sample of 250 independent observations under several assumptions about the true level of coverage, as well as type 1 or type 2 error probabilities derived from these exac probabilities. The left-hand portion of the table pertains to the case where the model is accurate and its true level of coverage is 99%. Thus, th probability of any given observation being an exception is 1% (100% – 99% = 1%). The column labelled "exact" reports the probability of obtainin exactly the number of exceptions shown under this assumption in a sample of 250 independent observations. The column labelled "type 1" report the probability that using a given number of exceptions as the cut-off for rejecting a model will imply erroneous rejection of an accurate model usin a sample of 250 independent observations. For example, if the cut-off level is set at five or more exceptions, the type 1 column reports the probabilit of falsely rejecting an accurate model with 250 independent observations is 10.8%. The right-hand portion of the table pertains to models that ar inaccurate. In particular, the table concentrates of four specific inaccurate models, namely models whose true levels of coverage are 98%, 97%, 96% and 95% respectively. For each inaccurate model, the exact column reports the probability of obtaining exactly the number of exceptions show under this assumption in a sample of 250 independent observations. The type 2 columns report the probability that using a given number o exceptions as the cut-off for rejecting a model will imply erroneous acceptance of an inaccurate model with the assumed level of coverage using sample of 250 independent observations. For example, if the cut-off level is set at five or more exceptions, the type 2 column for an assume coverage level of 97% reports the probability of falsely accepting a model with only 97% coverage with 250 independent observations is 12.8%.

99.12 The right-hand portion of the table reports probabilities associated with several possible inaccurate models, namely models whose true levels of coverage are 98%, 97%, 96%, and 95%, respectively. Thus, the column labelled “exact” under an assumed coverage level of 97% shows that five exceptions would then be expected in 10.9% of the samples.

99.13 Table 1 also reports several important error probabilities. For the assumption that the model covers 99% of outcomes (the desired level of coverage), the table reports the probability that selecting a given number of exceptions as a threshold for rejecting the accuracy of the model will result in an erroneous rejection of an accurate model (type 1 error). For example, if the threshold is set as low as one exception, then accurate models will be rejected fully 91.9% of the time, because they will escape rejection only in the 8.1% of cases where they generate zero exceptions. As the threshold number of exceptions is increased, the probability of making this type of error declines.

99.14 Under the assumptions that the model’s true level of coverage is not 99%, the table reports the probability that selecting a given number of exceptions as a threshold for rejecting the accuracy of the model will result in an erroneous acceptance of a model with the assumed (inaccurate) level of coverage (type 2 error). For example, if the model’s actual level of coverage is 97%, and the threshold for rejection is set at seven or more exceptions, the table indicates that this model would be erroneously accepted 37.5% of the time.

99.15 The results in Table 1 also demonstrate some of the statistical limitations of backtesting. In particular, there is no threshold number of exceptions that yields both a low probability of erroneously rejecting an accurate model and a low probability of erroneously accepting all of the relevant inaccurate models. It is for this reason that the Committee has rejected an approach that contains only a single threshold.

99.16 Given these limitations, the Committee has classified outcomes for the backtesting of the bank-wide model into three categories. In the first category, the test results are consistent with an accurate model, and the possibility of erroneously accepting an inaccurate model is low (ie backtesting ”green zone”). At the other extreme, the test results are extremely unlikely to have resulted from an accurate model, and the probability of erroneously

rejecting an accurate model on this basis is remote (ie backtesting "red zone"). In between these two cases, however, is a zone where the backtesting results could be consistent with either accurate or inaccurate models, and the supervisor should encourage a bank to present additional information about its model before taking action (ie backtesting "amber zone").

99.17 Table 2 sets out the Committee's agreed boundaries for these zones and the presumptive supervisory response for each backtesting outcome, based on a sample of 250 observations. For other sample sizes, the boundaries should be deduced by calculating the binomial probabilities associated with true coverage of 99%, as in Table 1. The backtesting amber zone begins at the point such that the probability of obtaining that number or fewer exceptions equals or exceeds 95%. Table 2 reports these cumulative probabilities for each number of exceptions. For 250 observations, it can be seen that five or fewer exceptions will be obtained 95.88% of the time when the true level of coverage is 99%. Thus, the backtesting amber zone begins at five exceptions. Similarly, the beginning of the backtesting red zone is defined as the point such that the probability of obtaining that number or fewer exceptions equals or exceeds 99.99%. Table 2 shows that for a sample of 250 observations and a true coverage level of 99%, this occurs with 10 exceptions.

Backtesting zone boundaries — Table 2

Backtesting zone	Number of exceptions	Backtesting-dependent multiplier (to be added to any qualitative add-on per MAR33.44)	Cumulative probability
Green	0 1 2 3 4	1.50 1.50 1.50 1.50 1.50	8.11% 28.58% 54.32% 75.81% 89.22%
Amber	5 6 7 8 9	1.70 1.76 1.83 1.88 1.92	95.88% 98.63% 99.60% 99.89% 99.97%
Red	10 or more	2.00	99.99%

Notes to Table 2: The table defines the backtesting green, amber and red zones that supervisors will use to assess backtesting results in conjunction with the internal models approach to market risk capital requirements. The boundaries shown in the table are based on a sample of 250 observations. For other sample sizes, the amber zone begins at the point where the cumulative probability equals or exceeds 95%, and the red zone begins at the point where the cumulative probability equals or exceeds 99.99%. The cumulative probability is simply the probability of obtaining a given number or fewer exceptions in a sample of 250 observations when the true coverage level is 99%. For example, the cumulative probability shown for four exceptions is the probability of obtaining between zero and four exceptions. Note that these cumulative probabilities and the type 1 error probabilities reported in Table 1 do not sum to one because the cumulative probability for a given number of exceptions includes the possibility of obtaining exactly that number of exceptions, as does the type 1 error probability. Thus, the sum of these two probabilities exceeds one by the amount of the probability of obtaining exactly that number of exceptions.

99.18 The backtesting green zone needs little explanation. Since a model that truly provides 99% coverage would be quite likely to produce as many as four exceptions in a sample of 250 outcomes, there is little reason for concern raised by backtesting results that fall in this range. This is reinforced by the results in Table 1, which indicate that accepting outcomes in this range leads to only a small chance of erroneously accepting an inaccurate model.

99.19 The range from five to nine exceptions constitutes the backtesting amber zone. Outcomes in this range are plausible for both accurate and inaccurate models, although Table 1 suggests that they are generally more likely for inaccurate models than for accurate models. Moreover, the results in Table 1 indicate that the presumption that the model is inaccurate should grow as the number of exceptions increases in the range from five to nine.

99.20 Table 2 sets out the Committee's agreed guidelines for increases in the multiplication factor applicable to the internal models capital requirement, resulting from backtesting results in the backtesting amber zone.

99.21 These particular values reflect the general idea that the increase in the multiplication factor should be sufficient to return the model to a 99th percentile standard. For example, five exceptions in a sample of 250 imply only 98% coverage. Thus, the increase in the multiplication factor should be sufficient to transform a model with 98% coverage into one with 99% coverage. Needless to say, precise calculations of this sort require additional statistical assumptions that are not likely to hold in all cases. For example, if the distribution of trading outcomes is assumed to be normal, then the ratio of the 99th percentile to the 98th percentile is approximately 1.14, and the increase needed in the multiplication factor is therefore approximately 1.13 for a multiplier of 1. If the actual distribution is not normal, but instead has "fat tails", then larger increases may be required to reach the 99th percentile standard. The concern about fat tails was also an important factor in the choice of the specific increments set out in Table 2.

Examples of the application of the principles for risk factor modellability

99.22 Although supervisors may use discretion regarding the types of evidence required of banks to provide risk factor modellability, the following are examples of the types of evidence that banks may be required to provide.

(1) Regression diagnostics for multi-factor beta models. In addition to showing that indices or other regressors are appropriate for the region, asset class and credit quality (if applicable) of an instrument, banks must be prepared to demonstrate that the coefficients used in multi-factor models are adequate to capture both general market risk and idiosyncratic risk. If the bank assumes that the residuals from the multi-factor model are uncorrelated with each other, the bank should be prepared to demonstrate that the modellable residuals are uncorrelated. Further, the factors in the multi-factor model must be appropriate for the region and asset class of the instrument and must explain the general market risk of the instrument. This must be demonstrated through goodness-of-fit statistics (eg an adjusted-R2 coefficient) and other diagnostics on the coefficients. Most importantly, where the estimated coefficients are not used (ie the parameters are judgment-based), the bank must describe how the coefficients are chosen and why they cannot be estimated, and demonstrate that the choice does not underestimate risk. In general, risk factors are not considered modellable in cases where parameters are set by judgment.

(2) Recovery of price from risk factors. The bank must periodically demonstrate and document that the risk factors used in its risk model can be fed into front office pricing models and recover the actual prices of the assets. If the recovered prices substantially deviate from the actual prices, this can indicate a problem with prices used to derive the risk factors and call into question the validity of data inputs for risk purposes. In such cases, supervisors may determine that the risk factor is non-modellable.

(3) Risk pricing is periodically reconciled with front office and back office prices. While banks are free to use price data from external sources, these external prices should periodically be reconciled with internal prices (from both front office and back office) to ensure they do not deviate substantially, and that they are not consistently biased in any fashion. Results of these reconciliations should be made available to supervisors, including statistics on the differences of the risk price from front office and back office prices. It is standard practice for banks to conduct reconciliation of front office and back office prices; the risk prices must be included as part of the reconciliation of the front office and whenever there is a potential for discrepancy. If the discrepancy is

large, supervisors may determine that the risk factor is non-modellable.

(4) Risk factor backtesting. Banks must periodically demonstrate the appropriateness of their modelling methodology by comparing the risk factor returns forecast produced by the risk management model with actual returns produced by front office prices. Alternatively, a bank could backtest hypothetical portfolios that are substantively dependent on key risk factors (or combinations thereof). This risk factor backtesting is intended to confirm that risk factors accurately reflect the volatility and correlations of the instruments in the risk model. Hypothetical backtesting can be effective in identifying whether risk factors in question adequately reflect volatility and correlations when the portfolio of instruments is chosen to highlight specific products.

(5) Risk factors generated from parameterised models. For options, implied volatility surfaces are often built using a parameterised model based on single-name underlyings and/or option index RPOs and/or market quotes. Liquid options at moneyness, tenor and option expiry points may be used to calibrate level, volatility, drift and correlation parameters for a single-name or benchmark volatility surface. Once these parameters are set, they are derived risk factors in their own right that must be updated and recalibrated periodically as new data arrive and trades occur. In the event that these risk factors are used to proxy for other single-name option surface points, there must be an additional-basis non-modellable risk factor overlay for any potential deviations.